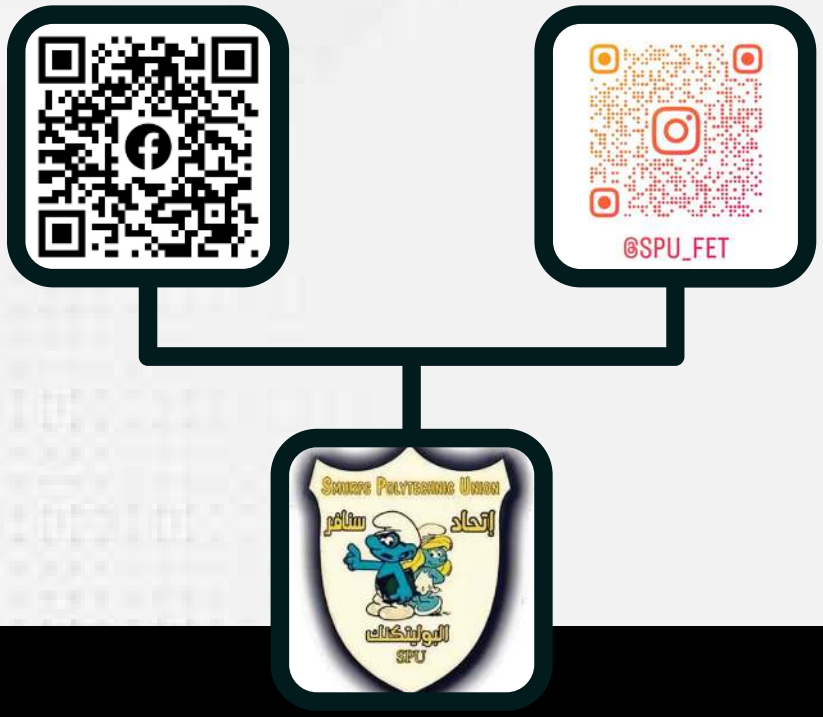
سلايدات الدكتورة هبة
كيمياء 101

▼ من له شغف في طلب العلم، لا يتعبه بحث ولا يرهقه جهد، نحن أمة اقرأ.. أبناء القلم



GENERAL CHEMISTRY I
30206101
I ST 2023-2024

CH7
LECTURE 1-3
WEEK 1

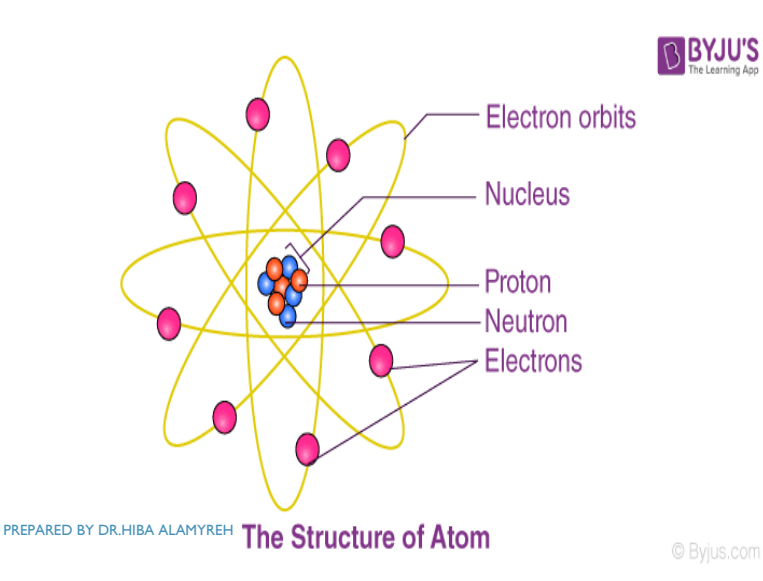
PREPARED BY DR.HIBA ALAMYREH

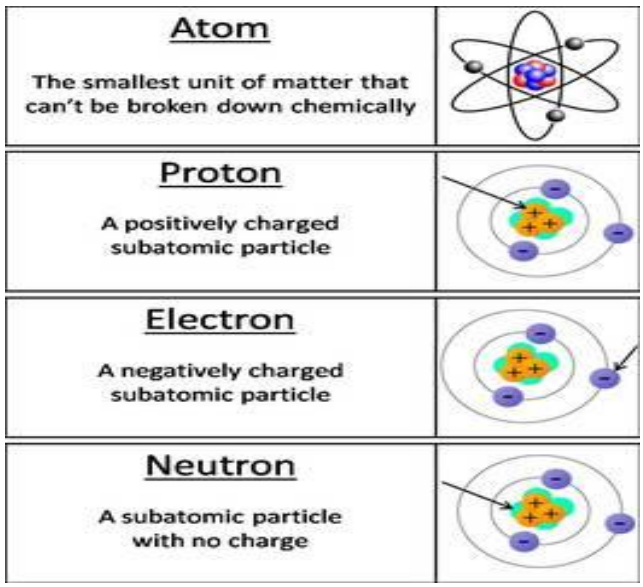
TOPICS

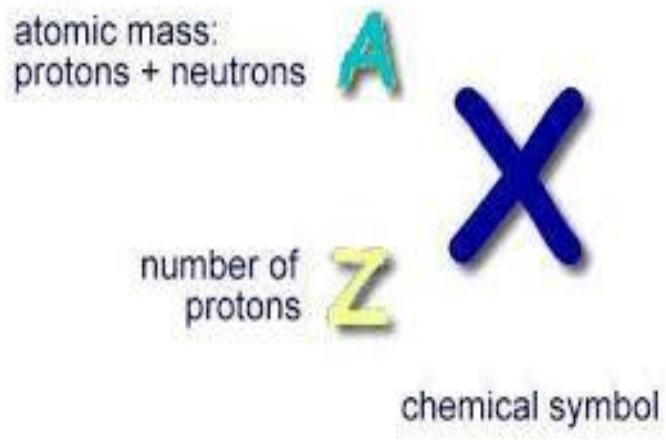
- Quantum Numbers
- Atomic Orbitals
- Building up principle
- Electronic configuration of atom

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STRUCTURE OF ATOM






atomic mass:
protons + neutrons
A
X
number of
protons
Z
chemical symbol

ORBITALS CHARACTERIZED BY THREE QUANTUM NUMBERS:

Quantum Numbers:

- Shorthand
- Describes characteristics of electron's position
- Predicts its behavior

***n* = principal quantum number**

- All orbitals with same ***n*** are in same **shell**

ℓ = secondary quantum number

- Divides shells into smaller groups called **subshells**

m_ℓ = magnetic quantum number

- Divides subshells into individual **orbitals**

m_s = magnetic spin quantum number

PRINCIPAL QUANTUM NUMBER n

- Allowed values: positive integers from **1 to** ∞
  - ***n*** **= 1, 2, 3, 4, 5, … ∞**
- Determines:
  - **Size** of orbital
  - Total **energy** of orbital
  - Number of **nodes** (points where $\psi\psi^* = 0$)
- **$R_H hc = 2.18 \times 10^{-18}$ J/atom**
- For given atom,
  - Lower **n** = Lower (more negative) **E**

    = More stable

$$E = -\frac{Z^2 R_H hc}{n^2}$$

ℓ = ORBITAL ANGULAR MOMENTUM QN

- Allowed values: **0, 1, 2, 3, 4, 5…(*n* – 1)**
- Letters: **s, p, d, f, g, h**

Orbital designation

- number → ***nℓ*** ← **letter**
- Possible values of ℓ depend on ***n***
- ***n*** different values of ℓ for given ***n***
- Specifies orbital angular momentum of e^-
- Determines : **Shape** of orbital
- Angular variation of e^- path
- **Kinetic energy** of orbital

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M_ℓ = MAGNETIC QUANTUM NUMBER

- Allowed values:
- $m_\ell = -\ell, -\ell+1, -\ell+2, \ldots, 0, \ldots, \ell-2, \ell-1, \ell$
- Possible values of m_ℓ depend on ℓ
  - There are $2\ell + 1$ different values of m_ℓ for given ℓ
- **z** axis component of orbital angular momentum
- Determines **orientation** of orbital in space
- To designate specific orbital, you need three quantum numbers (n, ℓ, m_ℓ)

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TABLE SUMMARY OF RELATIONSHIPS

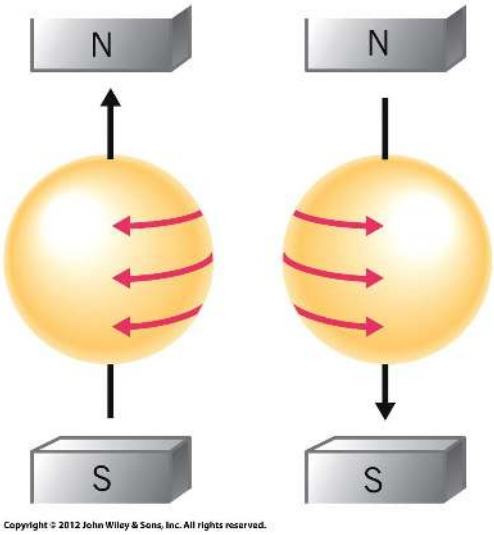
Table 8.1 **Summary of Relationships among the Quantum Numbers n, ℓ, and m_ℓ**

Value of n	Value of ℓ	Value of m_ℓ	Subshell	Number of Orbitals
1	0	0	$1s$	1
2	0	0	$2s$	1
	1	$-1, 0, 1$	$2p$	3
3	0	0	$3s$	1
	1	$-1, 0, 1$	$3p$	3
	2	$-2, -1, 0, 1, 2$	$3d$	5
4	0	0	$4s$	1
	1	$-1, 0, 1$	$4p$	3
	2	$-2, -1, 0, 1, 2$	$4d$	5
	3	$-3, -2, -1, 0, 1, 2, 3$	$4f$	7

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SPIN QUANTUM NUMBER, M_S

- Arises out of behavior of e^- in magnetic field
- e^- acts like a top
- Spinning charge is like a magnet
  - e^- behave like tiny magnets
- Leads to 2 possible directions of e^- spin
  - up and down
  - north and south



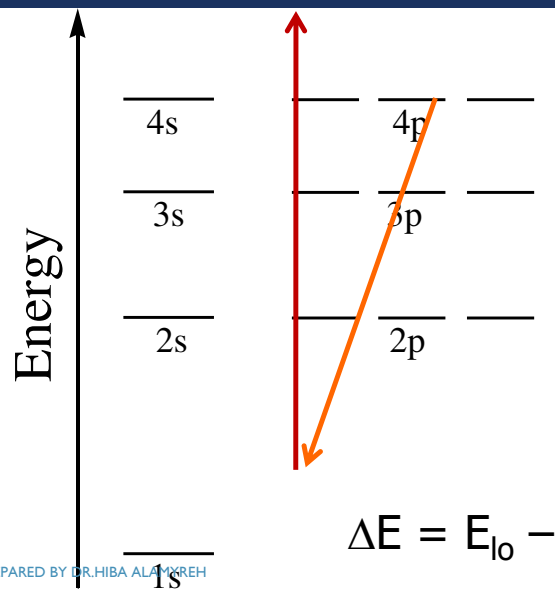
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Possible Values:

$+\frac{1}{2}$ $-\frac{1}{2}$

↑ ↓

ENERGY LEVEL DIAGRAM FOR H ATOM OR OTHER 1 E^- ION



- All orbital subshells with same *n* value have same Energy

$$E = -\frac{Z^2 R_H hc}{n^2}$$

$$\Delta E = E_{lo} - E_{hi}$$

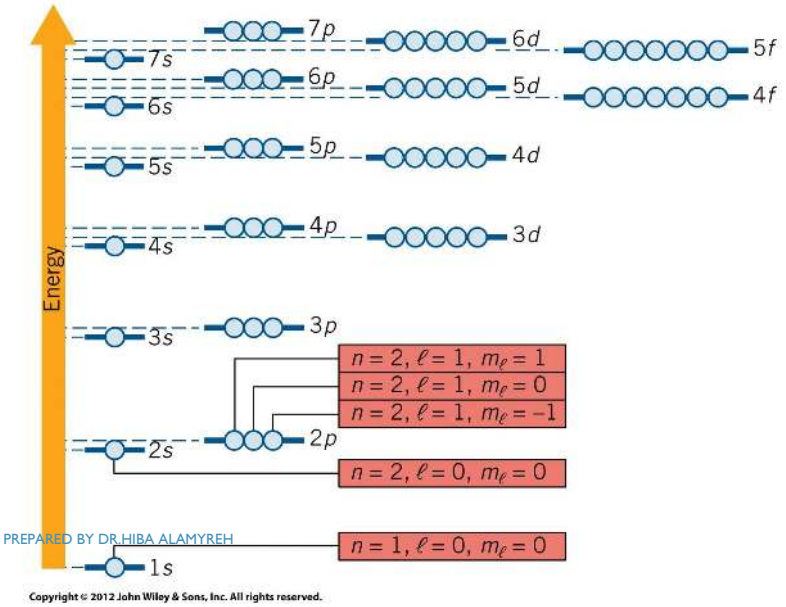
$$\Delta E = Z^2 R_H hc\left(\frac{1}{n_{hi}^2} - \frac{2}{n_{lo}^2}\right)$$

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ORBITAL ENERGIES IN MANY ELECTRON ATOMS

1. Each orbital represented by circle or line
2. Now different subshells (ℓ values) have different E
3. All orbitals of same subshell = same E
4. As you go up in energy, spacing between successive shells (***n*** values) decreases as number of subshells increases
   - Leads to overlapping of several subshells 4s/3d 5s/4d 6s/4f/5d

ORBITALS OF MANY ELECTRONS



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Orbital Designation

- Based on first 2 quantum numbers
- number for n and letter for ℓ
- How many e^- can go in each orbital?
  - Need another quantum number

PAULI EXCLUSION PRINCIPLE

- No two e^- in same atom can have same set of all four quantum numbers (n, ℓ, m_ℓ, m_s)
- **Can only have 2 e^- per orbital**
- 2 e^- s in same orbital must have opposite spin (e^- s are **paired)**
- Odd number of e^-s
  - Not all spins paired
  - Have **unpaired** e^-s
- Even number of e^-s
  - Depends on number of orbitals

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CONSEQUENCES OF PAULI EXCLUSION

Subshell	Number of Orbitals	Maximum Number of Electrons
s	1	2
p	3	6
d	5	10
f	7	14

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Shell	Subshells	Maximum Shell Population	
1	$1s$	2	
2	$2s\ 2p$	8	$(2 + 6)$
3	$3s\ 3p\ 3d$	18	$(2 + 6 + 10)$
4	$4s\ 4p\ 4d\ 4f$	32	$(2 + 6 + 10 + 14)$

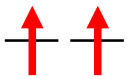
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KNOW FROM MAGNETIC PROPERTIES

- Two e⁻ s in **same** orbital with **different** spin
  - Spins paired—**diamagnetic**
  - Sample not attracted to magnetic field
  - Magnetic effects tend to cancel each other



- Two e⁻ s in **different** orbital with **same** spin
  - Spins unpaired—**paramagnetic**
  - Sample pulled into magnetic field
  - Magnetic effects add



- Measure extent of attraction

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  - Gives number of unpaired spins

YOUR TURN!

Which of the following is a **valid** set of four quantum numbers ($\mathbf{n, \ell, m_\ell, m_s}$)?

A. 3, 2, 3, +½

B. 3, 2, 1, 0

C. 3, 0, 0, -½

D. 3, 3, 0, +½

E. 0, -1, 0, -½

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YOUR TURN!

What is the maximum number of electrons allowed in a set of **4*p*** orbitals?

A. 14

B. 6

C. 0

D. 2

E. 10

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Electron Configurations

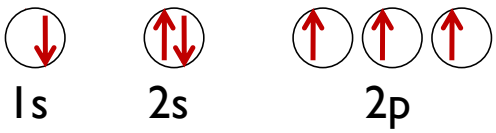
- Distribution of **e⁻**s among orbitals of atom
  1. List subshells that contain electrons
  2. Indicate their electron population with superscript

Ex. N is $1s^2\ 2s^2\ 2p^3$

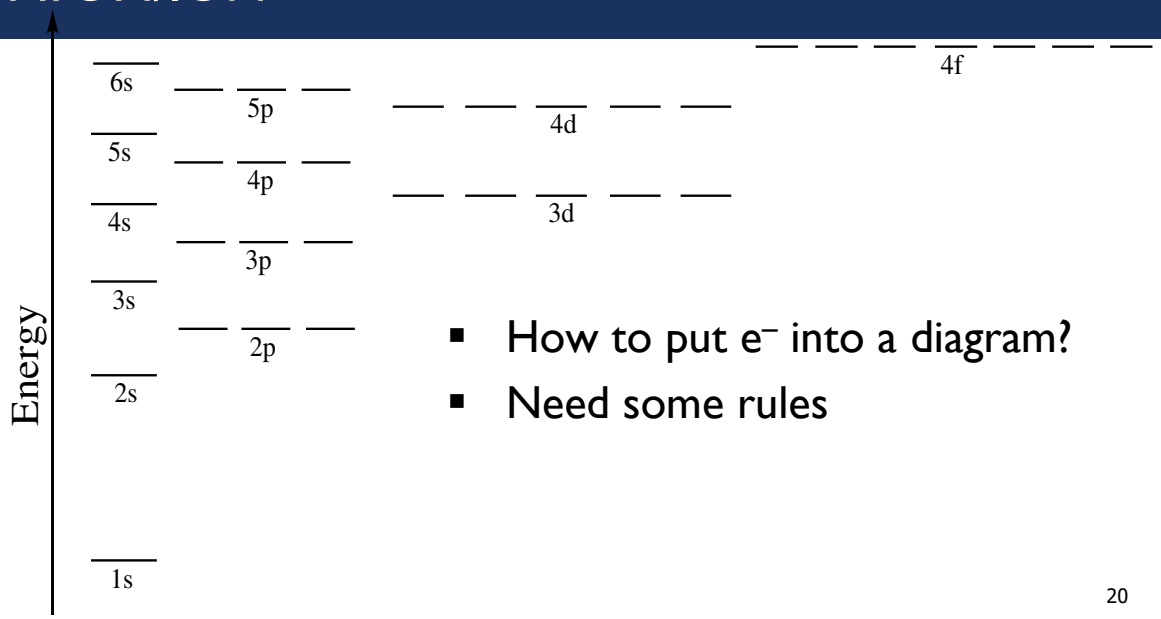
Orbital Diagrams

- Way to represent **e⁻**s in orbitals
  1. Represent each orbital with circle (or line)
  2. Use arrows to indicate spin of each electron

Ex. N is



ENERGY LEVEL DIAGRAM FOR MULTI ELECTRON ATOM/ION



- How to put e^- into a diagram?
- Need some rules

AUFBAU PRINCIPLE

- Building-up principle
  - Fill lowest energy subshell before going to next highest energy subshell
  - Fill lowest ***n*** first
  - Within given ***n*,** fill lowest ℓ first
  - Within given ℓ**,** fill highest $\mathbf{m}_\ell$ first
  - Within given $\mathbf{m}_\ell$, fill highest $\mathbf{m}_s$ (+ ½ , ↑) first

Pauli Exclusion Principle

- 2 e^- per orbital
- Spins opposite

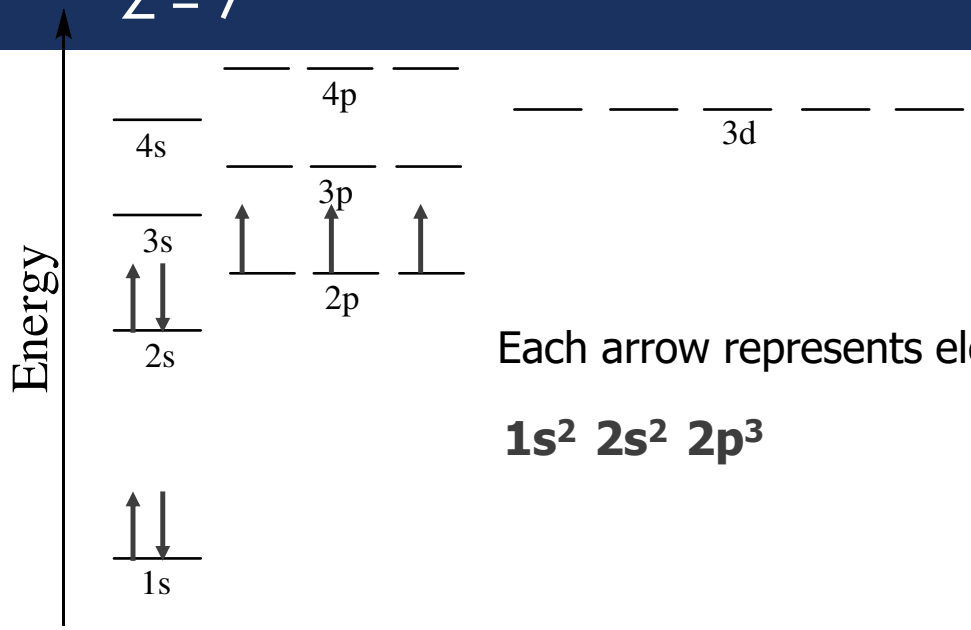
HUND'S RULE

- If you have more than 1 orbital all at the same energy
  - **Put 1e⁻ into each orbital with spins parallel (all up)** until all are half filled
  - **Before pair up e⁻s** in same orbital

Why?

- Repulsion of e⁻ in same region of space
- Empirical observation based on magnetic properties

ORBITAL DIAGRAM AND E⁻ CONFIGURATIONS:
N Z = 7

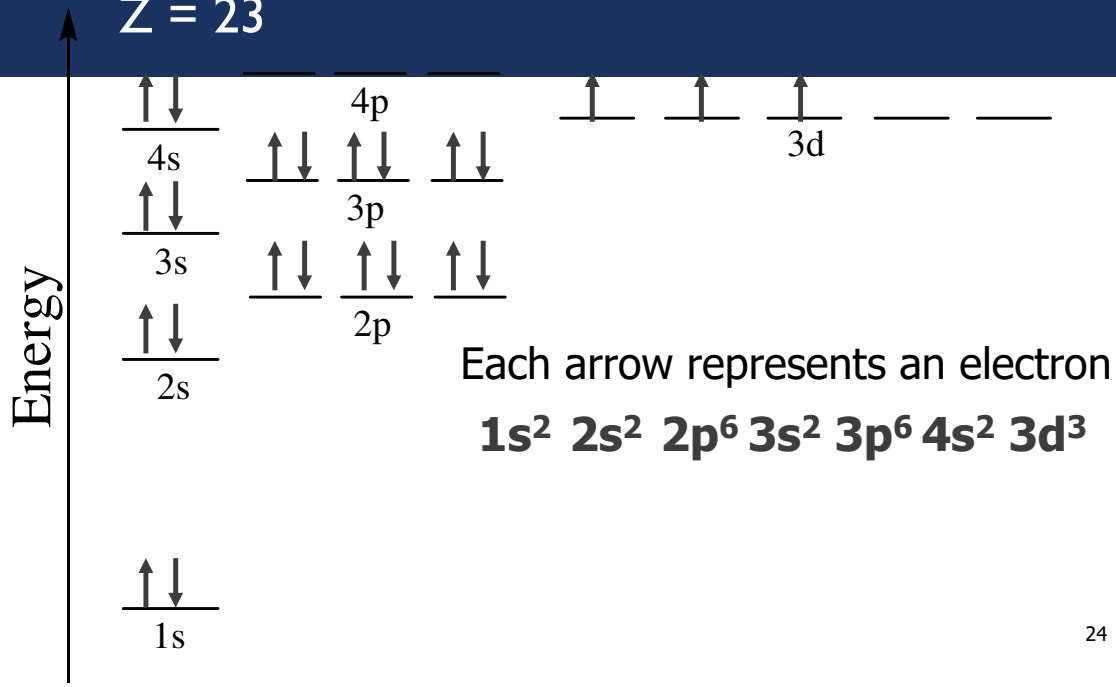


Each arrow represents electron

$1s^2$ $2s^2$ $2p^3$

ORBITAL DIAGRAM AND E- CONFIGURATIONS:

V Z = 23

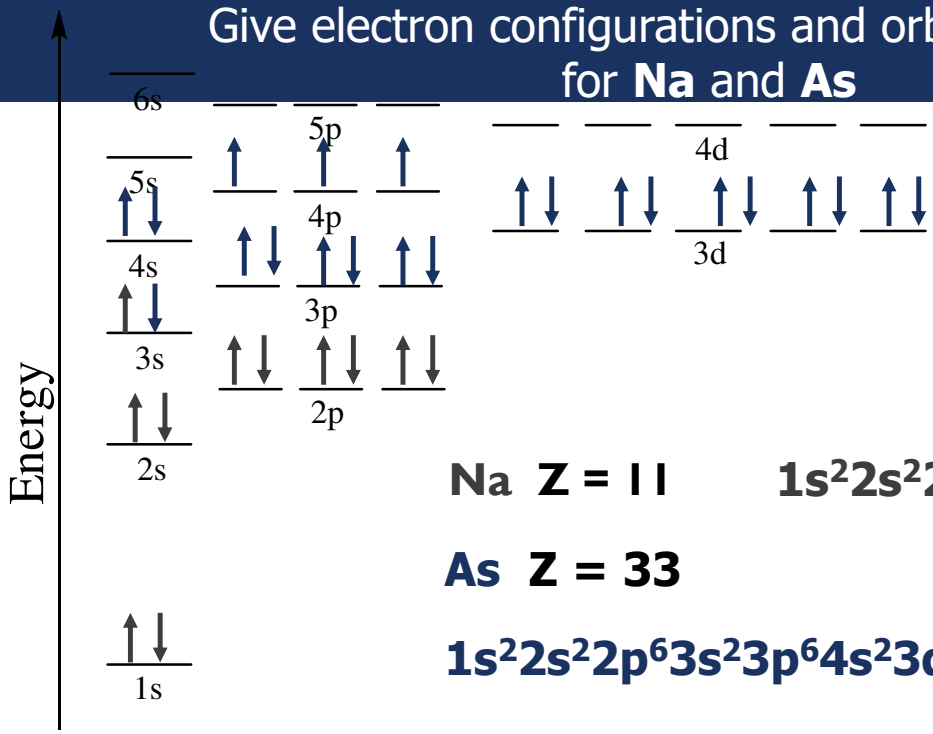


Each arrow represents an electron

$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^3$

LEARNING CHECK

Give electron configurations and orbital diagrams for **Na** and **As**



Na Z = 11 $1s^2 2s^2 2p^6 3s^1$

As Z = 33

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$

YOUR TURN!

The ground state electron configuration for Ca is:

A. [Ar] $3s^1$

B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5\ 4s^2$

C. [Ar] $4s^2$

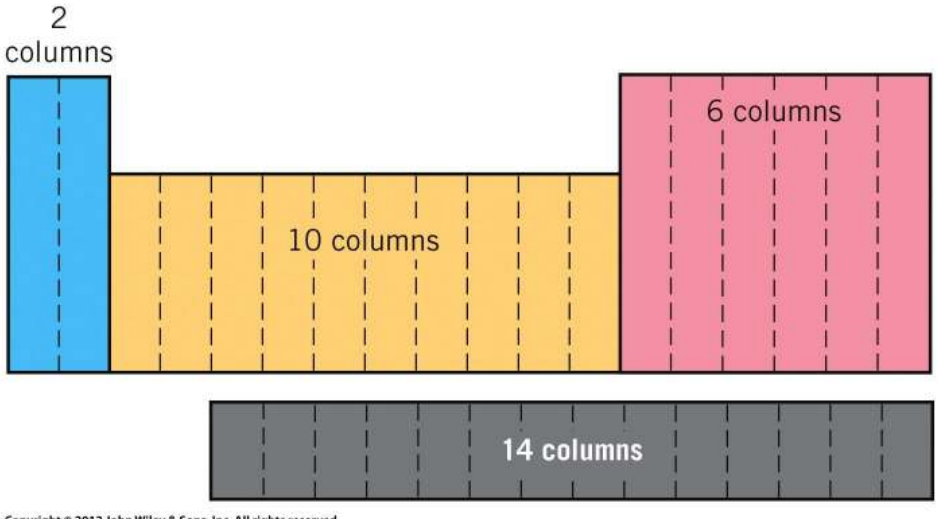
D. [Kr] $4s^1$

E. [Kr] $4s^2$

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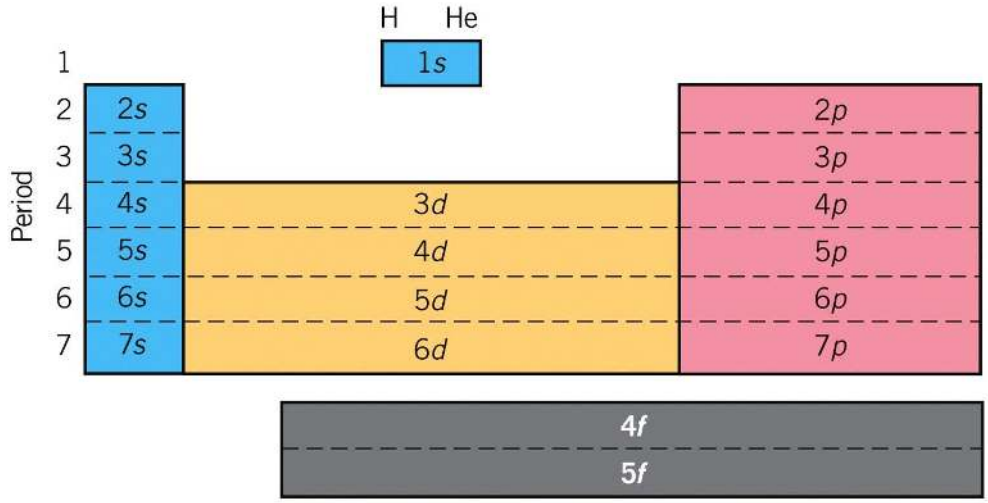
PERIODIC TABLE

- Divided into regions of 2, 6, 10, and 14 columns
- = maximum number of electrons in *s*, *p*, *d*, and *f* sublevels



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- Each row (period) represents different energy level
- Each region of chart represents different type of sublevel



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NOW READY TO PUT E⁻S INTO ATOMS

Electron configurations must be consistent with:

Pauli Exclusion principle

- 2 e^- per orbital, spins opposite

Aufbau principle

- Start at lowest energy orbital
- Fill, then move up

Hund's rule

- 1 e^- in each orbital of same, spins parallel
- Only pair up if have to

WHERE ARE THE ELECTRONS?

- Each box represents room for electron
- Read from left to right

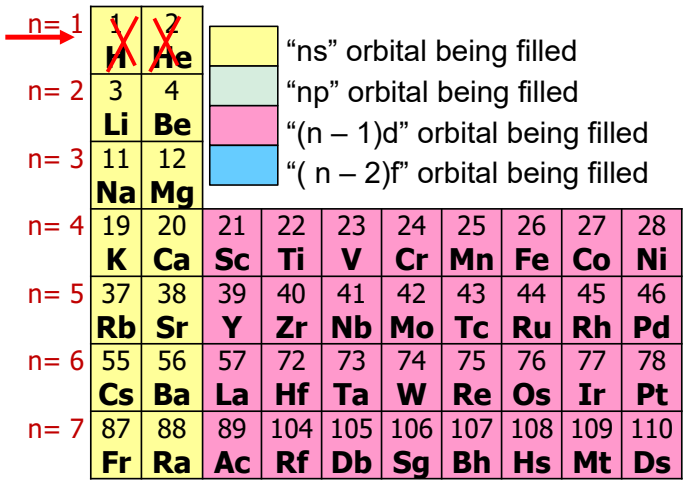
n= 1	1 **H**	2 **He**															
n= 2	3 **Li**	4 **Be**											5 **B**	6 **C**	7 **N**	8 **O**	9 **F**	**10 Ne**
n= 3	11 **Na**	12 **Mg**											13 **Al**	14 **Si**	15 **P**	16 **S**	17 **Cl**	18 **Ar**
n= 4	19 **K**	20 **Ca**	21 **Sc**	22 **Ti**	23 **V**	24 **Cr**	25 **Mn**	26 **Fe**	27 **Co**	28 **Ni**	29 **Cu**	30 **Zn**	31 **Ga**	32 **Ge**	33 **As**	34 **Se**	35 **Br**	36 **Kr**
n= 5	37 **Rb**	38 **Sr**	39 **Y**	40 **Zr**	41 **Nb**	42 **Mo**	43 **Tc**	44 **Ru**	45 **Rh**	46 **Pd**	47 **Ag**	48 **Cd**	49 **In**	50 **Sn**	51 **Sb**	52 **Te**	53 **I**	54 **Xe**
n= 6	55 **Cs**	56 **Ba**	57 **La**	72 **Hf**	73 **Ta**	74 **W**	75 **Re**	76 **Os**	77 **Ir**	78 **Pt**	79 **Au**	80 **Hg**	81 **Tl**	82 **Pb**	83 **Bi**	84 **Po**	85 **At**	86 **Rn**
n= 7	87 **Fr**	88 **Ra**	89 **Ac**	104 **Rf**	105 **Db**	106 **Sg**	107 **Bh**	108 **Hs**	109 **Mt**	110 **Ds**	111 **Rg**							

"ns" orbital being filled
"np" orbital being filled
"(n – 1)d" orbital being filled
"(n – 2)f" orbital being filled

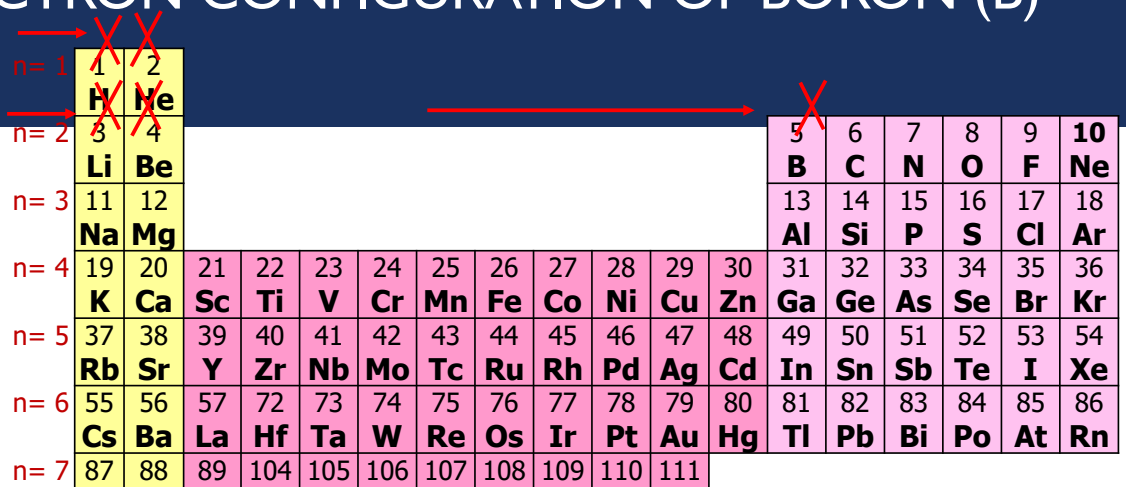
58 **Ce**	59 **Pr**	60 **Nd**	61 **Pm**	62 **Sm**	63 **Eu**	64 **Gd**	65 **Tb**	66 **Dy**	67 **Ho**	68 **Er**	69 **Tm**	70 **Yb**	71 **Lu**
90 **Th**	91 **Pa**	92 **U**	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lr**

READ PERIODIC TABLE TO DETERMINE E-

- Read from left to right
- 1st e⁻ goes into period 1
- 1st type of sublevel to fill = "1s"
- He has 2 e⁻
- e⁻ configuration for He is: **$1s^2$**



ELECTRON CONFIGURATION OF BORON (B)



- B has 5 e^-
- Fill first shell…
- Fill two subshells in 2nd shell, in order of increasing E
- Electron Configuration B = **$1s^2 2s^2 2p^1$**

GENERAL CHEMISTRY 1
30206101

CH8
LECTURE 4-7

PREPARED BY:DR HIBA ALAMYREH WITH SPECIAL THANKS TO DR.REEM NASER ENCAUSTIC NOTES

Although in principle an electron can be found anywhere, we know that most of the time it is quite close to the nucleus. Figure 7.15(a) shows the distribution of electron density in a hydrogen 1*s* orbital moving outward from the nucleus. As you can see, the electron density falls off rapidly as the distance from the nucleus increases. Roughly speaking, there is about a 90 percent probability of finding the electron within a sphere of radius 100 pm (1 pm $= 1 \times 10^{-12}$ m) surrounding the nucleus. Thus, we can represent the 1*s* orbital by drawing a ***boundary surface diagram*** that *encloses about 90 percent of the total electron density in an orbital,* as shown in Figure 7.15(b). A 1*s* orbital represented in this manner is merely a sphere.

Figure 7.16 shows boundary surface diagrams for the 1*s*, 2*s*, and 3*s* hydrogen atomic orbitals. All *s* orbitals are spherical in shape but differ in size, which increases as the principal quantum number increases. Although the details of electron density variation within each boundary surface are lost, there is no serious disadvantage. For us the most important features of atomic orbitals are their shapes and *relative* sizes, which are adequately represented by boundary surface diagrams.

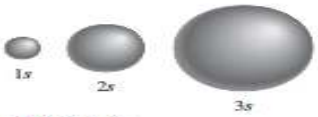


Figure 7.16
Boundary surface diagrams of the hydrogen 1s, 2s, and 3s orbitals. Each sphere contains about 90 percent of the total electron density. All s orbitals are spherical. Roughly speaking, the size of an orbital is proportional to n^2, where n is the principal quantum number.

p Orbitals

It should be clear that the *p* orbitals start with the principal quantum number $n = 2$. If $n = 1$, then the angular momentum quantum number ℓ can assume only the value of zero; therefore, there is only a 1*s* orbital. As we saw earlier, when $\ell = 1$, the magnetic quantum number m_ℓ can have values of -1, 0, and 1. Starting with $n = 2$ and $\ell = 1$, we therefore have three 2*p* orbitals: $2p_x$, $2p_y$, and $2p_z$ (Figure 7.17). The letter subscripts indicate the axes along which the orbitals are oriented. These three

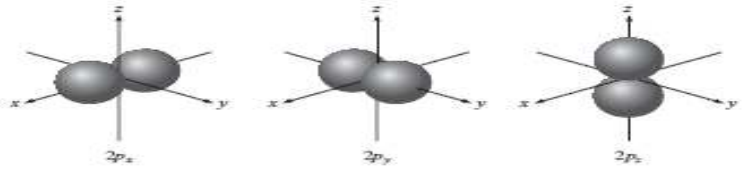


Figure 7.17
The boundary surface diagrams of the three 2p orbitals. These orbitals are identical in shape and energy, but their orientations are different. The p orbitals of higher principal quantum numbers have a similar shape.

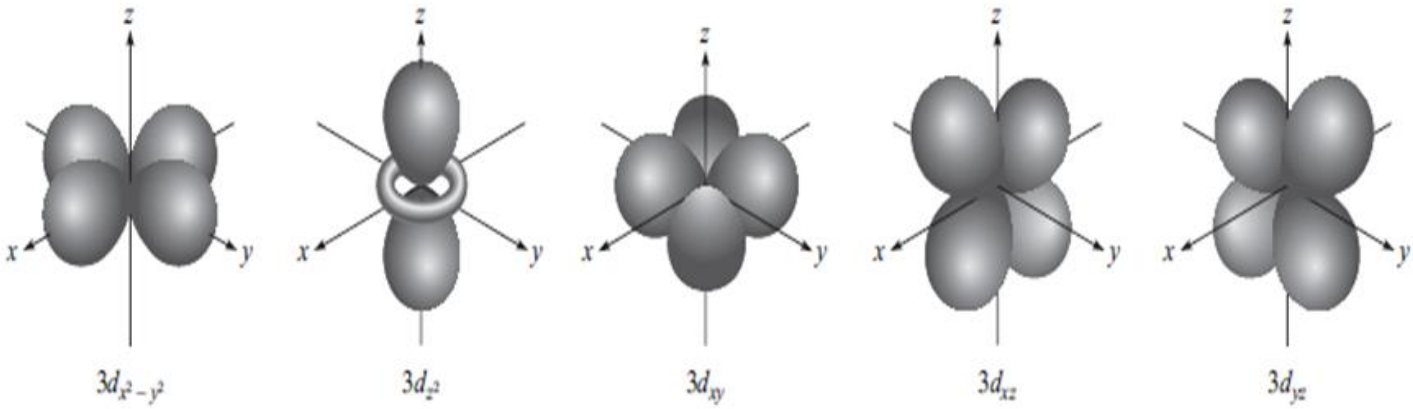
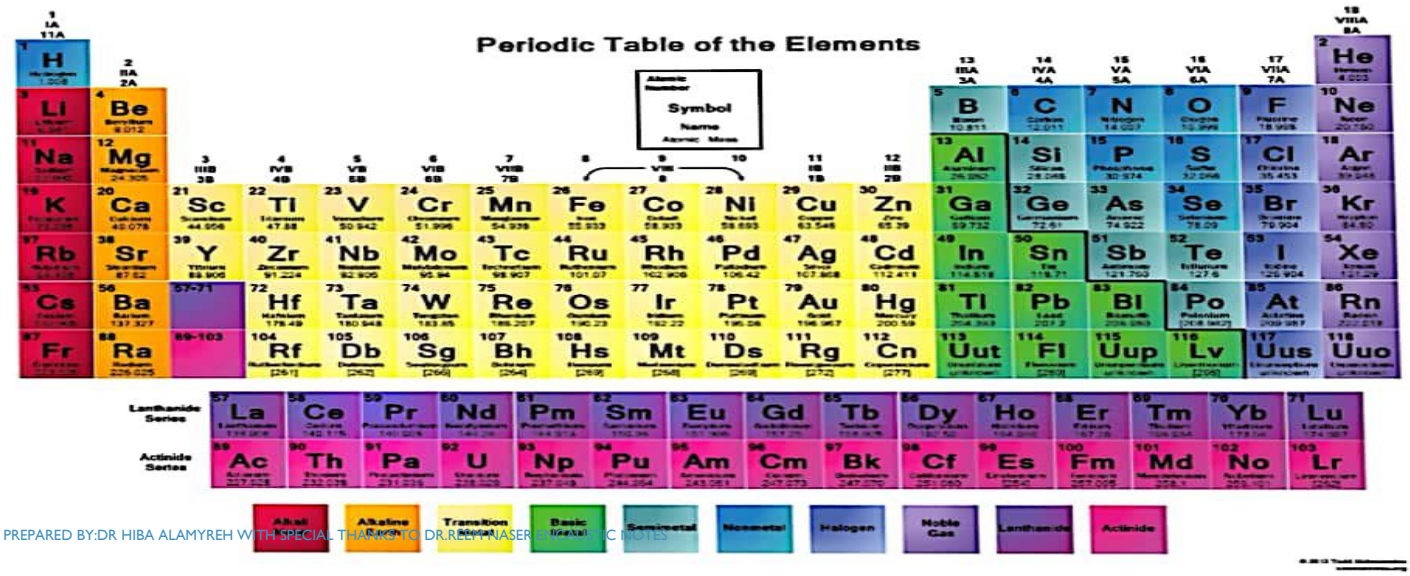


Figure 7.18

Boundary surface diagrams of the five 3d orbitals. Although the $3d_{z^2}$ orbital looks different, it is equivalent to the other four orbitals in all other respects. The d orbitals of higher principal quantum numbers have similar shapes.

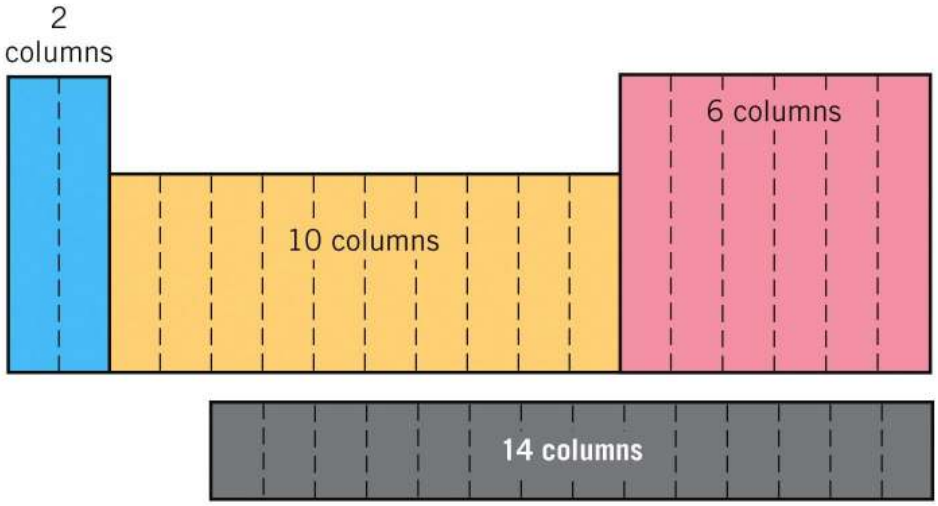
CH8 PERIODIC TABLE



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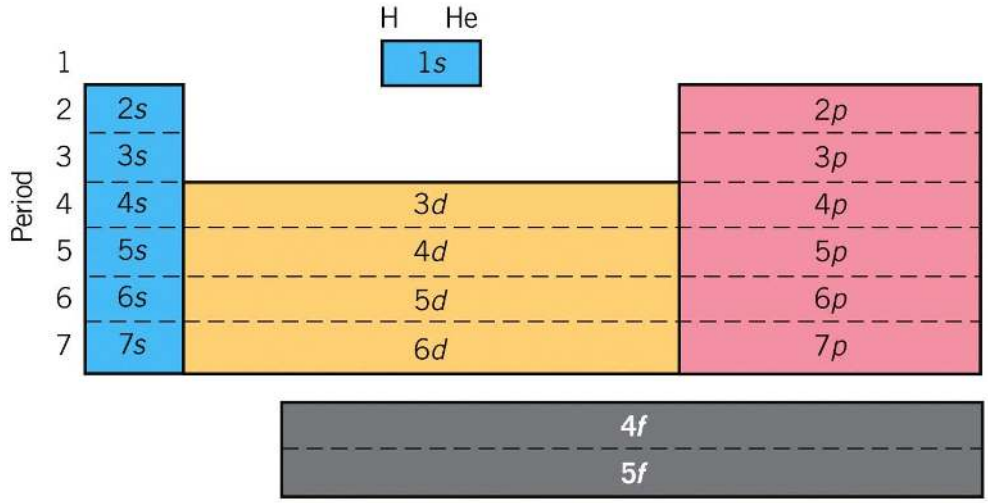
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- Each region of chart represents different type of sublevel



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WHERE ARE THE ELECTRONS?

- Each box represents room for electron
- Read from left to right

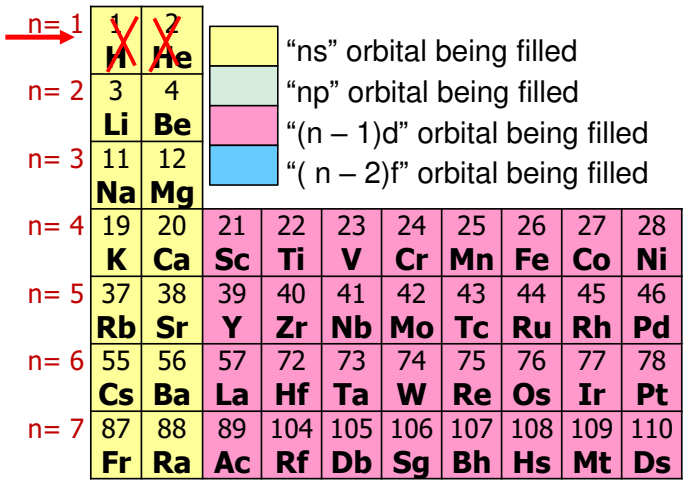
n= 1	1 **H**	2 **He**															
n= 2	3 **Li**	4 **Be**											5 **B**	6 **C**	7 **N**	8 **O**	9 **F**	**10 Ne**
n= 3	11 **Na**	12 **Mg**											13 **Al**	14 **Si**	15 **P**	16 **S**	17 **Cl**	18 **Ar**
n= 4	19 **K**	20 **Ca**	21 **Sc**	22 **Ti**	23 **V**	24 **Cr**	25 **Mn**	26 **Fe**	27 **Co**	28 **Ni**	29 **Cu**	30 **Zn**	31 **Ga**	32 **Ge**	33 **As**	34 **Se**	35 **Br**	36 **Kr**
n= 5	37 **Rb**	38 **Sr**	39 **Y**	40 **Zr**	41 **Nb**	42 **Mo**	43 **Tc**	44 **Ru**	45 **Rh**	46 **Pd**	47 **Ag**	48 **Cd**	49 **In**	50 **Sn**	51 **Sb**	52 **Te**	53 **I**	54 **Xe**
n= 6	55 **Cs**	56 **Ba**	57 **La**	72 **Hf**	73 **Ta**	74 **W**	75 **Re**	76 **Os**	77 **Ir**	78 **Pt**	79 **Au**	80 **Hg**	81 **Tl**	82 **Pb**	83 **Bi**	84 **Po**	85 **At**	86 **Rn**
n= 7	87 **Fr**	88 **Ra**	89 **Ac**	104 **Rf**	105 **Db**	106 **Sg**	107 **Bh**	108 **Hs**	109 **Mt**	110 **Ds**	111 **Rg**							

“ns” orbital being filled
“np” orbital being filled
“(n – 1)d” orbital being filled
“(n – 2)f” orbital being filled

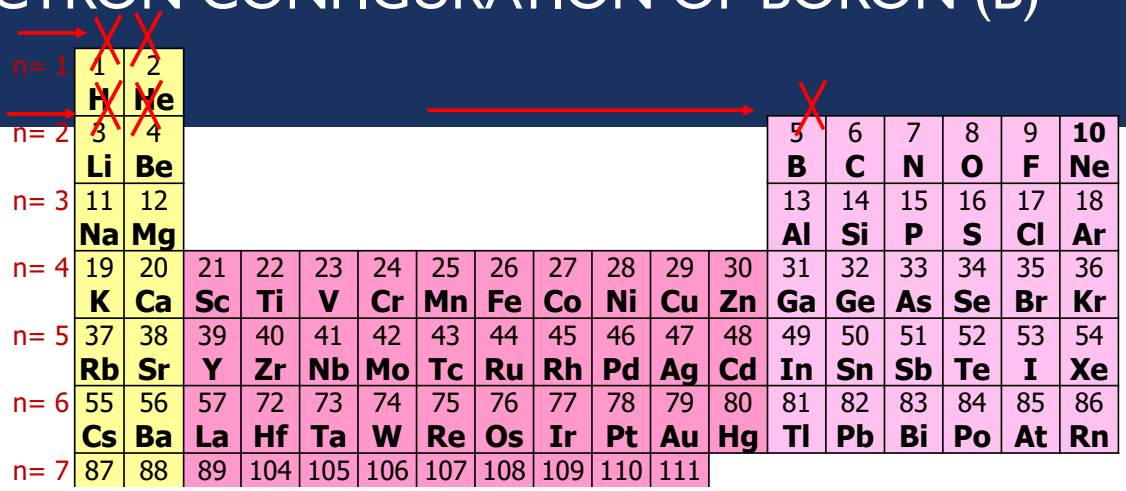
58 **Ce**	59 **Pr**	60 **Nd**	61 **Pm**	62 **Sm**	63 **Eu**	64 **Gd**	65 **Tb**	66 **Dy**	67 **Ho**	68 **Er**	69 **Tm**	70 **Yb**	71 **Lu**
90 **Th**	91 **Pa**	92 **U**	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lr**

READ PERIODIC TABLE TO DETERMINE E-

- Read from left to right
- 1st e⁻ goes into period 1
- 1st type of sublevel to fill = "1s"
- He has 2 e⁻
- e⁻ configuration for He is: **$1s^2$**



ELECTRON CONFIGURATION OF BORON (B)



n= 1	1 **H**																2 **He**
n= 2	3 **Li**	4 **Be**											5 **B**	6 **C**	7 **N**	8 **O**	9 **F**	**10 Ne**
n= 3	11 **Na**	12 **Mg**											13 **Al**	14 **Si**	15 **P**	16 **S**	17 **Cl**	18 **Ar**
n= 4	19 **K**	20 **Ca**	21 **Sc**	22 **Ti**	23 **V**	24 **Cr**	25 **Mn**	26 **Fe**	27 **Co**	28 **Ni**	29 **Cu**	30 **Zn**	31 **Ga**	32 **Ge**	33 **As**	34 **Se**	35 **Br**	36 **Kr**
n= 5	37 **Rb**	38 **Sr**	39 **Y**	40 **Zr**	41 **Nb**	42 **Mo**	43 **Tc**	44 **Ru**	45 **Rh**	46 **Pd**	47 **Ag**	48 **Cd**	49 **In**	50 **Sn**	51 **Sb**	52 **Te**	53 **I**	54 **Xe**
n= 6	55 **Cs**	56 **Ba**	57 **La**	72 **Hf**	73 **Ta**	74 **W**	75 **Re**	76 **Os**	77 **Ir**	78 **Pt**	79 **Au**	80 **Hg**	81 **Tl**	82 **Pb**	83 **Bi**	84 **Po**	85 **At**	86 **Rn**
n= 7	87	88	89	104	105	106	107	108	109	110	111							

- B has 5 e^-
- Fill first shell…
- Fill two subshells in 2nd shell, in order of increasing E
- Electron Configuration B = **$1s^2 2s^2 2p^1$**

- Summarizes periodic properties of elements

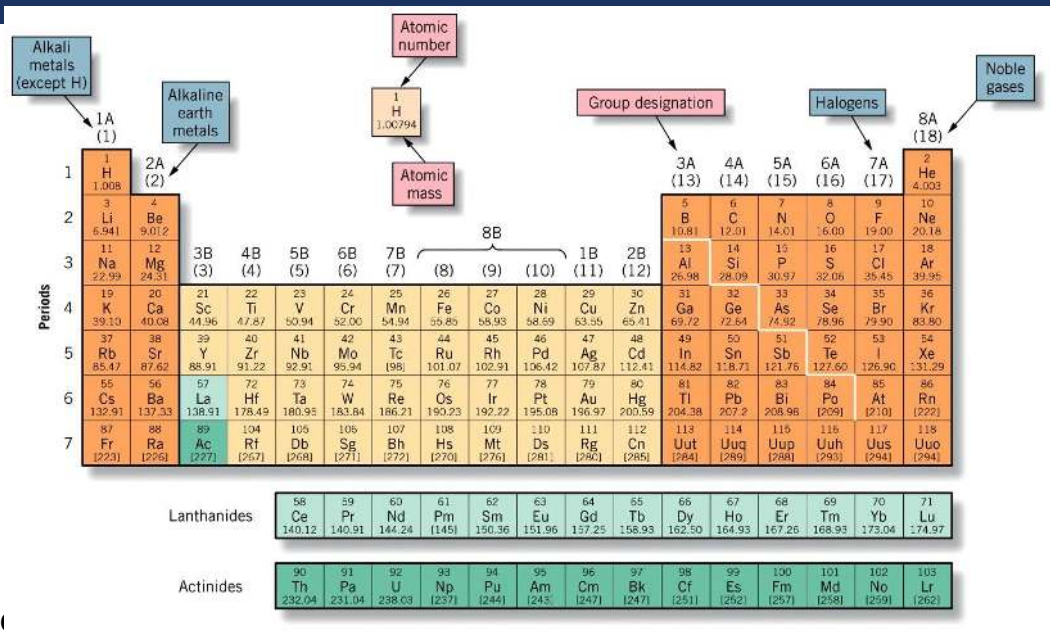
Early Versions of Periodic Tables

- Arranged by increasing **atomic mass**
- Mendeleev (Russian) & Meyer (German) in 1869
- Noted repeating (periodic) properties

Modern Periodic Table

- Arranged by increasing **atomic number (Z)**:
- **Rows** called **periods**
- **Columns** called **groups** or **families**
  - Identified by numbers
  - 1 – 18 standard international

with group labels and chemical families identified



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A groups—Longer columns

- **Alkali Metals**
  - **IA** = first group
  - Very reactive
  - All Metals except for H
  - Tend to form **+1** ions
  - React with oxygen
    - Form compounds that dissolve in water
    - Yield strongly caustic or alkaline solution (M_2O)

A groups—Longer columns

- **Alkaline Earth Metals**
  - **2A** = second group
  - Reactive
  - Tend to form **+2** ions
  - Oxygen compounds are strongly alkaline (MO)
  - Many are not water soluble
  - Found in the deposits in the ground

A groups—Longer columns

- **Halogens** (derived from Greek word meaning sea or salt)
  - **7A** = next to last group on right
  - Reactive
  - Form **diatomic** molecules in elemental state
    - 2 gases
    - 1 liquid
    - 2 solids
  - Form **–1** ions with alkali metals—salts

A groups—Longer columns

- **Noble Gases**
  - **8A** = last group on right
  - Inert—very unreactive
  - Only heavier elements of group react & then very limited
  - Don't form charged ions
  - Monatomic gases

B groups—shorter columns

- All are metals
- In center of table
- Begin in fourth row
- Tend to form ions with several different charges

Ex.

- Fe^{2+} and Fe^{3+}
- Cu^{+} and Cu^{2+}
- Mn^{2+}, Mn^{3+}, Mn^{4+}, Mn^{5+}, Mn^{6+}, Mn^{7+}

Note: Last 3 columns all have 8B designation

Lanthanide elements

- Elements 58 – 71

Actinide elements

- Elements 90 – 103

- At bottom of periodic table
- Tend to form **+2** and **+3** ions.
- All Actinides are radioactive

METALS, NONMETALS, OR METALLOIDS

- Elements break down into 3 broad categories
- Organized by regions of periodic table

Metals

- Left-hand side
- Sodium, lead, iron, gold

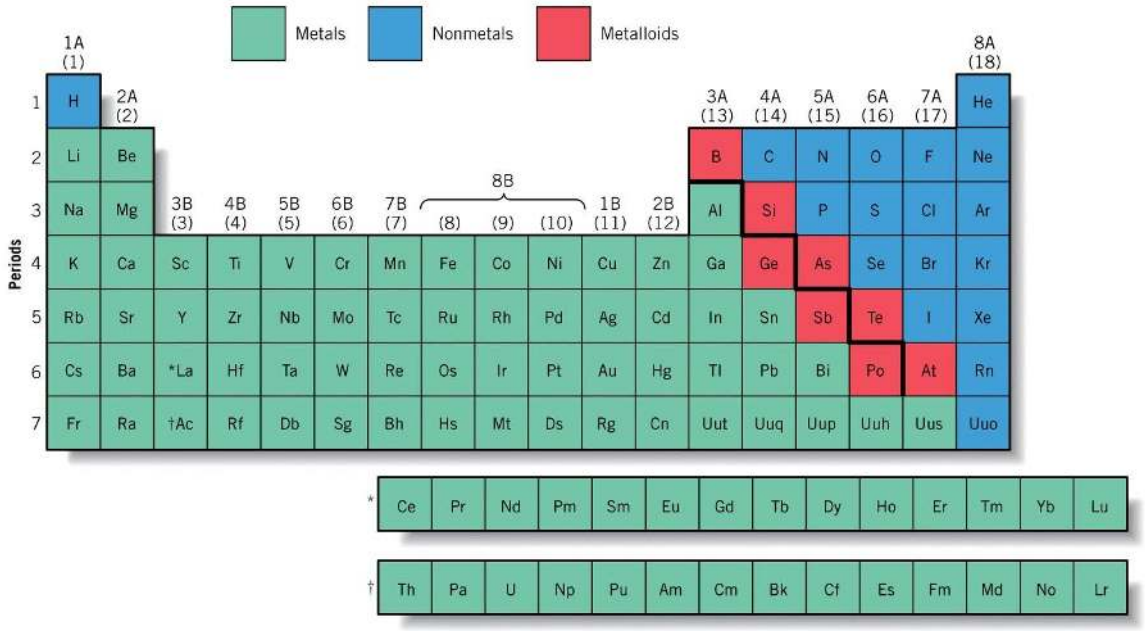
Nonmetals

- Upper right hand corner
- Oxygen, nitrogen, chlorine
- Boron to astatine

Metalloids

Diagonal line between metals & nonmetals

METALS, NONMETALS, OR METALLOIDS



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IONIC COMPOUNDS

Cations

- Positively charged ions
- Formed from metals
- Atoms **lose** electrons

Ex. **Na** has 11 e^- & 11 p+

Na^+ has 10 e^- & 11 p+

Anions

- Negatively charged ions
- Formed from non-metals
- Atoms **gain** electrons

Ex. **Cl** has 17 e^- & 17 p^+

Cl^- has 16 e^- & 17 p+

IONS OF REPRESENTATIVE ELEMENTS

- Can use periodic table to predict ion charges

Table 3.3 Some Ions Formed from the Representative Elements

Group Number						
1A	2A	3A	4A	5A	6A	7A
H^+						
Li^+	Be^{2+}		C^{4-}	N^{3-}	O^{2-}	F^-
Na^+	Mg^{2+}	Al^{3+}	Si^{4-}	P^{3-}	S^{2-}	Cl^-
K^+	Ca^{2+}				Se^{2-}	Br^-
Rb^+	Sr^{2+}				Te^{2-}	I^-
Cs^+	Ba^{2+}					

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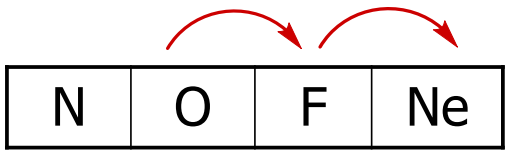
- When we use North American numbering of groups: **Cation positive charge = group #**

IONS OF REPRESENTATIVE ELEMENTS

- Noble gases are especially stable

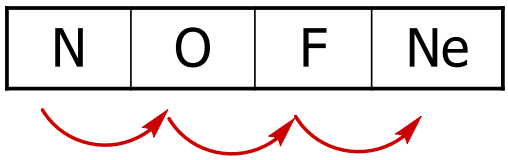
Nonmetals

- **Negative (–)** charge on anion



- = # spaces you have to move to right to get to noble gas
- Expected charge on O is
  - Move 2 spaces to right
  - O^{2-}

- What is expected charge on N?
  - Move 3 spaces to right
  - N^{3-}



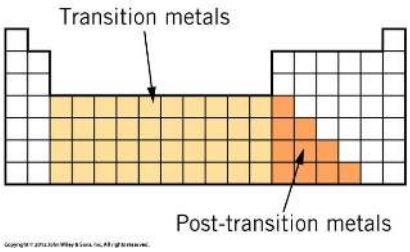
CATIONS OF TRANSITION METALS

Transition metals

- Center (shorter) region of periodic table
- Much less reactive than group 1A & 2A
- Still transfer electrons to nonmetals to form ionic compounds
- # of electrons removed from S the from d
- Form more than 1 positive ion

Ex. Fe + Cl

$FeCl_2$ & $FeCl_3$



EX:

PREPARED BY DR.HIBA ALAMYREH

PREPARED BY DR.HIBA ALAMYREH

PREPARED BY DR.HIBA ALAMYREH

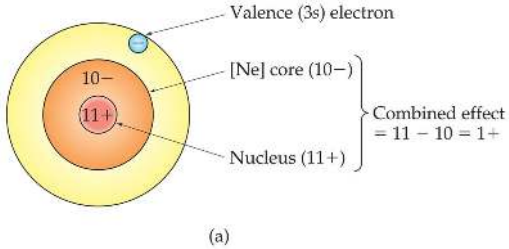
PREPARED BY DR.HIBA ALAMYREH

PERIODIC VARIATION IN PHYSICAL PROPERTIES

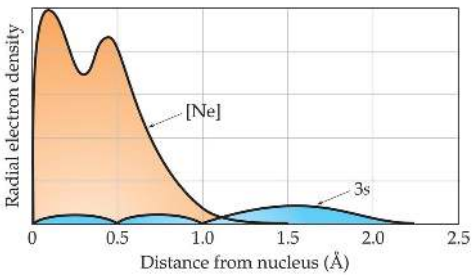
- Sizes of atoms and ions.
- Ionization energy.
- Electron affinity.

Effective Nuclear Charge

Na atom looks like this:

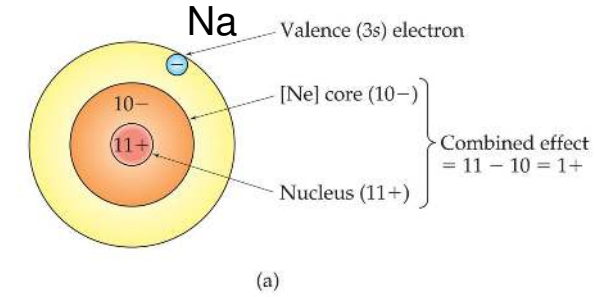


(a)

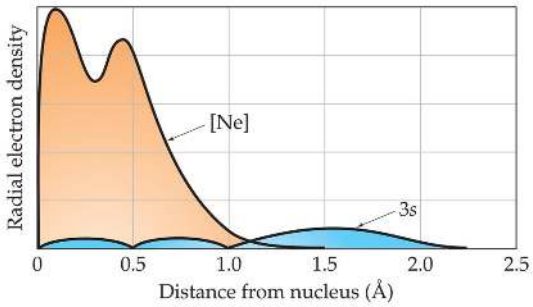


- In a many-electron atom, electrons are both attracted to the nucleus and repelled by other electrons.
- The nuclear charge that an electron "feels" depends on both factors.
- It's called Effective nuclear charge.
- electrons in lower energy levels "shield" outer electrons from positive charge of nucleus.

Effective Nuclear Charge



(a)



The effective nuclear charge, Z_{eff}, is:

$$Z_{eff} = Z - S$$

Where:

Z = atomic number

S = screening constant, usually close to the number of inner (n-1) electrons.

- Example: Which element's outer shell or "valence" electrons is predicted to have the largest Effective nuclear charge? Kr, Cl or O?

- Example: Which element's outer shell or "valence" electrons is predicted to have the largest Effective nuclear charge? Kr, Cl or O?

- Cl: $Z_{eff} \approx 17 - 10 = 7$
- O: $Z_{eff} \approx 8 - 2 = 6$
- N: $Z_{eff} \approx 7 - 2 = 5$
- Ca: $Z_{eff} \approx 20 - 18 = 2$

valence electrons.

Many chemical properties depend on the valence electrons.

Valence electrons: The outer electrons, that are involved in bonding and most other chemical changes of elements.

Rules for defining valence electrons.

1. In outer most energy level (or levels)
2. For main group (representative) elements (elements in s world or p world) electrons in filled d or f shells are not valence electrons
3. For transition metals, electrons in full f shells are not valence electrons.

Examples: (valence electrons in blue)

P: [Ne]$3s^2 3p^3$

As: [Ar] $4s^2 3d^{10} 4p3$

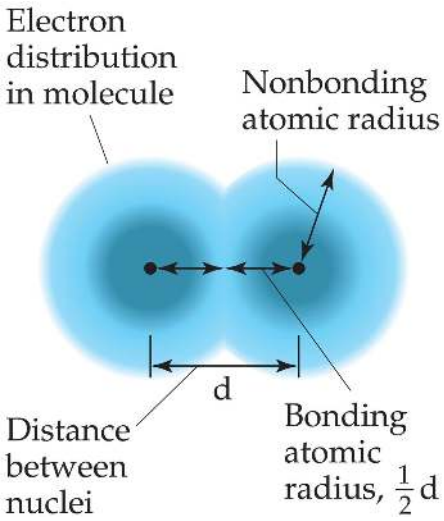
I: [Kr]$5s^2 4d^{10} 5p^5$

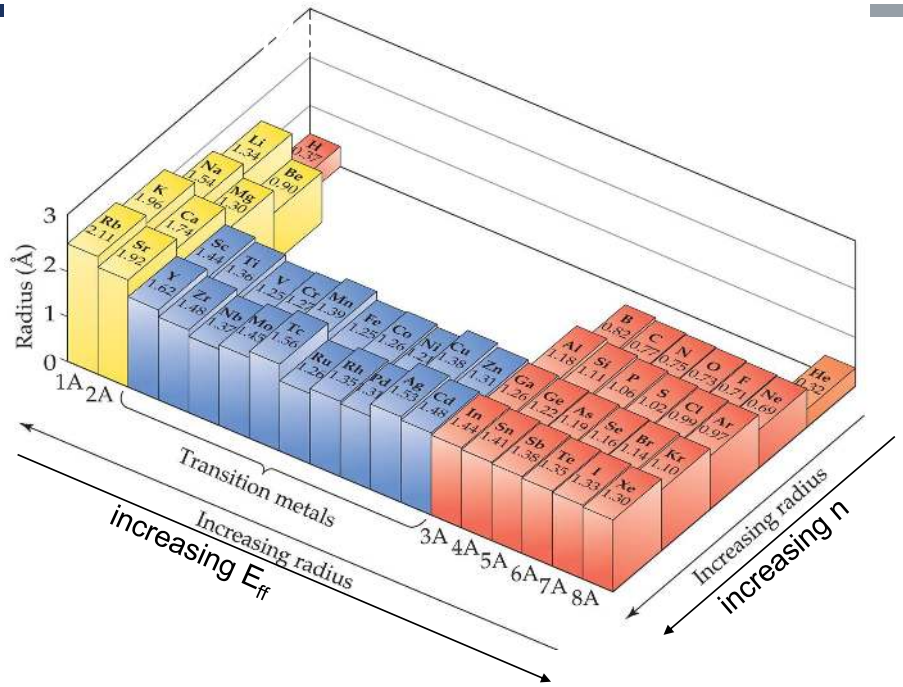
Ta: [Kr]$6s^2 4f^{14} 5d^3$

Zn: [Ar]$4s^2 3d^{10}$

SIZES OF ATOMS

The bonding atomic radius is defined as one-half of the distance between covalently bonded nuclei.

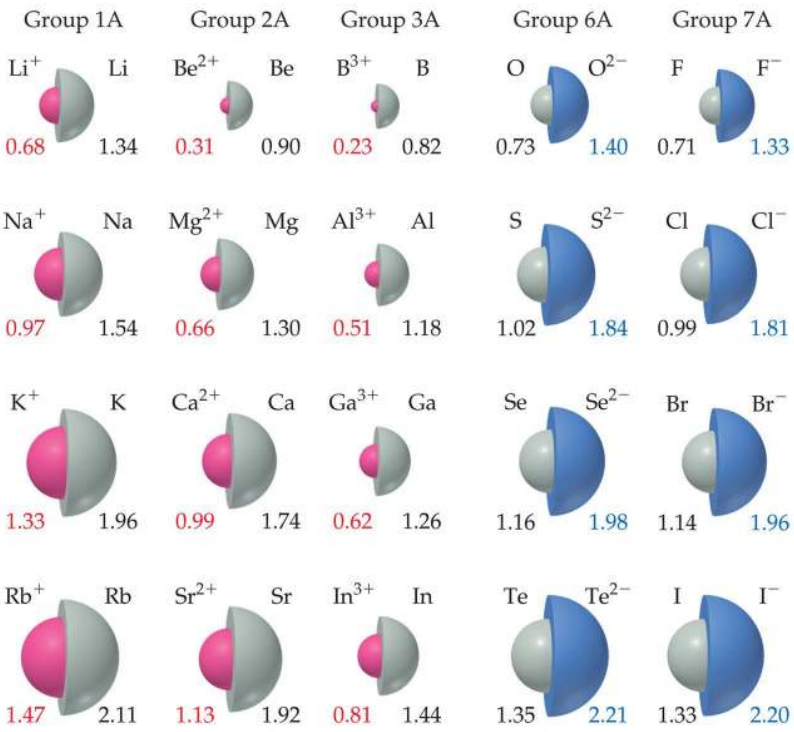




Bonding atomic radius tends to…

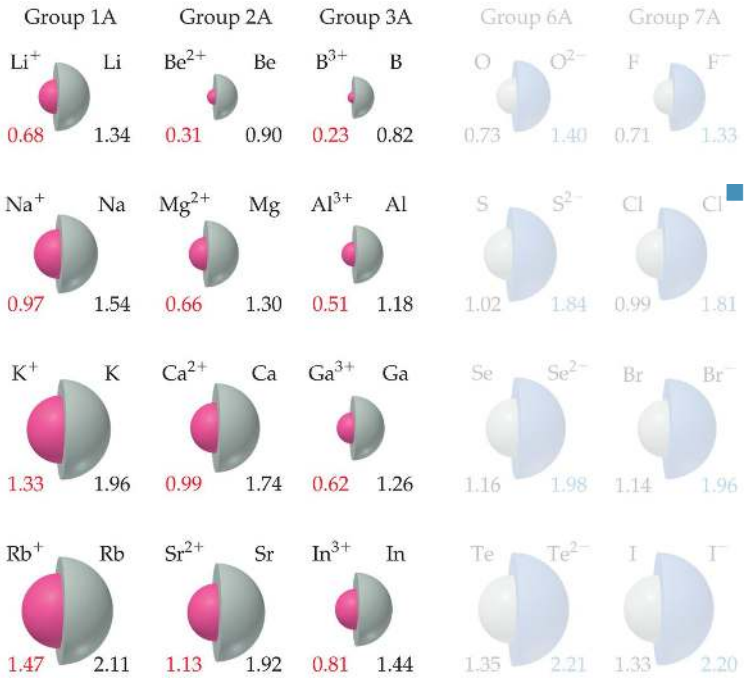
…decrease from left to right across a row due to increasing Z_{eff}.

…increase from top to bottom of a column due to increasing value of *n*

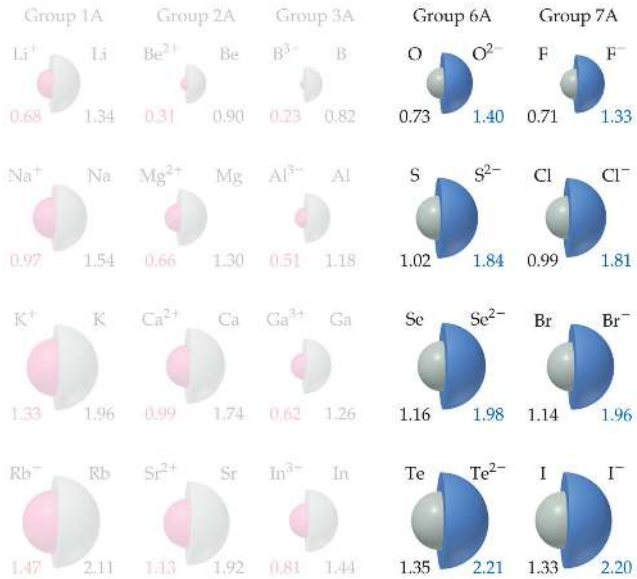


Ionic size depends upon:

- Nuclear charge.
- Number of electrons.
- Orbitals in which electrons reside.

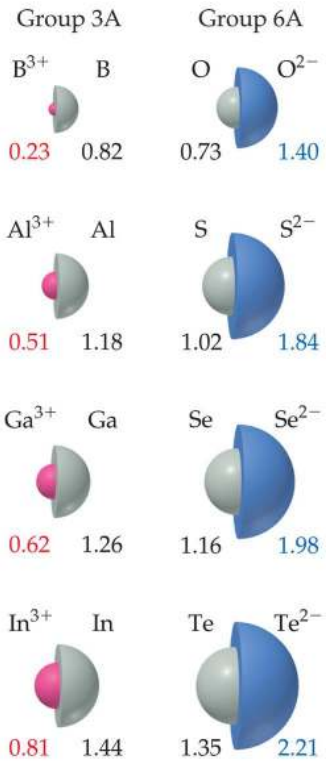


- Cations are smaller than their parent atoms.
  - The outermost electron is removed and repulsions are reduced.

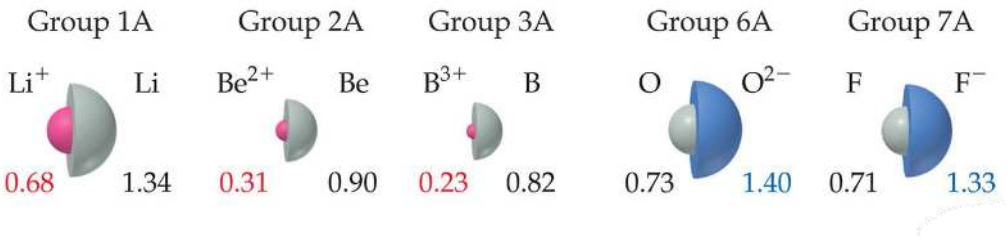


- Anions are larger than their parent atoms.
  - Electrons are added and repulsions are increased.

- Ions increase in size as you go down a column.
  - Due to increasing value of *n*.



- In an isoelectronic series, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.



- EX:
- Put the following in order of size, smallest to largest:

- $Na, Na^{+}, Mg, Mg^{2+}, Al, Al^{3+}, S, S^{2-}, Cl, Cl^{-}$

Al^{3+}, Mg^{2+}, Na^{+}, Cl, S, Al, Mg, Na, Cl^{-}, S^{2-}

Start with atoms with no n=3 electrons, order isoelectronic by nuclear charge.

Next, neutral atoms highest E_{ff} first

Last, anions, highest E_{ff} first

Ambiguity: anions versus neutrals (is Cl^{-} really larger than Na?)

IONIZATION ENERGY

El -------> $El^+ + e^-$

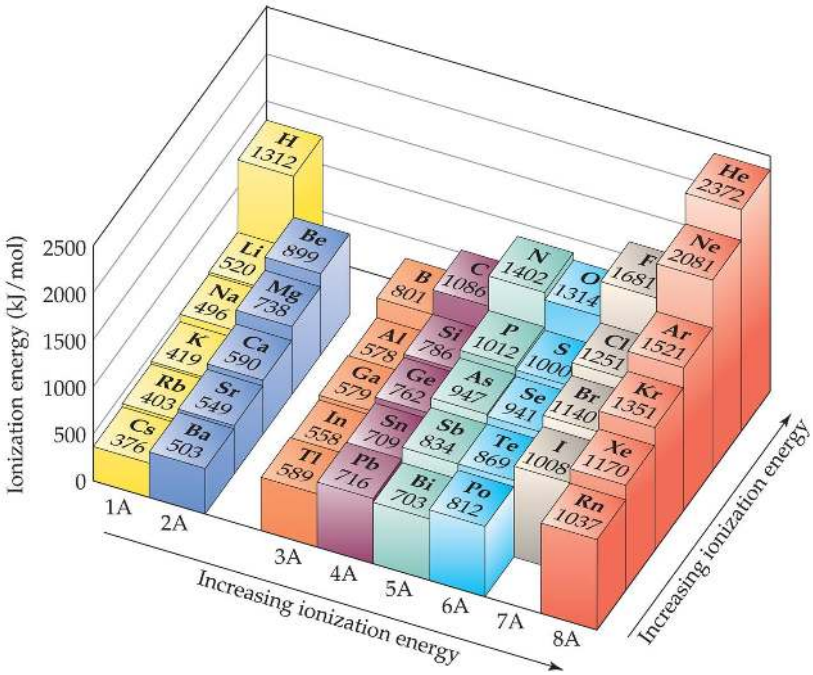
Na -------> $Na^+ + e^-$

- Amount of energy required to remove an electron from the ground state of a gaseous atom or ion.
  - First ionization energy is that energy required to remove first electron.
  - Second ionization energy is that energy required to remove second electron, etc.

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, the ionization energy takes a quantum leap.

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4562			(inner-shell electrons)		
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

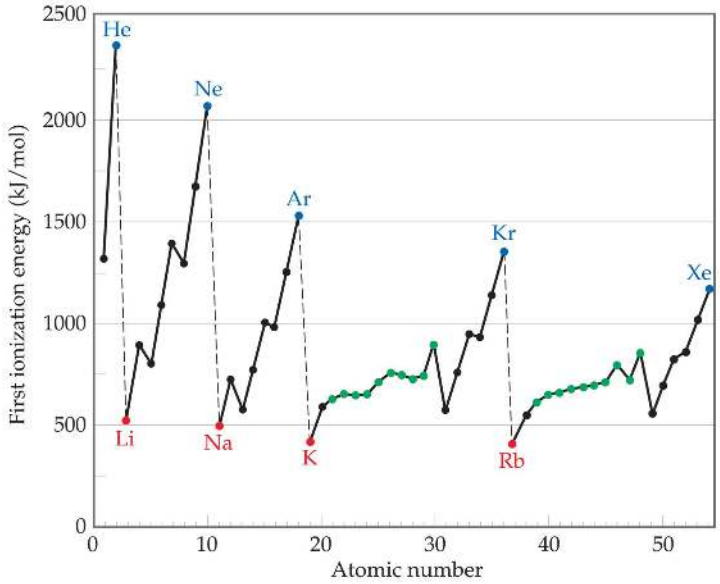
TRENDS IN FIRST IONIZATION ENERGIES



- going down a column, less energy to remove the first electron.
  - For atoms in the same group, Z_{eff} is essentially the same, but the valence electrons are farther from the nucleus.

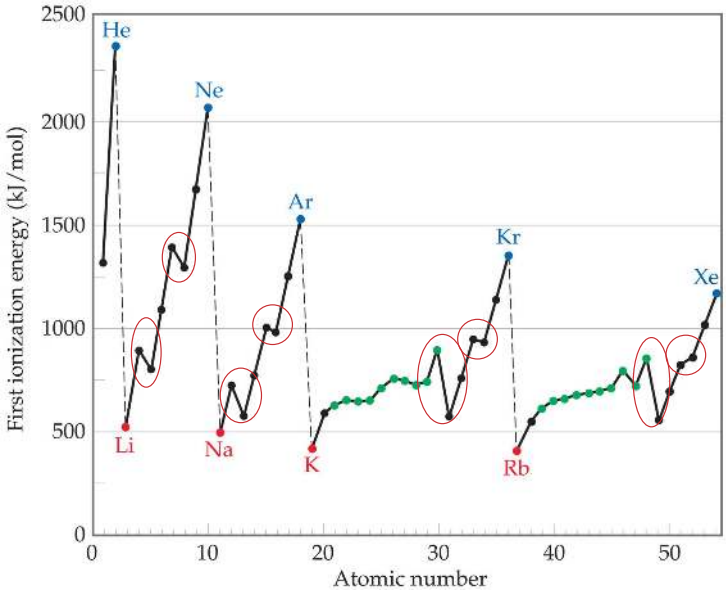
TRENDS IN FIRST IONIZATION ENERGIES

- Generally, it gets harder to remove an electron going across.
  - As you go from left to to right, Z_{eff} increases.



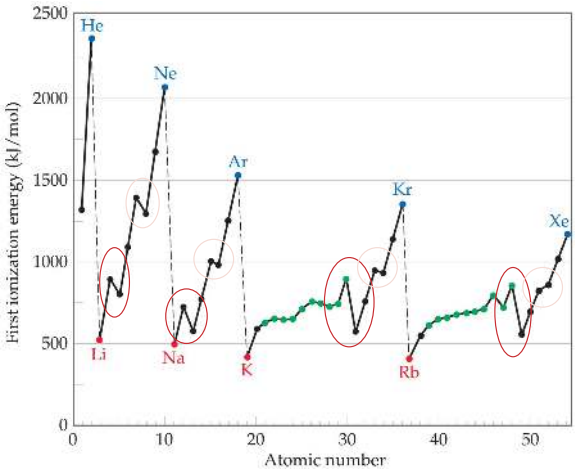
TRENDS IN FIRST IONIZATION ENERGIES

On a smaller scale, there are two jags in each line. Why?



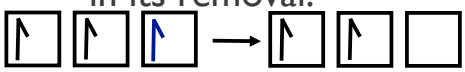
TRENDS IN FIRST IONIZATION ENERGIES

- The first occurs between Groups IIA and IIIA.
- Electron removed from *p*-orbital rather than *s*-orbital
  - Electron farther from nucleus
  - Small amount of repulsion by *s* electrons.



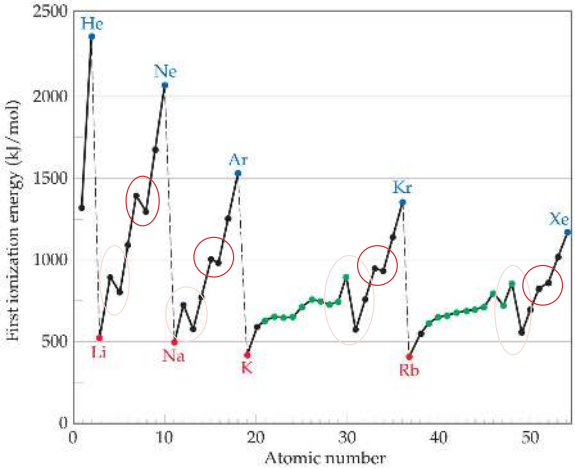
TRENDS IN FIRST IONIZATION ENERGIES

- The second occurs between Groups VA and VIA.
  - Electron removed comes from doubly occupied orbital.
  - Repulsion from other electron in orbital helps in its removal.



versus:



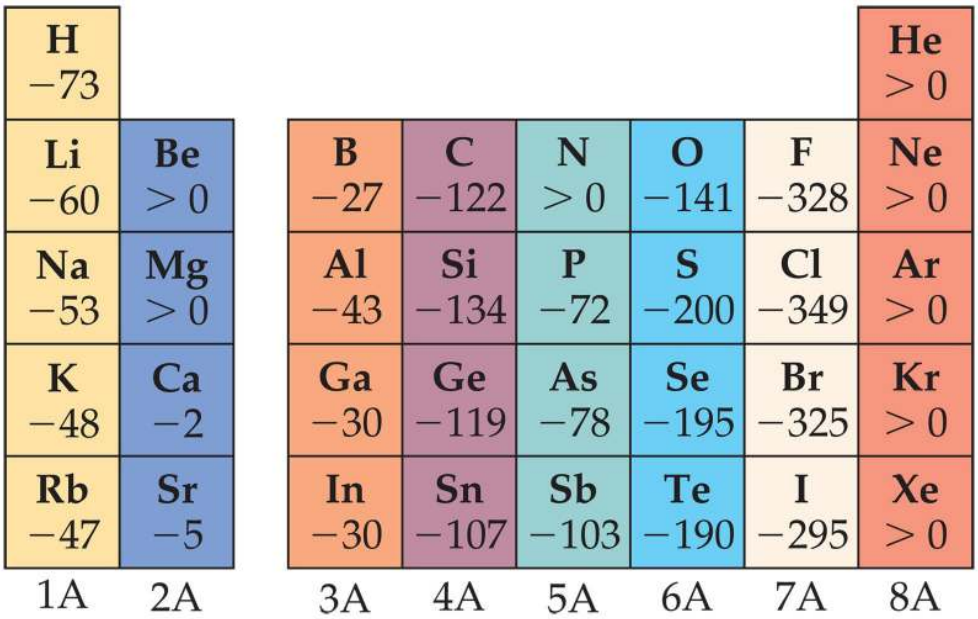


ELECTRON AFFINITY

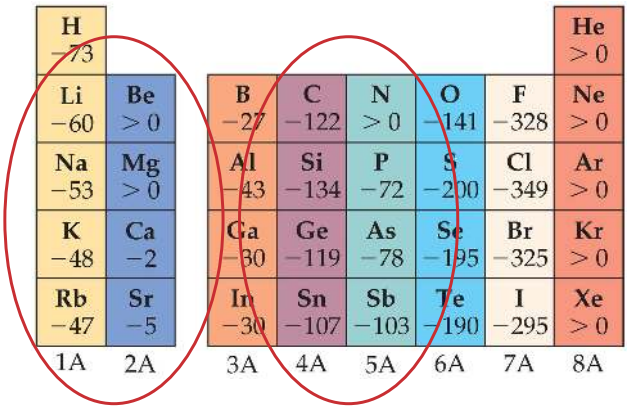
Energy change accompanying addition of electron to gaseous atom:

$$Cl + e^- \longrightarrow Cl^-$$

TRENDS IN ELECTRON AFFINITY

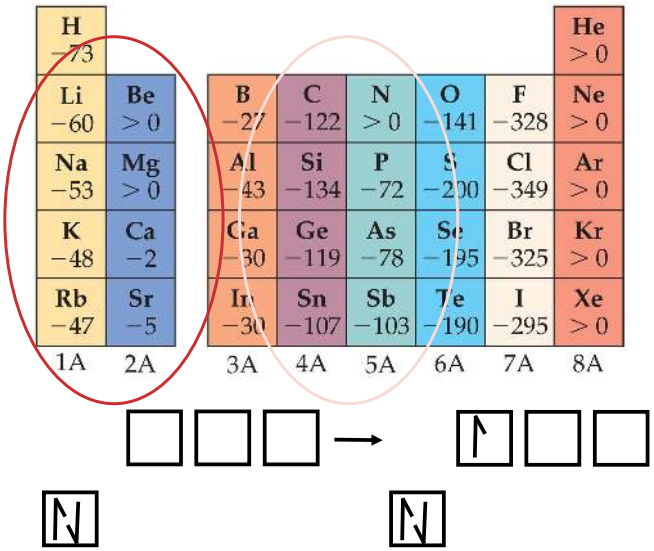


TRENDS IN ELECTRON AFFINITY



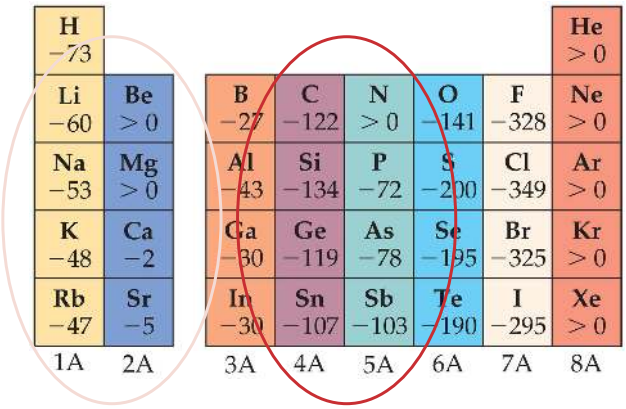
There are also two discontinuities in this trend.

TRENDS IN ELECTRON AFFINITY



- The first occurs between Groups IA and IIA.
  - Added electron must go in *p*-orbital, not *s*-orbital.
  - Electron is farther from nucleus and feels repulsion from *s*-electrons.

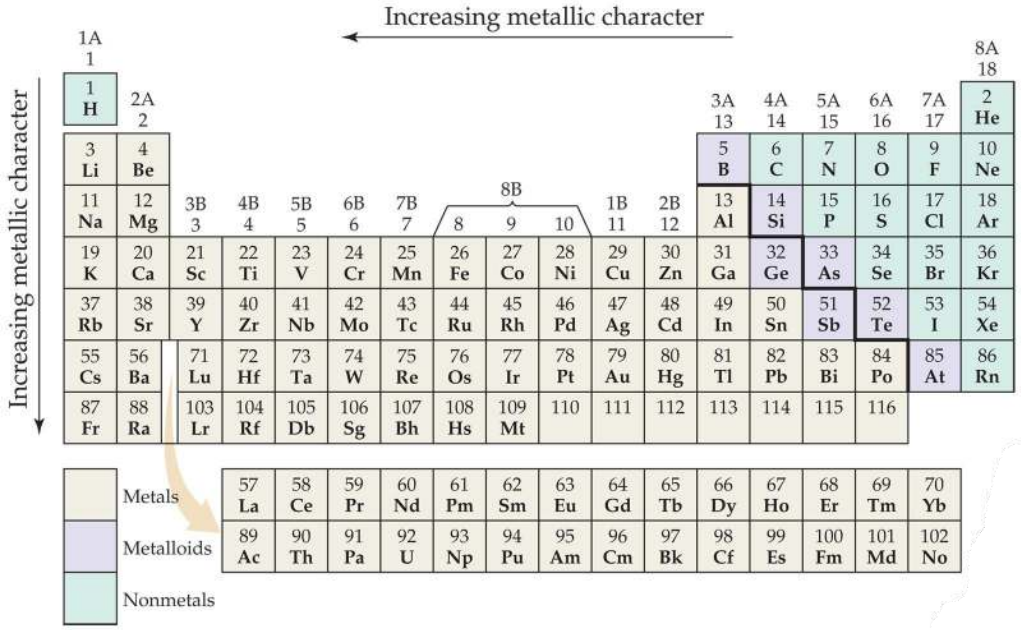
TRENDS IN ELECTRON AFFINITY



H −73						He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

- The second occurs between Groups IVA and VA.
  - Group VA has no empty orbitals.
  - Extra electron must go into occupied orbital, creating repulsion.

PROPERTIES OF METALS, NONMETALS, AND METALLOIDS



Metals	Nonmetals
Have a shiny luster; various colors, although most are silvery	Do not have a luster; various colors
Solids are malleable and ductile	Solids are usually brittle; some are hard, some are soft
Good conductors of heat and electricity	Poor conductors of heat and electricity
Most metal oxides are ionic solids that are basic	Most nonmetal oxides are molecular substances that form acidic solutions
Tend to form cations in aqueous solution	Tend to form anions or oxyanions in aqueous solution

Differences between metals and nonmetals tend to revolve around these properties.

- All neutral atoms are larger in size than their cations.
- As positive charge of ion increase size decrease.

$$Fe > Fe^{2+} > Fe^{3+}$$

- On the otherside, all neutral atoms are smaller in size than their anions.

$$Cl < Cl^{-}$$

- As negative sign increase, size decrease.

$$O < O^{-} < O^{2-}$$

How many unpaired electrons does $_{26}Fe^{3+}$ have ?

$_{26}Fe$: $[Ar]\ 4s^2\ 3d^6$

Fe^{3+}: $[Ar]\ 3d^5$ ↓ ↓ ↓ ↓ ↓ (5)

Transtion elements lose $\bar{e}_s$ ↑ from the "s" subshell, before losing $\bar{e}_s$ from "d" subshell.

GENERAL CHEMISTRY 1
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CHAPTER 3: STOICHIOMETRY

LECTURE 13-17

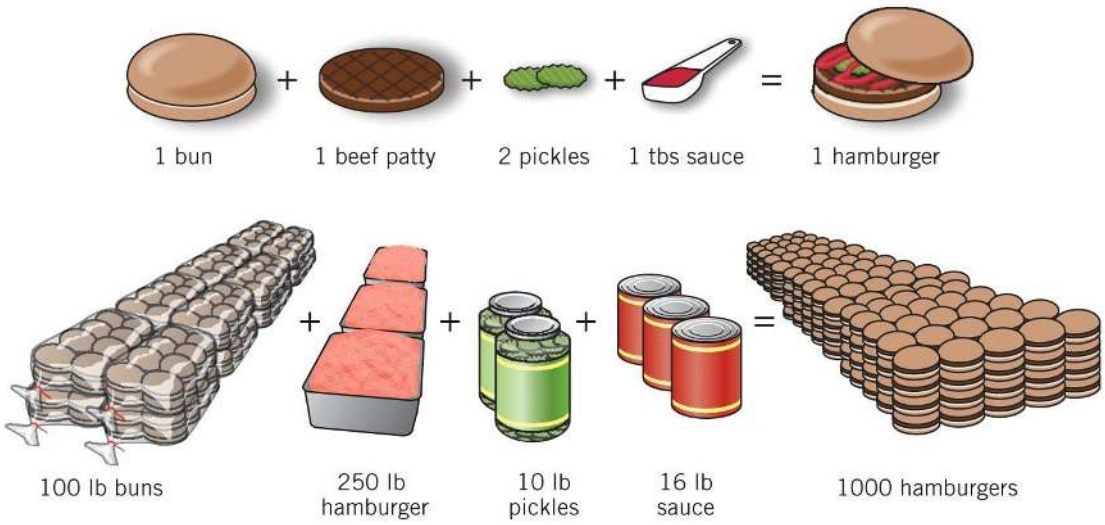
STOICHIOMETRY

- Mass balance of all formulas involved in chemical reactions

Stoichiometric Calculations

- Conversions from one set of units to another using Dimensional Analysis
- Need to know:
  1. Equalities to make conversion factors
  2. Steps to go from starting units to desired units

USING MASS TO COUNT



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MOLECULAR TO LABORATORY SCALE

- So far, we have looked at chemical formulas & reactions at a molecular scale.
- It is known from experiments that:
  - Electrons, neutrons & protons have set masses.
  - Atoms must also have characteristic masses
  - Just extremely small
- Need a way to scale up chemical formulas & reactions to carry out experiments in laboratory
- **Mole** is our conversion factor

THE MOLE

- Number of atoms in exactly 12 grams of ^{12}C atoms

How many atoms in 1 mole of ^{12}C ?

- Based on experimental evidence **1 mole of $^{12}C = 6.022 \times 10^{23}$ atoms = 12.011 g**

Avogadro's number = N_A

- Number of atoms, molecules or particles in one mole

- **1 mole of X = 6.022×10^{23} units of X**
  - 1 mole Xe = 6.022×10^{23} Xe atoms
  - 1 mole NO_2 = 6.022×10^{23} NO_2 molecules

MOLES OF COMPOUNDS

Atoms

- **Atomic Mass**
  - Mass of atom (from periodic table)

- **1 mole of atoms = gram atomic mass = 6.022×10^{23} atoms**

Molecules

- **Molecular Mass**
  - Sum of atomic masses of all atoms in compound's formula

1 mole of molecule X = gram molecular mass of X= 6.022×10^{23} molecules

MOLES OF COMPOUNDS

Ionic compounds

- **Formula Mass**
  - Sum of atomic masses of all atoms in ionic compound's formula

1 mole ionic compound X = gram formula mass of X= 6.022×10^{23} formula units

General

Molar mass (MM)

- Mass of 1 mole of substance (element, molecule, or ionic compound) under consideration

1 mol of X = gram molar mass of X
= 6.022×10^{23} formula units

SI UNIT FOR AMOUNT = MOLE

1 mole of substance X = gram molar mass of X



- 1 mole S = 32.06 g S
- 1 mole NO_2= 46.01 g NO_2

- Molar mass is our conversion factor between g & moles



- **1 mole of X = 6.022×10^{23} units of X**
- N_A is our conversion factor between moles & molecules
  - 1 mole H_2O = 6.022×10^{23} molecules H_2O
  - 1 mole NaCl = 6.022×10^{23} formula units NaCl

LEARNING CHECK: USING MOLAR MASS

Ex. How many moles of iron (Fe) are in 15.34 g Fe?

- What do we know?

  1 mol Fe = 55.85 g Fe

- What do we want to determine?

  15.34 g Fe = ? Mol Fe

Start → End

- Set up ratio so that what you want is on top & what you start with is on the bottom

$$15.34\ \text{g Fe} \times \left(\frac{1\ \text{mol Fe}}{55.85\ \text{g Fe}} \right) = \textbf{0.2747 mole Fe}$$

LEARNING CHECK: USING MOLAR MASS

Ex. If we need 0.168 mole $Ca_3(PO_4)_2$ for an experiment, how many grams do we need to weigh out?

- Calculate MM of $Ca_3(PO_4)_2$

  3 × mass Ca = 3 × 40.08 g = 120.24 g

  2 × mass P = 2 × 30.97 g = 61.94 g

  8 × mass O = 8 × 16.00 g = 128.00 g

  1 mole $Ca_3(PO_4)_2$ = 310.18 g $Ca_3(PO_4)_2$

- What do we want to determine?

  0.168 g $Ca_3(PO_4)_2$ = ? Mol Fe

Start → End

LEARNING CHECK: USING MOLAR MASS

$$0.160 \text{ mol } Ca_3(PO_4)_2 \times \left(\frac{310.18 \text{ g } Ca_3(PO_4)_2}{1 \text{ mol } Ca_3(PO_4)_2} \right)$$

- Set up ratio so that what you want is on the top & what you start with is on the bottom

= 52.11 g $Ca_3(PO_4)_2$

YOUR TURN!

How many moles of CO_2 are there in 10.0 g?

A. 1.00 mol

B. 0.0227 mol

C. 4.401 mol

D. 44.01 mol

E. 0.227 mol

Molar mass of CO_2

$1 \times 12.01\ g = 12.01\ g\ C$

$2 \times 16.00\ g = 32.00\ g\ O$

$1\ mol\ CO_2 = 44.01\ g\ CO_2$

$$10.0\ g\ CO_2 \left(\frac{1\ mol\ CO_2}{44.01\ g\ CO_2} \right)$$

$= 0.227\ mol\ CO_2$

YOUR TURN!

How many grams of platinum (Pt) are in 0.475 mole Pt?

A. 195 g

B. 0.0108 g

C. 0.000513 g

D. 0.00243 g

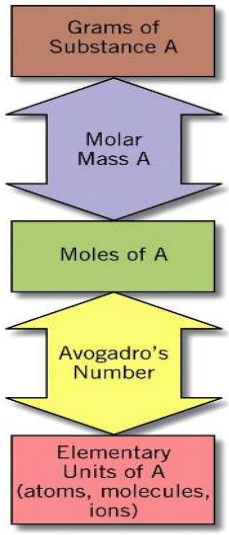
E. 92.7 g

Molar mass of Pt = 195.08 g/mol

$$0.475 \cancel{\text{mol Pt}} \times \left(\frac{195.08 \text{ g Pt}}{1 \cancel{\text{mol Pt}}} \right)$$

= 92.7 g Pt

USING MOLES IN CALCULATIONS



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- Start with either
  - Grams (Macroscopic)
  - Elementary units (Microscopic)
- Use molar mass to convert from grams to mole
- Use Avogadro's number to convert from moles to elementary units

MACROSCOPIC TO MICROSCOPIC

How many silver atoms are in a 85.0 g silver bracelet?

- What do we know?

  107.87 g Ag = 1 mol Ag

  1 mol Ag = 6.022×10^{23} Ag atoms

$$85.0\ \text{g Ag} \times \left(\frac{1\ \text{mol Ag}}{107.87\ \text{g Ag}}\right) \times \left(\frac{6.022 \times 10^{23}\ \text{atoms Ag}}{1\ \text{mol Ag}}\right)$$

- What do we want to determine?

85.0 g silver = ? atoms silver

g Ag ⟶ mol Ag ⟶ atoms Ag

USING AVOGADRO'S NUMBER

What is the mass, in grams, of one molecule of octane, C_8H_{18}?

Molecules octane ⟶ mol octane ⟶ g octane

1. Calculate molar mass of octane

Mass C = 8 × 12.01 g = 96.08 g

Mass H = 18 × 1.008 g = 18.14 g

1 mol octane = 114.22 g octane

2. Convert 1 molecule of octane to grams

$$\left(\frac{114.22\ \text{g octane}}{1\ \text{mol octane}}\right)\times\left(\frac{1\ \text{mol octane}}{6.022\times10^{23}\ \text{molecules octane}}\right)$$

= **1.897 × 10⁻²² g octane**

LEARNING CHECK: MOLE CONVERSIONS

- Calculate the number of formula units of Na_2CO_3 in 1.29 moles of Na_2CO_3.

$$1.29 \text{ mol } Na_2CO_3 \left(\frac{6.0223 \times 10^{23} \text{ formula units } Na_2CO_3}{1 \text{ mol } Na_2CO_3} \right)$$

$= 7.77 \times 10^{23}$ particles Na_2CO_3

- How many moles of Na_2CO_3 are there in 1.15×10^5 formula units of Na_2CO_3 ?

$$1.15 \times 10^5 \text{ formula units } Na_2CO_3 \left(\frac{1 \text{ mol } Na_2CO_3}{6.0223 \times 10^{23} \text{ formula units } Na_2CO_3} \right)$$

$= 1.91 \times 10^{-19}$ mol Na_2CO_3

YOUR TURN!

How many atoms are in 1.00×10^{-9} g of U (1 ng)? Molar mass U = 238.03 g/mole.

A. 6.02×10^{14} atoms
B. 4.20×10^{11} atoms
C. 2.53×10^{12} atoms
D. 3.95×10^{-31} atoms
E. 2.54×10^{21} atoms

$$\left(1.00 \times 10^{-9}\ \text{g U}\right)\left(\frac{1\ \text{mol U}}{238.03\ \text{g U}}\right)\left(\frac{6.022 \times 10^{23}\ \text{atoms U}}{1\ \text{mol U}}\right)$$

$= 2.53 \times 10^{12}$ atoms U

YOUR TURN!

Calculate the mass in grams of $FeCl_3$ in 1.53×10^{23} formula units. (molar mass = 162.204 g/mol)

A. 162.2 g

B. 0.254 g

C. 1.661×10^{-22} g

D. 41.2 g

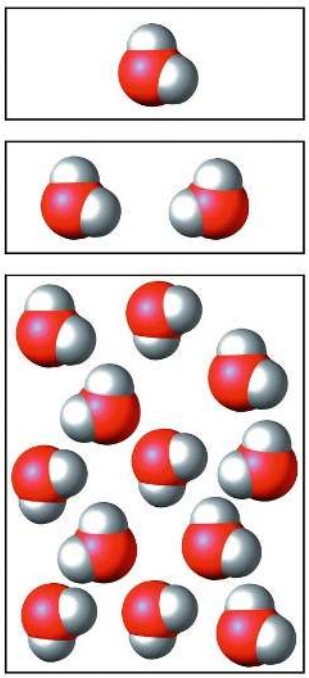
E. 2.37×10^{-22}

$$1.53 \times 10^{23}\ \text{units FeCl}_3 \left(\frac{1\ \text{mol FeCl}_3}{6.022 \times 10^{23}\ \text{units FeCl}_3} \right) \times \left(\frac{162.2\ \text{g FeCl}_3}{1\ \text{mol FeCl}_3} \right)$$

= 41.2 g $FeCl_3$

MOLE-TO-MOLE CONVERSION FACTORS

- Can use chemical formula to relate amount of each atom to amount of compound
- In H_2O there are 3 relationships:
  - 2 mol H ⇔ 1 mol H_2O
  - 1 mol O ⇔ 1 mol H_2O
  - 2 mol H ⇔ 1 mol O
- Can also use these on atomic scale
  - 2 atom H ⇔ 1 molecule H_2O
  - 1 atom O ⇔ 1 molecule H_2O
  - 2 atom H ⇔ 1 molecule O



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STOICHIOMETRIC EQUIVALENCIES

- Within chemical compounds, moles of atoms always combine in the same ratio as the individual atoms themselves
- Ratios of atoms in chemical formulas must be **whole numbers**!!
- These ratios allow us to convert between moles of each quantity

Ex. N_2O_5

2 mol N ⇔ 1 mol N_2O_5

5 mol O ⇔ 1 mol N_2O_5

2 mol N ⇔ 5 mol O

STOICHIOMETRIC EQUIVALENCIES

Equivalency	Mole Ratio	Mole Ratio
2 mol N $\Leftrightarrow$ 1 mol N_2O_5	$\frac{2 \text{ mol N}}{1 \text{ mol } N_2O_5}$	$\frac{1 \text{ mol } N_2O_5}{2 \text{ mol N}}$
5 mol O $\Leftrightarrow$ 1 mol N_2O_5	$\frac{5 \text{ mol O}}{1 \text{ mol } N_2O_5}$	$\frac{1 \text{ mol } N_2O_5}{5 \text{ mol O}}$
2 mol N $\Leftrightarrow$ 5 mol O	$\frac{5 \text{ mol O}}{2 \text{ mol N}}$	$\frac{2 \text{ mol N}}{5 \text{ mol O}}$

CALCULATING THE AMOUNT OF A COMPOUND BY ANALYZING ONE ELEMENT

Calcium phosphate is widely found in natural minerals, bones, and some kidney stones. A sample is found to contain 0.864 moles of phosphorus. How many moles of $Ca_3(PO_4)_2$ are in that sample?

- What do we want to find?

  0.864 mol P = ? mol $Ca_3(PO_4)_2$

- What do we know?

  2 mol P ⇔ 1 mol $Ca_3(PO_4)_2$

- Solution

$$0.864 \text{ mol P} \left(\frac{1 \text{ mol } Ca_3(PO_4)_2}{2 \text{ mol P}} \right)$$

$$= \mathbf{0.432 \text{ mol } Ca_3(PO_4)_2}$$

YOUR TURN!

Calculate the number of **moles** of calcium in 2.53 moles of $Ca_3(PO_4)_2$

A. 2.53 mol Ca

B. 0.432 mol Ca

C. 3.00 mol Ca

D. 7.59 mol Ca

E. 0.843 mol Ca

$$2.53 \text{ mol } Ca_3(PO_4)_2 \left(\frac{3 \text{ mol Ca}}{1 \text{ mol } Ca_3(PO_4)_2} \right)$$

= 7.59 mol Ca

2.53 moles of $Ca_3(PO_4)_2$ = ? mol Ca

3 mol Ca $\Leftrightarrow$ 1 mol $Ca_3(PO_4)_2$

MASS-TO-MASS CALCULATIONS

- Common laboratory calculation
- Need to know what mass of reagent B is necessary to completely react given mass of reagent A to form a compound
- Stoichiometry comes from chemical formula of compounds
  - Subscripts
- Summary of steps

mass A → moles A → moles B → mass B

MASS-TO-MASS CALCULATIONS

Chlorophyll, the green pigment in leaves, has the formula $C_{55}H_{72}MgN_4O_5$. If 0.0011 g of Mg is available to a plant for chlorophyll synthesis, how many grams of carbon will be required to completely use up the magnesium?

- Analysis

  0.0011 g Mg ⇔ ? g C

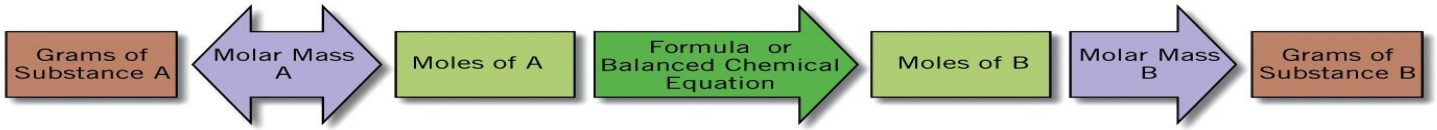
  0.0011 g Mg → mol Mg → mol C → g C

- Assembling the tools

  24.3050 g Mg = 1 mol Mg

  1 mol Mg ⇔ 55 mol C 1 mol C = 12.011 g C

EX. MASS-TO-MASS CONVERSION



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1 mol Mg ⇔ 24.3 g Mg

1 mol C ⇔ 12.0 g C

0.0011 g Mg ⟶ mol Mg ⟶ mol C ⟶ g C

1 mol Mg ⇔ 55 mol C

$$0.0011\ \text{g Mg} \times \left(\frac{1\ \text{mol Mg}}{24.3\ \text{g Mg}}\right) \times \left(\frac{55\ \text{mol C}}{1\ \text{mol Mg}}\right) \times \left(\frac{12.0\ \text{g C}}{1\ \text{mol C}}\right)$$

= 0.030 g C

YOUR TURN!

How many g of iron are required to use up all of 25.6 g of oxygen atoms (O) to form Fe_2O_3?

A. 59.6 g

B. 29.8 g

C. 89.4 g

D. 134 g

E. 52.4 g

mass O → mol O → mol Fe → mass Fe

25.6 g O → ? g Fe

3 mol O ⇔ 2 mol Fe

$$25.6\ \text{g O} \times \left(\frac{1\ \text{mol O}}{16.0\ \text{g O}}\right) \times \left(\frac{2\ \text{mol Fe}}{3\ \text{mol O}}\right) \times \left(\frac{55.845\ \text{g Fe}}{1\ \text{mol Fe}}\right)$$

= **59.6 g Fe**

PERCENTAGE COMPOSITION

- Way to specify relative masses of each element in a compound
- List of percentage by mass of each element

Percentage by Mass

$$\%\text{ by mass of element} = \frac{\text{mass of element}}{\text{mass of sample}} \times 100\%$$

Ex. Na_2CO_3 is

- 43.38% Na
- 11.33% C
- 45.29% O

100.00%

EX. PERCENT COMPOSITION

- Determine percentage composition based on chemical analysis of substance

Ex. A sample of a liquid with a mass of 8.657 g was decomposed into its elements and gave 5.217 g of carbon, 0.9620 g of hydrogen, and 2.478 g of oxygen. What is the percentage composition of this compound?

Analysis:

- Calculate % by mass of each element in sample

Tools:

- Eqn for % by mass
- Total mass = 8.657 g
- Mass of each element

EX. % COMPOSITION OF COMPOUND

For C: $\left(\frac{g\ \text{C}}{g\ total}\right)\times 100\% = \frac{5.217\ \text{g C}}{8.657\ \text{g}}\times 100\% = 60.26\%\ \text{C}$

For H: $\left(\frac{g\ \text{H}}{g\ total}\right)\times 100\% = \frac{0.9620\ \text{g H}}{8.657\ \text{g}}\times 100\% = 11.11\%\ \text{H}$

For O: $\left(\frac{g\ \text{O}}{g\ total}\right)\times 100\% = \frac{2.478\ \text{g O}}{8.657\ \text{g}}\times 100\% = 28.62\%\ \text{O}$

Sum of percentages: 99.99%

- % composition tells us mass of each element in 100.00 g of substance
- In 100.00 g of our liquid
  - 60.26 g C, 11.11 g H & 28.62 g O

YOUR TURN!

A sample was analyzed and found to contain 0.1417 g nitrogen and 0.4045 g oxygen. What is the percentage composition of this compound?

1. Calculate total mass of sample

Total sample mass = 0.1417 g + 0.4045 g = 0.5462 g

2. Calculate % Composition of N

$$\left(\frac{\text{g N}}{\text{g total}}\right)\times 100\% = \left(\frac{0.1417\ \text{g N}}{0.5462\ \text{g}}\right)\times 100\% \quad \textbf{= 25.94\% N}$$

3. Calculate % Composition of O

$$\left(\frac{\text{g O}}{\text{g total}}\right)\times 100\% = \left(\frac{0.4045\ \text{g O}}{0.5462\ \text{g}}\right)\times 100\% \quad \textbf{= 74.06\% O}$$

PERCENT COMPOSITIONS & CHEMICAL IDENTITY

Theoretical or Calculated % Composition

- Calculated from molecular or ionic formula.
- Lets you distinguish between multiple compounds formed from the same 2 elements

- If experimental percent composition is known
  - Calculate **Theoretical % Composition** from proposed **Chemical Formula**
  - Compare with experimental composition

Ex. N & O form multiple compounds

- N_2O, NO, NO_2, N_2O_3, N_2O_4, & N_2O_5

EX. USING PERCENT COMPOSITION

Are the mass percentages 30.54% N & 69.46% O consistent with the formula N_2O_4?

Procedure:

1. Assume 1 mole of compound
2. Subscripts tell how many moles of each element are present
   - 2 mol N & 4 mol O
3. Use molar masses of elements to determine mass of each element in 1 mole
   - Molar Mass of N_2O_4 = 92.14 g N_2O_4 / 1 mol
4. Calculate % by mass of each element

EX. USING PERCENT COMPOSITION (CONT)

$$2\ \text{mol N} \times \frac{14.07\ \text{g N}}{1\ \text{mol N}} = 28.14\ \text{g N}$$

$$4\ \text{mol O} \times \frac{16.00\ \text{g O}}{1\ \text{mol O}} = 64.00\ \text{g O}$$

$$\%\text{N} = \frac{28.14\ \text{g N}}{92.14\ \text{g N}_2\text{O}_4} \times 100\% = \textbf{30.54\% N in } \mathbf{N_2O_4}$$

$$\%\text{O} = \frac{64.00\ \text{g O}}{92.14\ \text{g N}_2\text{O}_4} \times 100\% = \textbf{69.46\% N in } \mathbf{N_2O_4}$$

- The experimental values match the theoretical percentages for the formula N_2O_4.

YOUR TURN

If a sample containing only phosphorous & oxygen has percent composition 56.34% P & 43.66% O, is this P_4O_{10}?

A. Yes

B. No

$$\%P = \frac{123.9\text{g P}}{283.9\text{g } P_4O_{10}} \times 100\%$$

$$\%O = \frac{160.0\text{g O}}{283.9\text{g } P_4O_{10}} \times 100\%$$

10 mol O $\Leftrightarrow$ 1 mol P_4O_{10}

4 mol P = 4 × 30.97 g/mol P = 123.9 g P

4 mol P $\Leftrightarrow$ 1 mol P_4O_{10}

= 43.64 % P

10 mol O = 10 ×16.00 g/mol O = 160.0 g O

= 56.36 % O

1 mol P_4O_{10} = **283.9 g** P_4O_{10}

GENERAL CHEMISTRY 1
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CHAPTER 3: STOICHIOMETRY

LECTURE 17-20

EXPERIMENTAL DETERMINATION OF EMPIRICAL FORMULAS

- When making or isolating new compounds one must characterize them to determine structure & Formula

Molecular Formula

- Exact composition of one molecule
- Exact whole # ratio of atoms of each element in molecule

Empirical Formula

- Simplest ratio of atoms of each element in compound
- Obtained from experimental analysis of compound

Glucose **Empirical formula** CH_2O

Molecular formula $C_6H_{12}O_6$

THREE WAYS TO CALCULATE EMPIRICAL FORMULAS

1. **From Masses of Elements**

   Ex. 2.448 g sample of which 1.771 g is Fe and 0.677 g is O.

2. **From Percentage Composition**

   Ex. 43.64 % P and 56.36 % O.

3. **From Combustion Data**

   - Given masses of combustion products

   Ex. The combustion of a 5.217 g sample of a compound of C, H, and O in pure oxygen gave 7.406 g CO_2 and 4.512 g of H_2O.

STRATEGY FOR DETERMINING EMPIRICAL FORMULAS

1. Determine mass in **g** of each element
2. Convert mass in **g** to **moles**
3. Divide all quantities by smallest number of moles to get smallest ratio of moles
4. Convert any non-integers into integer numbers.
   - If number ends in decimal equivalent of fraction, multiply all quantities by least common denominator
   - Otherwise, round numbers using rules

1. EMPIRICAL FORMULA FROM MASS DATA

When a 0.1156 g sample of a compound was analyzed, it was found to contain 0.04470 g of C, 0.01875 g of H, and 0.05215 g of N. Calculate the empirical formula of this compound.

Step 1: Calculate moles of each substance

$$0.04470\ \text{g C} \times \frac{1\ \text{mol C}}{12.011\ \text{g C}} = 3.722 \times 10^{-3}\ \text{mol C}$$

$$0.01875\ \text{g H} \times \frac{1\ \text{mol H}}{1.008\ \text{g H}} = 1.860 \times 10^{-2}\ \text{mol H}$$

$$0.05215\ \text{g N} \times \frac{1\ \text{mol N}}{14.0067\ \text{g N}} = 3.723 \times 10^{-3}\ \text{mol N}$$

1. EMPIRICAL FORMULA FROM MASS DATA

Step 2: Select the smallest # of moles.

- Lowest is 3.722×10^{-3} mole

		Mole ratio	Integer ratio
C =	$\frac{3.722 \times 10^{-3}\ \text{mol C}}{3.722 \times 10^{-3}\ \text{mol C}}$	1.000	= 1
H =	$\frac{1.860 \times 10^{-2}\ \text{mol H}}{3.722 \times 10^{-3}\ \text{mol C}}$	4.997	= 5
N =	$\frac{3.723 \times 10^{-3}\ \text{mol N}}{3.722 \times 10^{-3}\ \text{mol C}}$	1.000	= 1

Step 3: Divide all # of moles by the smallest one

Empirical formula = CH_5N

EMPIRICAL FORMULA FROM MASS COMPOSITION

One of the compounds of iron and oxygen, "black iron oxide," occurs naturally in the mineral magnetite. When a 2.448 g sample was analyzed it was found to have 1.771 g of Fe and 0.677 g of O. Calculate the empirical formula of this compound.

Assembling the tools:

1 mol Fe = 55.845 g Fe 1 mol O = 16.00 g O

1. Calculate moles of each substance

$$0.677\ \text{g O} \times \frac{1\ \text{mol O}}{16.00\ \text{g O}} = 0.0423\ \text{mol O}$$

$$1.771\ \text{g Fe} \times \frac{1\ \text{mol Fe}}{55.485\ \text{g Fe}} = 0.03171\ \text{mol Fe}$$

1. EMPIRICAL FORMULA FROM MASS DATA

2. Divide both by smallest #mol to get smallest whole # ratio.

$$\frac{0.03171 \text{ mol Fe}}{0.03171 \text{ mol Fe}} = 1.000 \text{ Fe} \times 3 = 3.000 \text{ Fe}$$

$$\frac{0.0423 \text{ mol O}}{0.03171 \text{ mol Fe}} = 1.33 \text{ O} \times 3 = 3.99 \text{ O}$$

Or $$Fe_{\frac{0.03171}{0.03171}}O_{\frac{0.0423}{0.03171}} = Fe_{1.00}O_{1.33}$$

$$Fe_{(1.00\times3)}O_{(1.33\times3)} = Fe_3O_{3.99}$$

Empirical Formula = Fe_3O_4

2. EMPIRICAL FORMULA FROM % COMPOSITION

- New compounds are characterized by elemental analysis, from which the percentage composition can be obtained
- Use percentage composition data to calculate empirical formula
- Must convert % composition to grams
- Assume 100.00 g sample
  - Convenient
  - Sum of % composition = 100%
  - Sum of masses of each element = 100 g

2. EMPIRICAL FORMULA FROM % COMPOSITION

Calculate the empirical formula of a compound whose % composition data is 43.64 % P and 56.36 % O. If the molar mass is determined to be 283.9 g/mol, what is the molecular formula?

1 mol P = 30.97 g

1 mol O = 16.00 g

Step 1: Assume 100 g of compound.

- 43.64 g P
- 56.36 g O

$$43.64 \text{ g P} \times \frac{1 \text{ mol P}}{30.97 \text{ g P}} = 1.409 \text{ mol P}$$

$$56.36 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 3.523 \text{ mol P}$$

2. EMPIRICAL FORMULA FROM % COMPOSITION

Step 2: Divide by smallest number of moles

$$\frac{1.409\ \text{mol P}}{1.409\ \text{mol P}} = 1.000 \quad \times 2 = 2$$

$$\frac{3.523\ \text{mol O}}{1.409\ \text{mol P}} = 2.500 \quad \times 2 = 5$$

Step 3: Multiple by **n** to get smallest integer ratio

Here **n = 2**

Empirical formula = P_2O_5

3. EMPIRICAL FORMULAS FROM INDIRECT ANALYSIS:

- In practice, compounds are not broken down into elements, but are changed into other compounds whose formula is known.

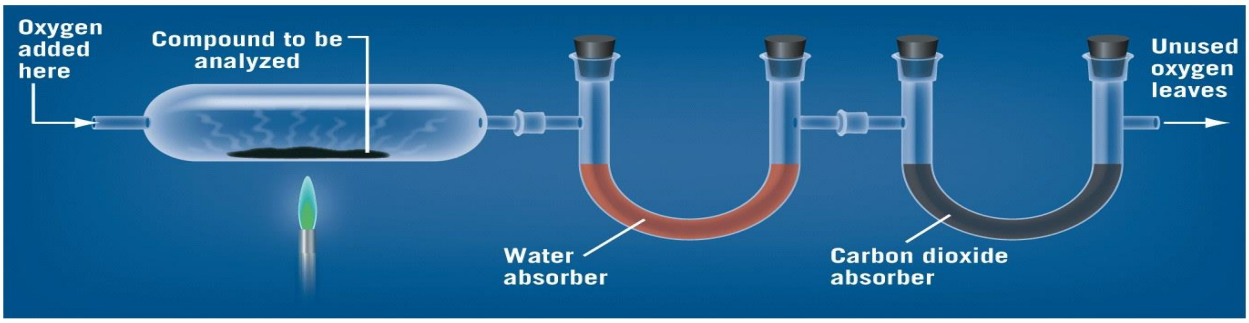
Combustion Analysis

- Compounds containing carbon, hydrogen, & oxygen, can be burned completely in pure oxygen gas
  - Only carbon dioxide & water are produced

Ex. Combustion of methanol (CH_3OH)

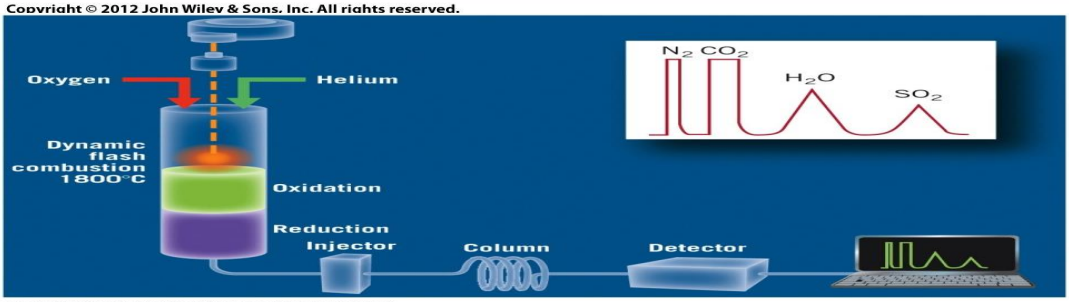
$$2CH_3OH + 3O_2 \longrightarrow 2CO_2 + 4H_2O$$

COMBUSTION ANALYSIS



Classic

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Modern CHN analysis

3. EMPIRICAL FORMULAS FROM INDIRECT ANALYSIS:

- Carbon dioxide & water separated & weighed separately
  - All C ends up as CO_2
  - All H ends up as H_2O
  - Mass of **C** can be derived from amount of **CO_2**
    - mass CO_2 → mol CO_2 → mol C → mass C
  - Mass of **H** can be derived from amount of **H_2O**
    - mass H_2O → mol H_2O → mol H → mass H
  - Mass of oxygen is obtained by difference

    mass O = mass sample – (mass C + mass H)

EX. INDIRECT OR COMBUSTION ANALYSIS

The combustion of a 5.217 g sample of a compound of C, H, and O in pure oxygen gave 7.406 g CO_2 and 4.512 g of H_2O. Calculate the empirical formula of the compound.

	C	H	O	CO_2
MM (g/mol)	12.011	1.008	15.999	44.01

1. Calculate mass of C from mass of CO_2.

mass CO_2 → mole CO_2 → mole C → mass C

$$7.406 \text{ g } CO_2 \left(\frac{1 \text{ mol } CO_2}{44.01 \text{ g } CO_2}\right)\left(\frac{1 \text{ mol C}}{1 \text{ mol } CO_2}\right)\left(\frac{12.011 \text{ g C}}{1 \text{ mol C}}\right)$$

= 2.021 g C

EX. INDIRECT OR COMBUSTION ANALYSIS

The combustion of a 5.217 g sample of a compound of C, H, and O gave 7.406 g CO_2 and 4.512 g of H_2O. Calculate the empirical formula of the compound.

2. Calculate mass of H from mass of H_2O.

mass H_2O → mol H_2O → mol H → mass H

$$4.512\ \text{g}\ H_2O \left(\frac{1\ \text{mol}\ H_2O}{18.015\ \text{g}\ H_2O}\right)\left(\frac{2\ \text{mol H}}{1\ \text{mol}\ H_2O}\right)\left(\frac{1.008\ \text{g H}}{1\ \text{mol H}}\right)$$

= 0.5049 g H

3. Calculate mass of O from difference.

5.217 g sample – 2.021 g C – 0.5049 g H **= 2.691 g O**

EX. INDIRECT OR COMBUSTION ANALYSIS

	C	H	O
MM	12.011	1.008	15.999
g	2.021	0.5049	2.691

4. Calculate mol of each element

$$\text{mol C} = \frac{\text{g C}}{\text{MM C}} = \frac{2.021\ \text{g}}{12.011\ \text{g/mol}} = 0.1683\ \text{mol C}$$

$$\text{mol H} = \frac{\text{g H}}{\text{MM H}} = \frac{0.5049\ \text{g}}{1.008\ \text{g/mol}} = 0.5009\ \text{mol H}$$

$$\text{mol O} = \frac{\text{g O}}{\text{MM O}} = \frac{2.691\ \text{g}}{15.999\ \text{g/mol}} = 0.1682\ \text{mol O}$$

EX. INDIRECT OR COMBUSTION ANALYSIS

- Preliminary empirical formula
  - $C_{0.1683}H_{0.5009}O_{0.1682}$

5. Calculate mol ratio of each element

$$C_{\frac{0.1683}{0.1682}}H_{\frac{0.5009}{0.1682}}O_{\frac{0.1682}{0.1682}} = C_{1.00}H_{2.97}O_{1.00}$$

- Since all values are close to integers, round to

Empirical Formula = CH_3O

DETERMINING MOLECULAR FORMULAS

- **Empirical formula**
  - Accepted formula unit for ionic compounds
- **Molecular formula**
  - Preferred for molecular compounds
- In some cases molecular & empirical formulas are the same
- When they are different, & the subscripts of molecular formula are integer multiples of those in empirical formula
  - If empirical formula is $\mathbf{A_xB_y}$
  - Molecular formula will be $\mathbf{A_{n\times x}B_{n\times y}}$

DETERMINING MOLECULAR FORMULA

- Need molecular mass & empirical formula
- Calculate ratio of molecular mass to mass predicted by empirical formula & round to nearest integer

$$n = \frac{\text{molecular mass}}{\text{empirical formula mass}}$$

Ex. Glucose

Molecular mass is 180.16 g/mol

Empirical formula = CH_2O

Empirical formula mass = 30.03 g/mol

$$n = \frac{180.16\ g}{30.03\ g} = 6$$

Molecular formula = $C_6H_{12}O_6$

LEARNING CHECK

The empirical formula of a compound containing phosphorous and oxygen was found to be P_2O_5. If the molar mass is determined to be 283.9 g/mol, what is the molecular formula?

Step 1: Calculate empirical mass

$$\begin{aligned}\text{empirical mass } P_2O_5 &= (2 \times \text{mass P}) + (5 \times \text{mass O}) \\ &= (2 \times 30.97\ \text{g/mol}) + (5 \times 16.00\ \text{g/mol}) \\ &= (61.94 + 80.00)\ \text{g/mol} \\ &= 141.94\ \text{g/mol}\ P_2O_5\end{aligned}$$

Step 2: Calculate ratio of molecular to empirical mass

$$n = \frac{283.9\ \text{g/mol}}{141.94\ \text{g/mol}} = 2$$

Molecular formula = P_4O_{10}

YOUR TURN!

The empirical formula of hydrazine is NH_2, and its molecular mass is 32.0. What is its molecular formula? **Atomic Mass**: N:14.007; H:1.008; O:15.999

A. NH_2

B. N_2H_4

C. N_3H_6

D. N_4H_8

E. $N_{1.5}H_3$

Molar mass of NH_2 =

$(1\times14.01)g + (2\times1.008)g = 16.017g$

$n = (32.0/16.02) = 2$

BALANCED CHEMICAL EQUATIONS

- Useful tool for problem solving
  - Prediction of reactants and products
  - All atoms present in reactants must also be present among products.
  - Coefficients are multipliers that are used to balance equations
- Two step process
  1. Write unbalanced equation
     - Given products & reactants
     - Organize with plus signs & arrow
  2. Adjust coefficients to get equal numbers of each kind of atom on both sides of arrow.

GUIDELINES FOR BALANCING EQUATIONS

1. Start balancing with the most complicated formula first.
   - Elements, particularly H_2 & O_2, should be left until the end.
2. Balance atoms that appear in only two formulas: one as a reactant & the other as a product.
   - Leave elements that appear in three or more formulas until later.
3. Balance as a group those polyatomic ions that appear unchanged on both sides of the arrow.

BALANCING EQUATIONS

- Use the inspection method

Step 1. Write unbalanced equation

$Zn(s) + HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ unbalanced

Step 2. Adjust coefficients to balance numbers of each kind of atom on both sides of arrow.

- Since $ZnCl_2$ has 2Cl on the product side, 2HCl on reactant side is needed to balance the equation.

$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$

- 1 Zn each side
- 2 H each side
- So balanced

LEARNING CHECK: BALANCING EQUATIONS

$AgNO_3(aq) + Na_3PO_4(aq) \rightarrow Ag_3PO_4(s) + NaNO_3(aq)$

- Count atoms

Reactants	Products
1 Ag	3 Ag
3 Na	1 Na

- Add in coefficients by multiplying Ag & Na by 3 to get 3 of each on both sides

$3AgNO_3(aq) + Na_3PO_4(aq) \rightarrow Ag_3PO_4(s) + 3NaNO_3(aq)$

- Now check polyatomic ions

3 NO_3^-	3 NO_3^-
1 PO_4^{3-}	1 PO_4^{3-} Balanced

BALANCE BY INSPECTION

$$__C_3H_8(g) + __O_2(g) \rightarrow __CO_2(g) + __H_2O(\ell)$$

Assume 1 in front of C_3H_8

3C 1C × 3

8H 2H × 4

$$1C_3H_8(g) + __O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$$

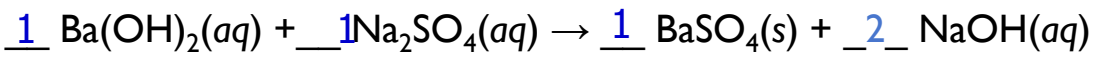
2O × 5 =10 O = (3 × 2) + 4 = 10

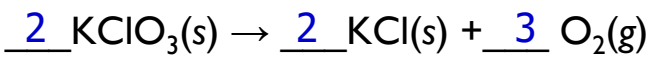
8H H = 2 × 4 = 8

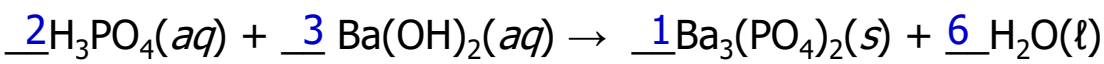
$$1C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$$

YOUR TURN!

Balance each of the following equations.
What are the coefficients in front of each compound?







USING BALANCED EQUATIONS: REACTION STOICHIOMETRY

- **Balanced equation**
  - Critical link between substances involved in chemical reactions
  - Gives relationship between amounts of reactants used & amounts of products likely to be formed
- **Numeric coefficient tells us**
  - The mole ratios for reactions
  - How many individual particles are needed in reaction on microscopic level
  - How many moles are necessary on macroscopic level

STOICHIOMETRIC RATIOS

- Consider the reaction

$$N_2 + 3H_2 \rightarrow 2NH_3$$

- Could be read as:

"When 1 **molecule** of nitrogen reacts with 3 **molecules** of hydrogen, 2 **molecules** of ammonia are formed."

- **Molecular relationships**
- 1 molecule $N_2 \Leftrightarrow$ 2 molecule NH_3
- 3 molecule $H_2 \Leftrightarrow$ 2 molecule NH_3
- 1 molecule $N_2 \Leftrightarrow$ 3 molecule H_2

STOICHIOMETRIC RATIOS

- Consider the reaction

$$N_2 + 3H_2 \rightarrow 2NH_3$$

- Could also be read as:

  "When 1 **mole** of nitrogen reacts with 3 **moles** of hydrogen, 2 **moles** of ammonia are formed."

- **Molar relationships**
  - 1 mole $N_2 \Leftrightarrow$ 2 mole NH_3
  - 3 mole $H_2 \Leftrightarrow$ 2 mole NH_3
  - 1 mole $N_2 \Leftrightarrow$ 3 mole H_2

USING STOICHIOMETRIC RATIOS

Ex. For the reaction $N_2 + 3\ H_2 \rightarrow 2NH_3$, how many moles of N_2 are used when 2.3 moles of NH_3 are produced?

- Assembling the tools
  - 2 moles NH_3 = 1 mole N_2
  - 2.3 mole NH_3 = ? moles N_2

$$2.3\ \text{mol NH}_3 \left(\frac{1\ \text{mol N}_2}{2\ \text{mol NH}_3} \right) = 1.2\ \text{mol N}_2$$

YOUR TURN!

If 0.575 mole of CO_2 is produced by the combustion of propane, C_3H_8, how many moles of oxygen are consumed? The balanced equation is

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

A. 0.575 mole
B. 2.88 mole
C. 0.192 mole
D. 0.958 mole
E. 0.345 mole

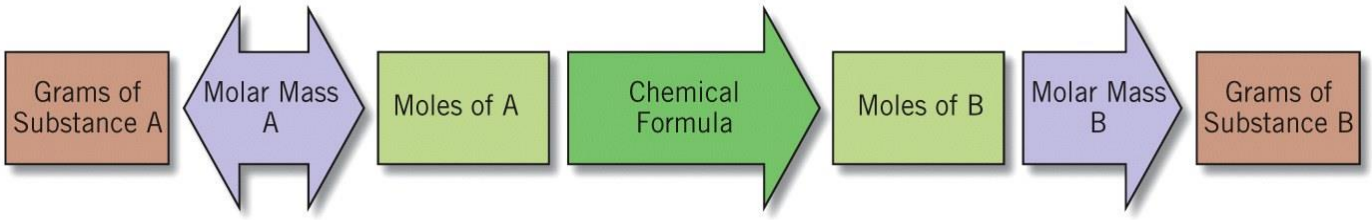
Assembling the tools

- 0.575 mole CO_2 = ? moles O_2
- 3 moles CO_2 = 5 mole O_2

$$0.575 \text{ mol } CO_2 \left(\frac{5 \text{ mol } O_2}{3 \text{ mol } CO_2} \right) = 0.958 \text{ mol } O_2$$

MASS-TO-MASS CONVERSIONS

- Most common stoichiometric conversions that chemists use involve converting mass of one substance to mass of another.



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- Use molar mass A to convert grams A to moles A
- Use chemical equations to relate moles A to moles B
- Use molar mass B to convert to moles B to grams B

USING BALANCED EQUATION TO DETERMINE STOICHIOMETRY

Ex. What mass of O_2 will react with 96.1 g of propane (C_3H_8) gas, to form gaseous carbon dioxide & water?

Strategy

1. Write the balanced equation

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

2. Assemble the tools

96.1 g C_3H_8 → moles C_3H_8 → moles O_2 → g O_2

1 mol C_3H_8 = 44.1 g C_3H_8

1 mol O_2 = 32.00 g O_2

1 mol C_3H_8 = 5 mol O_2

USING BALANCED EQUATION TO DETERMINE STOICHIOMETRY

Ex. What mass of O_2 will react with 96.1 g of propane in a complete combustion?

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

3. Assemble conversions so units cancel correctly

$$96.1\ \text{g}\ C_3H_8 \times \frac{1\ \text{mol}\ C_3H_8}{44.1\ \text{g}\ C_3H_8} \times \frac{5\ \text{mol}\ O_2}{1\ \text{mol}\ C_3H_8} \times \frac{32.0\ \text{g}\ O_2}{1\ \text{mol}\ O_2}$$

= 349 g of O_2 are needed

YOUR TURN!

How many grams of Al_2O_3 are produced when 41.5 g Al react?

$$2Al(s) + Fe_2O_3(s) \longrightarrow Al_2O_3(s) + 2Fe(\ell)$$

A. 78.4 g
B. 157 g
C. 314 g
D. 22.0 g
E. 11.0 g

$$41.5\ \text{g Al}\left(\frac{1\ \text{mol Al}}{26.98\ \text{g Al}}\right)\left(\frac{1\ \text{mol } Al_2O_3}{2\ \text{mol Al}}\right)\left(\frac{101.96\ \text{g } Al_2O_3}{1\ \text{mol } Al_2O_3}\right)$$

$$= 78.4\ \text{g } Al_2O_3$$

MOLECULAR LEVEL OF REACTIONS

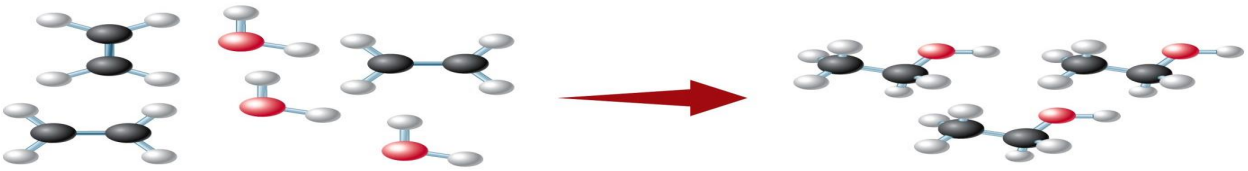
- Consider industrial synthesis of ethanol

$$C_2H_4 + H_2O \longrightarrow C_2H_5OH$$



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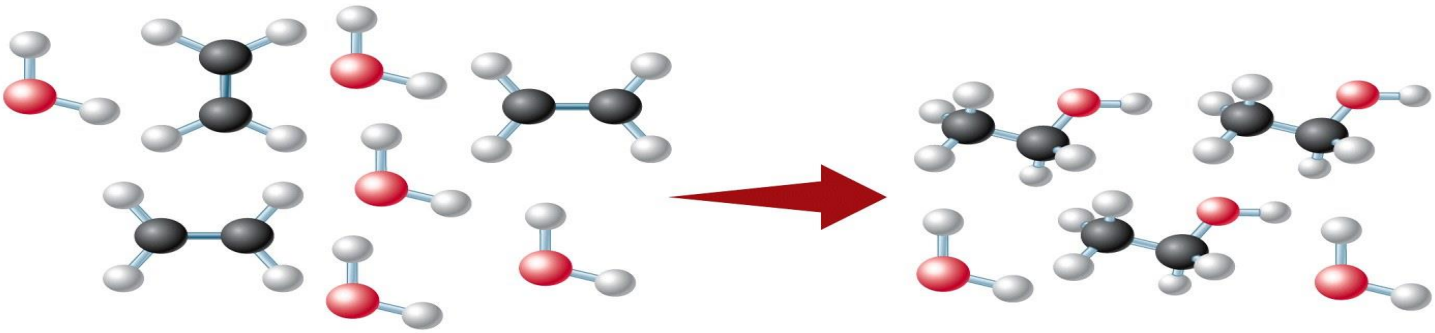
- **3 molecules ethylene + 3 molecules water react to form 3 molecules ethanol**



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MOLECULAR LEVEL OF REACTIONS

- What happens if these proportions are not met?
  - 3 molecules ethylene + 5 molecules of oxygen



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  - All ethylene will be consumed & some oxygen will be left over

LIMITING REACTANT

- Reactant that is completely used up in the reaction
- Present in lower # of moles
- It determines the amount of product produced
- For this reaction = ethylene

Excess reactant

- Reactant that has some amount left over at end
- Present in higher # of moles
- For this reaction = water

LIMITING REACTANT CALCULATIONS

1. Write the balanced equation.
2. Identify the limiting reagent.
   - Calculate amount of reactant B needed to react with reactant B

mass reactant A have → **mol reactant A** → **mol reactant B** → **Mass reactant B need**

   - Compare amount of B you need with amount of B you actually have.
     - If need **more** B than you have, then B is limiting
     - If need **less** B than you have, then A is limiting

LIMITING REACTANT CALCULATIONS

mass limiting reactant → **mol limiting reactant** → **mol product** → **mass product**

3. Calculate mass of desired product, using amount of **limiting reactant** & mole ratios.

EX. LIMITING REACTANT CALCULATION

How many grams of NO can form when 30.0 g NH_3 and 40.0 g O_2 react according to:

$$4\ NH_3 + 5\ O_2 \longrightarrow 4\ NO + 6\ H_2O$$

Solution: Step 1

mass NH_3 → mole NH_3 → mole O_2 → mass O_2

Only have 40.0 g O_2, O_2 limiting reactant

Assembling the tools

- 1 mol NH_3 = 17.03 g
- 1 mol O_2 = 32.00 g
- 4 mol NH_3 ⇔ 5 mol O_2

$$30.0\ g\ NH_3 \times \frac{1\ mol\ NH_3}{17.03\ g\ NH_3} \times \frac{5\ mol\ O_2}{4\ mol\ NH_3} \times \frac{32.00\ g\ O_2}{1\ mol\ O_2}$$

= 70.5 g O_2 needed

EX. LIMITING REACTANT CALCULATION

How many grams of NO can form when 30.0 g NH_3 and 40.0 g O_2 react according to:

$$4\ NH_3 + 5\ O_2 \longrightarrow 4\ NO + 6\ H_2O$$

Solution: Step 2

mass O_2 → mole O_2 → mole NO → mass NO

Can only form 30.0 g NO.

Assembling the tools

- 1 mol O_2 = 32.00 g
- 1 mol NO = 30.01 g
- 5 mol O_2 ⇔ 4 mol NO

$$40.0\ \text{g}\ O_2 \times \frac{1\ \text{mol}\ O_2}{32.00\ \text{g}\ O_2} \times \frac{4\ \text{mol NO}}{5\ \text{mol}\ O_2} \times \frac{30.01\ \text{g NO}}{1\ \text{mol NO}}$$

= 30.0 g NO formed

YOUR TURN!

If 18.1 g NH_3 is reacted with 90.4 g CuO, what is the maximum amount of Cu metal that can be formed?

$$2NH_3(g) + 3CuO(s) \rightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$$

(MM) (17.03) (79.55) (28.01) (64.55) (18.02) (g/mol)

A. 127 g
B. 103 g
C. 72.2 g
D. 108 g
E. 56.5 g

$$18.1\,g\,NH_3 \times \frac{1\,mol\,NH_3}{17.03\,g\,NH_3} \times \frac{3\,mol\,CuO}{2\,mol\,NH_3} \times \frac{79.55\,g\,CuO}{1\,mol\,CuO}$$

127 g CuO needed.
Only have 90.4g so CuO limiting

$$90.4\,g\,CuO \times \frac{1\,mol\,CuO}{79.55\,g\,CuO} \times \frac{3\,mol\,Cu}{3\,mol\,CuO} \times \frac{63.546\,g\,Cu}{1\,mol\,Cu}$$

72.2 g Cu can be formed

REACTION YIELD

- In many experiments, the amount of product is less than expected

- Losses occur for several reasons
  - Mechanical issues – sticks to glassware
  - Evaporation of volatile (low boiling) products.
  - Some solid remains in solution
  - Competing reactions & formation of by-products.
    - **Main reaction:**
      - $2\ P(s) + 3\ Cl_2(g) \rightarrow 2\ PCl_3(\ell)$
    - **Competing reaction:**
      - $PCl_3(\ell) + Cl_2(g) \rightarrow PCl_5(s)$ By-product

THEORETICAL VS. ACTUAL YIELD

- **Theoretical Yield**
  - Amount of product that must be obtained if no losses occur.
  - Amount of product formed if all of limiting reagent is consumed.
- **Actual Yield**
  - Amount of product that is actually isolated at end of reaction.
  - Amount obtained experimentally
  - How much is obtained in mass units or in moles.

PERCENTAGE YIELD

Useful to calculate % yield.

Percent yield

- Relates the actual yield to the theoretical yield
- It is calculated as:

$$\text{percentage } yield = \left(\frac{actual \text{ yield}}{\text{theoretical yield}} \right) \times 100$$

Ex. If a cookie recipe predicts a yield of 36 cookies and yet only 24 are obtained, what is the % yield?

$$\text{percentage yield} = \left(\frac{24}{36} \right) \times 100 = 67\%$$

EX. PERCENTAGE YIELD CALCULATION

When 18.1 g NH_3 and 90.4 g CuO are reacted, the theoretical yield is 72.2 g Cu. The actual yield is 58.3 g Cu. What is the percent yield?

$$2NH_3(g) + 3CuO(s) \rightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$$

$$\%\ \text{yield} = \frac{58.3\ \text{g Cu}}{72.2\ \text{g Cu}} \times 100\% \qquad \textbf{= 80.7\%}$$

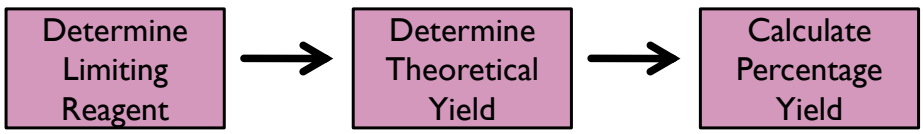
LEARNING CHECK: PERCENTAGE YIELD

A chemist set up a synthesis of solid phosphorus trichloride by mixing 12.0 g of solid phosphorus with 35.0 g chlorine gas and obtained 42.4 g of solid phosphorus trichloride. Calculate the percentage yield of this compound.

Analysis:

Write balanced equation

$$P(s) + Cl_2(g) \longrightarrow PCl_3(s)$$



LEARNING CHECK: PERCENTAGE YIELD

Assembling the Tools:

- 1 mol P = 30.97 g P
- 1 mol Cl_2 = 70.90 g Cl_2
- 3 mol Cl_2 ⇔ 2 mol P

Solution

1. Determine Limiting Reactant

$$12.0\ \text{g P} \times \frac{1\ \text{mol P}}{30.97\ \text{g P}} \times \frac{3\ \text{mol Cl}_2}{2\ \text{mol P}} \times \frac{70.90\ \text{g Cl}_2}{1\ \text{mol Cl}_2} = 41.2\ \text{g Cl}_2$$

- But you only have 35.0 g Cl_2, so Cl_2 is limiting reactant

LEARNING CHECK: PERCENTAGE YIELD

Solution

2. Determine Theoretical Yield

$$35.0\ \text{g } Cl_2 \times \frac{1\ \text{mol } Cl_2}{70.90\ \text{g } Cl_2} \times \frac{2\ \text{mol } PCl_3}{3\ \text{mol } Cl_2} \times \frac{137.32\ \text{g } PCl_3}{1\ \text{mol } PCl_3} = 45.2\ \text{g } PCl_3$$

3. Determine Percentage Yield

- Actual yield = 42.4 g

$$\text{percentage yield} = \left(\frac{42.2\ \text{g } PCl_3}{45.2\ \text{g } PCl_3}\right) \times 100 \quad \mathbf{= 93.8\ \%}$$

When 6.40 g of CH_3OH was mixed with 10.2 g of O_2 and ignited, 6.12 g of CO_2 was obtained. What was the percentage yield of CO_2?

$$2CH_3OH + 3O_2 \longrightarrow 2CO_2 + 4H_2O$$

MM(g/mol) (32.04) (32.00) (44.01) (18.02)

A. 6.12%

B. 8.79%

C. 100%

D. 142%

E. 69.6%

$$6.40\ g\ CH_3OH \times \frac{1\ mol\ CH_3OH}{32.04\ g\ CH_3OH} \times \frac{3\ mol\ CO_2}{2\ mol\ CH_3OH} \times \frac{32.00\ g\ O_2}{1\ mol\ O_2}$$

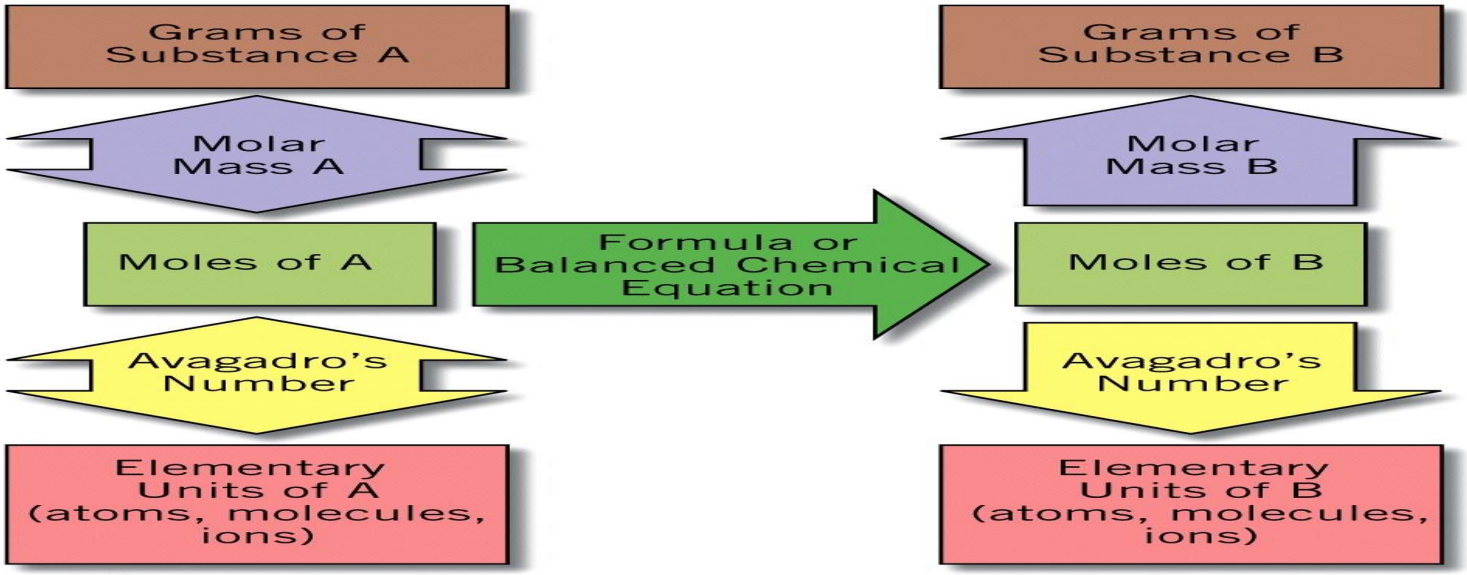
$= 9.59\ g\ O_2$ needed; CH_3OH limiting

$$6.40\ g\ CH_3OH \times \frac{1\ mol\ CH_3OH}{32.04\ g\ CH_3OH} \times \frac{2\ mol\ CO_2}{2\ mol\ CH_3OH} \times \frac{44.01\ g\ CO_2}{1\ mol\ CO_2}$$

$= 8.79\ g\ CO_2$ in theory

$$\frac{6.12\ g\ CH_3OH\ actual}{8.79\ g\ CH_3OH\ theory} \times 100\% = 69.6\%$$

STOICHIOMETRY SUMMARY



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GENERAL CHEMISTRY I
30206101

CHAPTER 4
REACTIONS IN AQUEOUS SOLUTIONS

LECTURE 19-22

REACTIONS IN SOLUTION

- For reaction to occur
  - Reactants needs to come into physical contact
- Happens best in gas or liquid phase
  - Movement occurs

Solution

- Homogeneous mixture
  - 2 or more components mix freely
  - Molecules or ions completely intermingled
  - Contains at least 2 substances

DEFINITIONS:

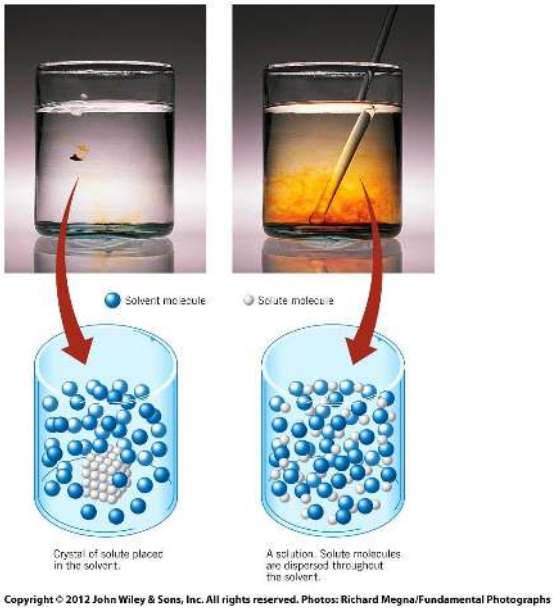
Solvent

- Medium that dissolves solutes
  - Component present in largest amount
  - Can be gas, liquid, or solid
  - Liquids most common
  - Aqueous solution—water is solvent

Solute

- Substance dissolved in solvent
  - Solution is named by solute
  - Can be gas—CO_2 in soda
  - Liquid—Ethylene glycol in antifreeze
  - Solid—Sugar in syrup

IODINE MOLECULES IN ETHANOL



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Crystal of solute placed in solvent

Solute molecules dispersed throughout solvent

SOLUTIONS

- May be characterized using

Concentration

- Solute-to-solvent ratio

$$\frac{g\ \text{solute}}{g\ \text{solvent}} \quad or \quad \frac{g\ \text{solute}}{g\ \text{solution}}$$

- Percent Concentration

$$\%\ \text{concentration} = \frac{g\ \text{solute}}{100\ g\ \text{solution}}$$

RELATIVE CONCENTRATION

Dilute solution

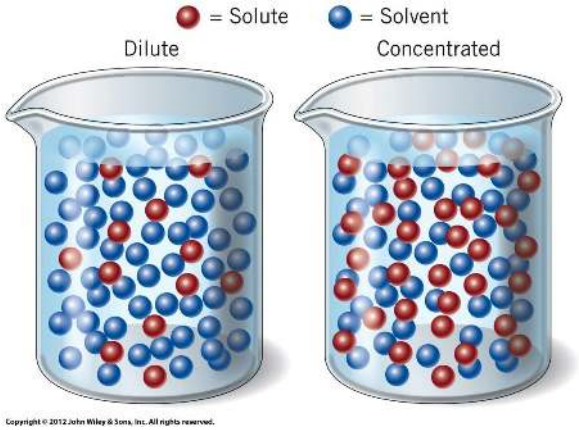
- Small solute to solvent ratio

Ex. Eyedrops

Concentrated solution

- Large solute to solvent ratio

Ex. Pickle brine



- Dilute solution contains less solute per unit volume than more concentrated solution

CONCENTRATION
Solubility

$$\text{Solubility} = \frac{\text{g } \textbf{solute} \text{ needed to make saturated solution}}{100 \text{ g solvent}}$$

- Temperature dependent

Saturated solution

- Solution in which no more solute can be dissolved at a given temperature

Unsaturated solution

- Solution containing **less** solute than maximum amount
- Able to dissolve more solute

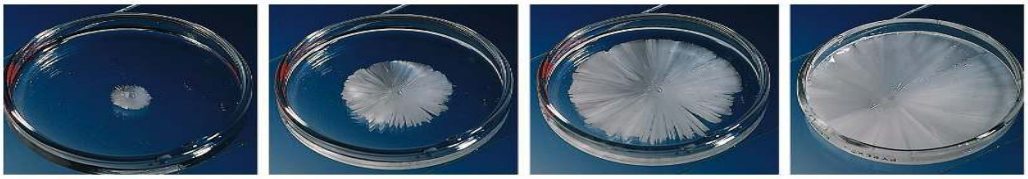
SOLUBILITIES OF SOME COMMON SUBSTANCES

Substance	Formula	Solubility (g/100 g water)
Sodium chloride	NaCl	35.7 at 0°C 39.1 at 100°C
Sodium hydroxide	NaOH	42 at 0°C 347 at 100°C
Calcium carbonate	$CaCO_3$	0.0015 at 25°C

CONCENTRATIONS

Supersaturated Solutions

- Contains more solute than required for saturation at a given temperature
- Formed by careful cooling of saturated solutions
- Unstable
- Crystallize out when add seed crystal – results in formation of solid or **precipitate (ppt.)**



Andy Washnik

PRECIPTATES

Precipitate

- Solid product formed when reaction carried out in solutions and one product has low solubility
- Insoluble product
- Separates out of solution

Precipitation reaction

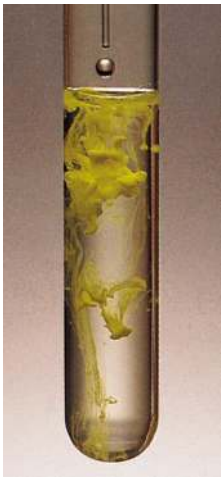
- Reaction that produces precipitate

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$$

- 1 mol $Pb(NO_3)_2$ $\Leftrightarrow$ 2 mol KI
- 0.100 mol $Pb(NO_3)_2$ $\Leftrightarrow$ 0.200 mol KI

PRECIPITATION REACTIONS

Precipitate – insoluble solid that separates from solution



PbI_2

precipitate

$$Pb(NO_3)_2\,(aq) + 2NaI\,(aq) \longrightarrow PbI_2\,(s) + 2NaNO_3\,(aq)$$

molecular equation

$$Pb^{2+} + \cancel{2NO_3^-} + \cancel{2Na^+} + 2I^- \longrightarrow PbI_2\,(s) + \cancel{2Na^+} + \cancel{2NO_3^-}$$

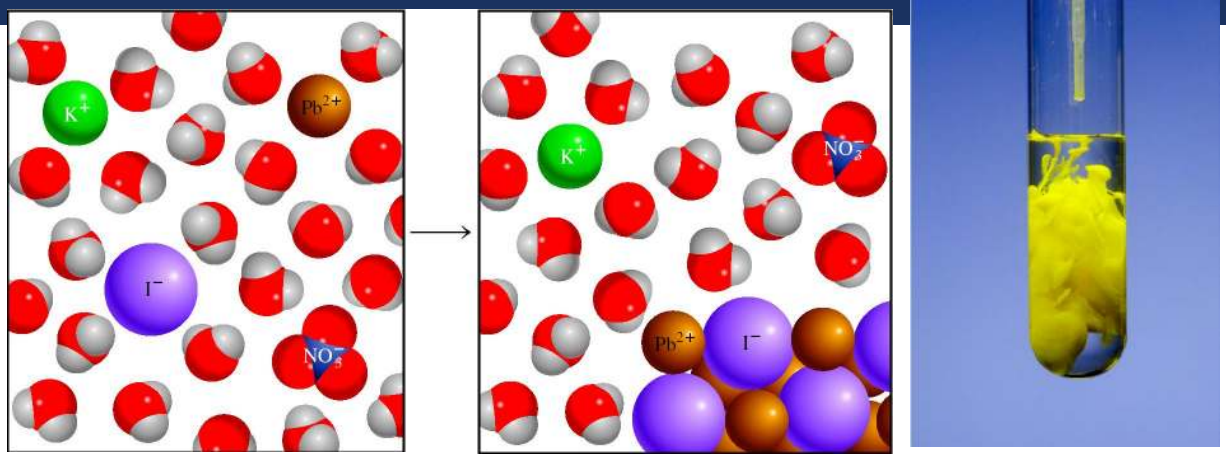
ionic equation

$$Pb^{2+} + 2I^- \longrightarrow PbI_2\,(s)$$

net ionic equation

Na^+ and NO_3^- are ***spectator*** ions

PRECIPITATION OF LEAD IODIDE



$Pb^{2+} + 2I^{-} \longrightarrow PbI_2\ (s)$

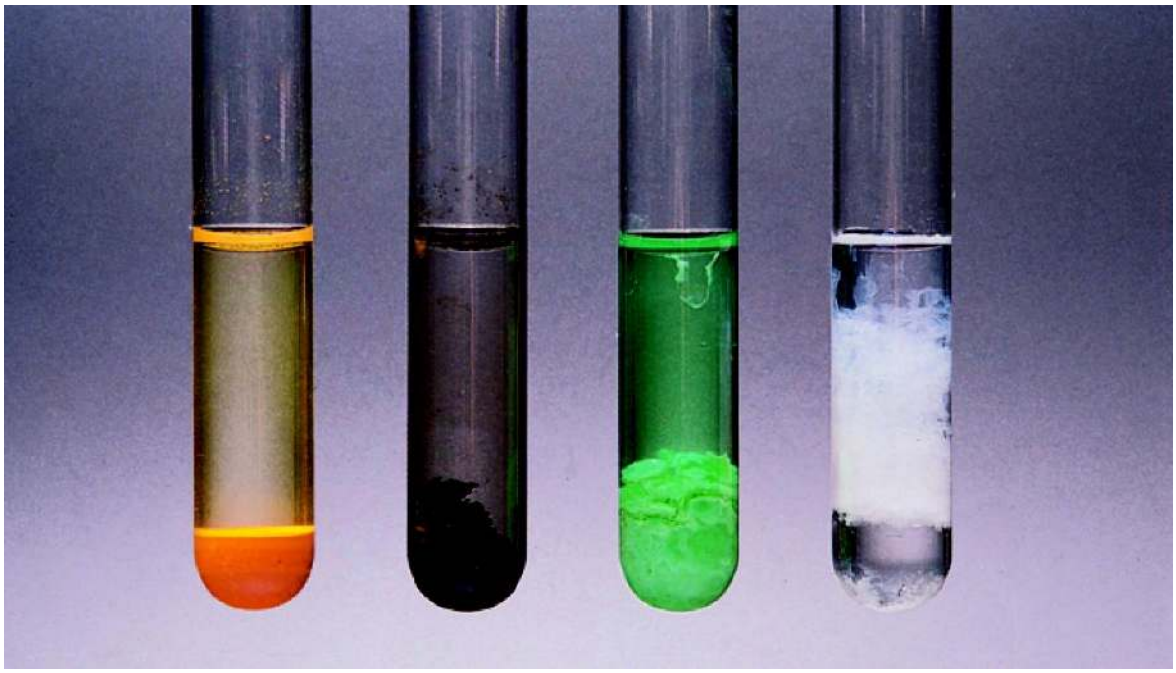
PbI_2

Solubility is the maximum amount of solute that will dissolve in a given quantity of solvent at a specific temperature.

TABLE 4.2 Solubility Rules for Common Ionic Compounds in Water at 25°C

Soluble Compounds	Insoluble Exceptions
Compounds containing alkali metal ions (Li^+, Na^+, K^+, Rb^+, Cs^+) and the ammonium ion (NH_4^+)	
Nitrates (NO_3^-), bicarbonates (HCO_3^-), and chlorates (ClO_3^-)	
Halides (Cl^-, Br^-, I^-)	Halides of Ag^+, Hg_2^{2+}, and Pb^{2+}
Sulfates (SO_4^{2-})	Sulfates of Ag^+, Ca^{2+}, Sr^{2+}, Ba^{2+}, Hg_2^{2+}, and Pb^{2+}
Insoluble Compounds	**Soluble Exceptions**
Carbonates (CO_3^{2-}), phosphates (PO_4^{3-}), chromates (CrO_4^{2-}), sulfides (S^{2-})	Compounds containing alkali metal ions and the ammonium ion
Hydroxides (OH^-)	Compounds containing alkali metal ions and the Ba^{2+} ion

Examples of Insoluble Compounds



CdS PbS $Ni(OH)_2$ $Al(OH)_3$

WRITING NET IONIC EQUATIONS

1. Write the balanced molecular equation.
2. Write the ionic equation showing the strong electrolytes completely dissociated into cations and anions.
3. Cancel the spectator ions on both sides of the ionic equation
4. Check that charges and number of atoms are balanced in the net ionic equation

Write the net ionic equation for the reaction of silver nitrate with sodium chloride.

$$AgNO_3\,(aq) + NaCl\,(aq) \longrightarrow AgCl\,(s) + NaNO_3\,(aq)$$

$$Ag^+ + \cancel{NO_3^-} + \cancel{Na^+} + Cl^- \longrightarrow AgCl\,(s) + \cancel{Na^+} + \cancel{NO_3^-}$$

$$Ag^+ + Cl^- \longrightarrow AgCl\,(s)$$

CONCENTRATION EXPRESSION

Way to express concentration of a solution:

- Remember: Concentration is a ratio between (solute/solution) or (solute/solvent.)

① Molarity = $\frac{\text{moles solute (mol)}}{\text{Liters solution}}$

$mol/L = mol.L^{-1} \equiv M$ (molar)

② (w/w) % = $\frac{\text{mass solute (g)}}{\text{mass solution (g)}} \times 100\%$

ex) 5g NaCl in 0.5 L water.

- density (H_2O) = 1 g/mL
- mass (H_2O) (g) = volume (H_2O) (mL)

% NaCl = $\frac{5g}{505g} \times 100\% = 1\%$

(وزنه المذاب + المذيب) = وزنه المحلول .

Scanned with CamScanner

- In molarity

  Volume (flask) $\cong$ V (water) = V solution.

  V: Volume.

- Calculate grams of KCl required to be dissolved in water in a 250mL flask with 0.32 M concentration.

$$\text{moles (KCl)} = M_{soln} \cdot V_{soln.}$$
$$= 0.32M * 0.25L$$
$$= 0.08 \text{ mol KCl}$$
$$= 5.96 \text{ gram.}$$

How many grams of KCl are found in a 6.5 % KCl solution given that Volume of solution = 300mL and density of solution = 1.13 g/mL

6.5 % → 6.5 g KCl ⇔ 100 g solution.

Solution ↗ 300mL
↘ density = 1.13 g/mL

mass (soln) = d x V = 1.13 g/mL x 300mL

= 339 gram. soln.

6.5 g KCl ⇔ 100 g soln.

? ⇔ 339 g soln

= 22.035 g KCl.

② % solute = w/w % = $\frac{\text{mass solute.}}{\text{mass soln}}$ x100 %

③ Molality = $\frac{\text{moles solute}}{\text{kilograms solvent}}$ mol/kg = m (molal)

$d_{(\text{solvent})} = \frac{\text{mass (solvent)}}{V_{(\text{solvent})}}$

لتحويل الكتله إلى حجم أو العكس

اتحاد سنافر البوليتكنك (spu)

Suppose : you have a 13.7 % of HCl solution. Calculate its molar concentration given that :

- density of HCl solution = 1.41 g/mL.
- F.wt of HCl = 36.5 g/mol

13.7 % HCl → 13.7 g HCl (solute)
→ 100g solution.

$$\text{Molarity} = \frac{\text{moles HCl}}{\text{V of soln (in L)}}$$

$$\text{Molarity} = \frac{(\text{mass}/\text{F.wt}) \Rightarrow \text{solute : HCl}}{(\text{Mass}/\text{density}) \Rightarrow \text{solution.}}$$

$$\text{Molarity} = \frac{(13.7\text{g}/36.5)}{(100\text{g}/1.41\text{g/mL})} = \frac{0.375\text{ mol}}{70.92}$$

$$\text{Molarity} = \frac{0.375\ \text{mol}}{(70.92\ \text{mL} / 1000)\ \text{L}}$$

$$= 5.28\ \text{M}.$$

$$\text{Molarity} = \frac{\% \times \text{density} \times 10}{\text{F.wt}}$$

$$= \frac{13.7 \times 1.41 \times 10}{36.5} = 5.29\ \text{M}.$$

MOLAR CONCENTRATION

Dissolve solutes. Make separate solutions.

↓

Mix Solutions.

↓

Allow Reaction to occur.

- Need to know Quantitatively HOW MUCH? of each solute we used.
- Define

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{liters of solution}} = \frac{\text{mole}}{\text{Volume}}$$

MOLARITY (*M*)

- Number of moles of solute per liter of solution.
- Allows us to express relationship between moles of solute and volume of solution
- Hence, 0.100*M* solution of NaCl contains 0.100 mole NaCl in 1.00 liter of solution
- Same concentration results if you dissolve 0.0100 mol of NaCl in 0.100 liter of solution

$$\frac{0.100 \text{ mol NaCl}}{1.00 \text{ L NaCl soln}} = \frac{0.0100 \text{ mol NaCl}}{0.100 \text{ L NaCl soln}} = 0.100M \text{ NaCl}$$

MOLARITY AS CONVERSION FACTOR

- Often have stoichiometry problems involving amount of chemical and volume of solution
  - Solve the problem using molarity



- Molarity provides conversion factors between moles and volume
- *M* = mole per liter

$$0.100M\ \text{NaCl} = \frac{0.100\ \text{mol NaCl}}{1.00\ \text{L NaCl soln}}$$

MOLARITY AS CONVERSION FACTOR

- Gives equivalence relationship between "mol NaCl" and "L soln"
- Forms two conversion factors

$$0.100\ \text{mol NaCl} \Leftrightarrow 1.00\ \text{L soln}$$

$$\frac{0.100\ \text{mol NaCl}}{1.00\ \text{L NaCl soln}} \qquad \frac{1.00\ \text{L NaCl soln}}{0.100\ \text{mol NaCl}}$$

- Three basic types of calculations:

$$M = \frac{\text{mol}}{\text{V}} \qquad M * \text{V} = \text{mol} \qquad \text{V} = \frac{\text{mol}}{M}$$

LEARNING CHECK: CALCULATING MOLARITY (FROM GRAMS AND VOLUME)

Calculate the molarity (*M*) of a solution prepared by dissolving 11.5 g NaOH (40.00 g/mol) solid in enough water to make 1.50 L of solution.

g NaOH ⟶ mol NaOH ⟶ *M* NaOH

$$M = \frac{\text{moles NaOH}}{\text{L soln}} = \frac{0.288 \text{ mol NaOH}}{1.50 \text{ L soln}}$$

$$11.5 \text{ g NaOH} \times \frac{1 \text{ mol NaOH}}{40.00 \text{ g NaOH}} = 0.288 \text{ mol NaOH}$$

$$= 0.192 M \text{ NaOH}$$

LEARNING CHECK: CALCULATING VOLUME

How many mL of 0.250 *M* NaCl solution are needed to obtain 0.100 mol of NaCl?

- Use *M* definition

$$0.250\ M\ \text{NaCl} = \frac{0.250\ \text{mol NaCl}}{1.00\ \text{L NaCl soln}}$$

- Given molarity and moles, need volume

$$0.100\ \text{mol NaCl} \times \frac{1.00\ \text{L NaCl soln}}{0.250\ \text{mol NaCl}} \times \frac{1000\ \text{mL NaCl soln}}{1\ \text{L NaCl soln}}$$

$$= 400\ \text{mL of } 0.250\ M\ \text{NaCl solution}$$

PREPARING SOLUTION OF KNOWN MOLARITY



a b c d e

a) Weigh solid and transfer to volumetric flask
b) Add part of the water
c) Dissolve solute completely
d) Add water to reach etched line
e) Stopper flask and invert to mix thoroughly

LEARNING CHECK: PREPARING SOLUTION OF

How many grams of strontium nitrate are required to prepare 250.0 mL of 0.100 *M* $Sr(NO_3)_2$ solution?

- **M × V ⟶ mol × MM ⟶ g**

1. **Convert Molarity and Volume to mole**

$$250\ \text{mL}\ Sr(NO_3)_2\ \text{soln} \times \frac{1\ \text{L}}{1000\ \text{mL}} \times \frac{0.100\ M}{1\ \text{L}}$$

$$= 0.0250\ \text{mol}\ Sr(NO_3)_2$$

2. Convert mol to g

$$0.0250\ \text{mol} \times \frac{211.62\ \text{g}}{1\ \text{mol}} = \mathbf{5.29\ g\ Sr(NO_3)_2}$$

YOUR TURN

How many grams of $KMnO_4$ must you weigh out if you want to make 250.mL of a 0.200 M $KMnO_4$ solution?

A. 7900 g

B. 50.0 g

C. 0.316 g

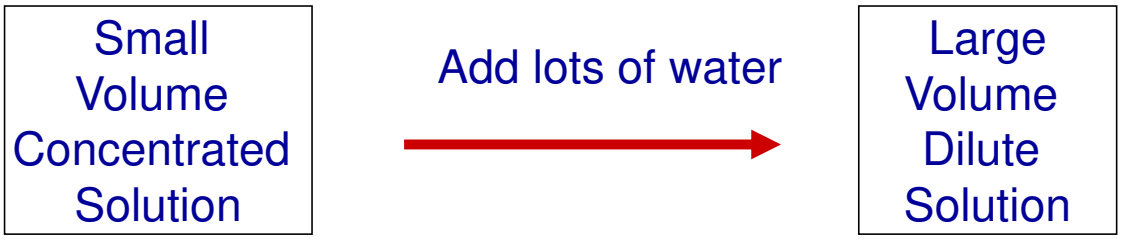
D. 7.90 g

E. 198 g

$$250\ \text{mL} \times \frac{1\ \text{L}}{1000\ \text{mL}} \times \frac{0.200\ \text{mol}}{1\ \text{L}} \times \frac{158.03\ \text{g}}{\text{mol}\ KMnO_4} = 7.90\ \text{g}$$

PREPARING A SOLUTION OF KNOWN MOLARITY BY DILUTION

- When making solutions, don't always begin with solute as pure solid, liquid or gas.
- Can take solution of high concentration and dilute to a lower concentration.



- Amount of MOLES does NOT change! Remains the same

DILUTING SOLUTIONS

- Moles of solute do not change upon dilution
- Just changing volume

moles in dilute = # moles in concentrated

$$\left(\begin{array}{l}\text{Volume of}\\ \text{dilute solution}\\ \textit{to be prepared}\end{array}\right) \times M_{dilute} = \left(\begin{array}{l}\text{Volume of}\\ \text{concentrated solution}\\ \textit{to be used}\end{array}\right) \times M_{concentrated}$$

Moles of solute in the dilute solution

Moles of solute in the concentrated solution

$$\mathbf{V_{dil} \times M_{dil} = V_{conc} \times M_{conc}}$$

LEARNING CHECK: DILUTIONS



What volume (in mL) of 16.0 *M* H_2SO_4 must be used to prepare 1.00 L of 2.00 *M* H_2SO_4?



$$1.00\ \text{L} \times 2.00\ M = \text{V} \times 16.0\ M$$

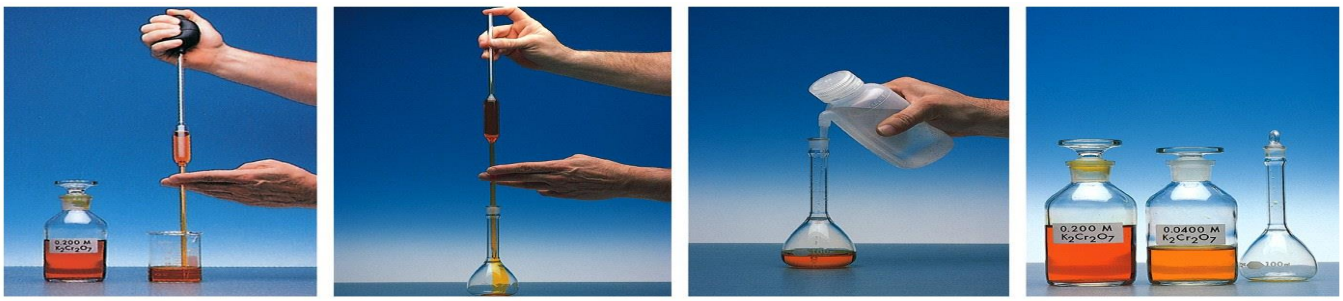
Rearranging gives

$$\text{V} = \frac{1.00\ \text{L} \times 2.00\ M}{16.0\ M} = \frac{2.00\ \text{mol}}{16.0\ \text{mol/L}}$$

$$\text{V} = 0.125\ \text{L} \times \frac{1000\ \text{mL}}{1.00\ \text{L}} \quad = 125\ mL$$

PREPARING SOLUTION OF KNOWN *M*

- Using volumetric glassware ensures that the volumes are known precisely
- Use a volumetric pipette to transfer the stock solution
- Use a volumetric flask to receive the final solution



OPC, Inc.

YOUR TURN

What volume of 12.1 *M* HCl is needed to create 250. mL of 3.2 *M* HCl?

A. 66 mL

B. 800 mL

C. 3025 mL

D. 945 mL

E. 9680 mL

$$V_{conc} \times 12.1\ M = 250.\ mL \times 3.2\ M$$

V_{conc} = 66 mL

YOUR TURN

25 mL of 6.0 *M* HCl are diluted to 500 mL with water. What is the molarity of the resulting solution?

A. 150 *M*

B. 3.0 *M*

C. 0.120 *M*

D. 120 *M*

E. 0.30 *M*

$25\ \text{mL} \times 6.0\ M = 500\ \text{mL} \times M_{dil}$

$M_{dil} = 0.30\ M$

YOUR TURN



What mass of KI is required to make 500. mL of a 2.80 *M* KI solution?

volume of KI solution $\xrightarrow{M\ \text{KI}}$ moles KI $\xrightarrow{\mathcal{M}\ \text{KI}}$ grams KI

$$500.\ \cancel{\text{mL}} \times \frac{1\ \cancel{\text{L}}}{1000\ \cancel{\text{mL}}} \times \frac{2.80\ \cancel{\text{mol KI}}}{1\ \cancel{\text{L soln}}} \times \frac{166\ \text{g KI}}{1\ \cancel{\text{mol KI}}} = 232\ \text{g KI}$$

YOUR TURN

**A 20.00 mL sample of 0.1015 M nitric acid is introduced into a flask, and water is added until the volume of the solution reaches 250. mL. What is the concentration of nitric acid in the final solution?

**A student measured 175 mL of 0.250 M HCl solution into a beaker.
How many moles of HCl were in the beaker?
How many grams of HCl are in the beaker?



How would you prepare 60.0 mL of 0.200 *M* HNO_3 from a stock solution of 4.00 *M* HNO_3?

$$M_i V_i = M_f V_f$$

$M_i = 4.00$ $M_f = 0.200$ $V_f = 0.06$ L $V_i = ?$ L

$$V_i = \frac{M_f V_f}{M_i} = \frac{0.200 \times 0.06}{4.00} = 0.003 \text{ L} = 3 \text{ mL}$$

3 mL of acid + 57 mL of water = 60 mL of solution

**How many grams of NaOH (MW = 40.0) are there in 500.0 mL of a 0.175 M NaOH solution?

**An aqueous ethanol solution (400 mL) was diluted to 4.00 L, giving a concentration of 0.0400 M. The concentration of the original solution was _____ M.

**The molarity (M) of an aqueous solution containing 22.5 g of sucrose($C_{12}H_{22}O_{11}$) in 35.5 mL of solution is _____.

**How many milliliters of water have to be added to 150 mL of 0.5 M HCl to reduce the concentration to 0.10 M HCl?

Gravimetric Analysis

1. Dissolve unknown substance in water
2. React unknown with known substance to form a precipitate
3. Filter and dry precipitate
4. Weigh precipitate
5. Use chemical formula and mass of precipitate to determine amount of unknown ion





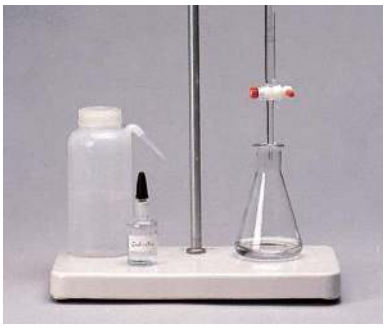


Titrations

In a ***titration*** a solution of accurately known concentration is added gradually added to another solution of unknown concentration until the chemical reaction between the two solutions is complete.

Equivalence point – the point at which the reaction is complete

Indicator – substance that changes color at (or near) the equivalence point



Slowly add base to unknown acid UNTIL

the indicator changes color



Titrations can be used in the analysis of



Acid-base reactions

$$H_2SO_4 + 2NaOH \longrightarrow 2H_2O + Na_2SO_4$$



Redox reactions

$$5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

What volume of a 1.420 *M* NaOH solution is required to titrate 25.00 mL of a 4.50 *M* H_2SO_4 solution?

WRITE THE CHEMICAL EQUATION!

$$H_2SO_4 + 2NaOH \longrightarrow 2H_2O + Na_2SO_4$$

volume acid $\xrightarrow[\text{acid}]{M}$ moles red $\xrightarrow[\text{coef.}]{\text{rxn}}$ moles base $\xrightarrow[\text{base}]{M}$ volume base

$$25.00\ \text{mL} \times \frac{4.50\ \text{mol}\ H_2SO_4}{1000\ \text{mL soln}} \times \frac{2\ \text{mol NaOH}}{1\ \text{mol}\ H_2SO_4} \times \frac{1000\ \text{ml soln}}{1.420\ \text{mol NaOH}} = 158\ \text{mL}$$

16.42 mL of 0.1327 *M* $KMnO_4$ solution is needed to oxidize 25.00 mL of an acidic $FeSO_4$ solution. What is the molarity of the iron solution?

WRITE THE CHEMICAL EQUATION!

$$5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

volume red $\xrightarrow[\text{red}]{M}$ moles red $\xrightarrow[\text{coef.}]{\text{rxn}}$ moles oxid $\xrightarrow[\text{oxid}]{V}$ *M* oxid

16.42 mL = 0.01642 L 25.00 mL = 0.02500 L

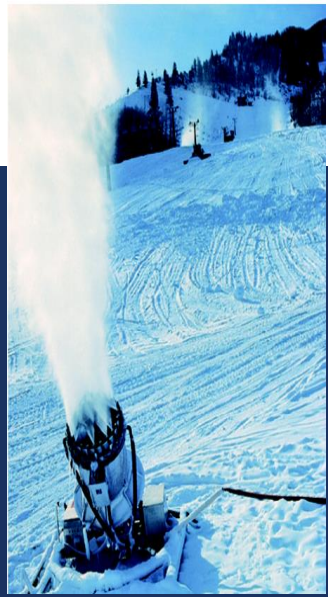
$$0.01642\ \text{L} \times \frac{0.1327\ \text{mol KMnO}_4}{1\ \text{L}} \times \frac{5\ \text{mol Fe}^{2+}}{1\ \text{mol KMnO}_4} \times \frac{1}{0.02500\ \text{L Fe}^{2+}} = 0.4358\ M$$

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CHAPTER 6
ENERGY RELATIONSHIPS IN CHEMICAL REACTIONS

LECTURE 23-30



Energy is the capacity to do work.

- *Radiant energy* comes from the sun and is earth's primary energy source

- *Thermal energy* is the energy associated with the random motion of atoms and molecules

- *Chemical energy* is the energy stored within the bonds of chemical substances

- *Nuclear energy* is the energy stored within the collection of neutrons and protons in the atom

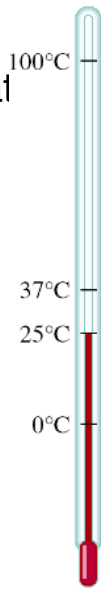
- *Potential energy* is the energy available by virtue of an object's position

Energy Changes in Chemical Reactions

Heat is the transfer of thermal energy between two bodies that are at different temperatures.

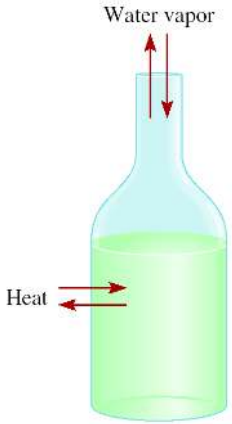
Temperature is a measure of the thermal energy.

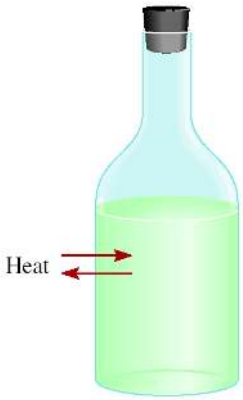
Temperature ≠ Thermal Energy



Thermochemistry is the study of heat change in chemical reactions.

The *system* is the specific part of the universe that is of interest in the study.







	open	closed	isolated
Exchange:	mass & energy	energy	nothing

Exothermic process is any process that gives off heat – transfers thermal energy from the system to the surroundings.

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O\,(l) + \text{energy}$$

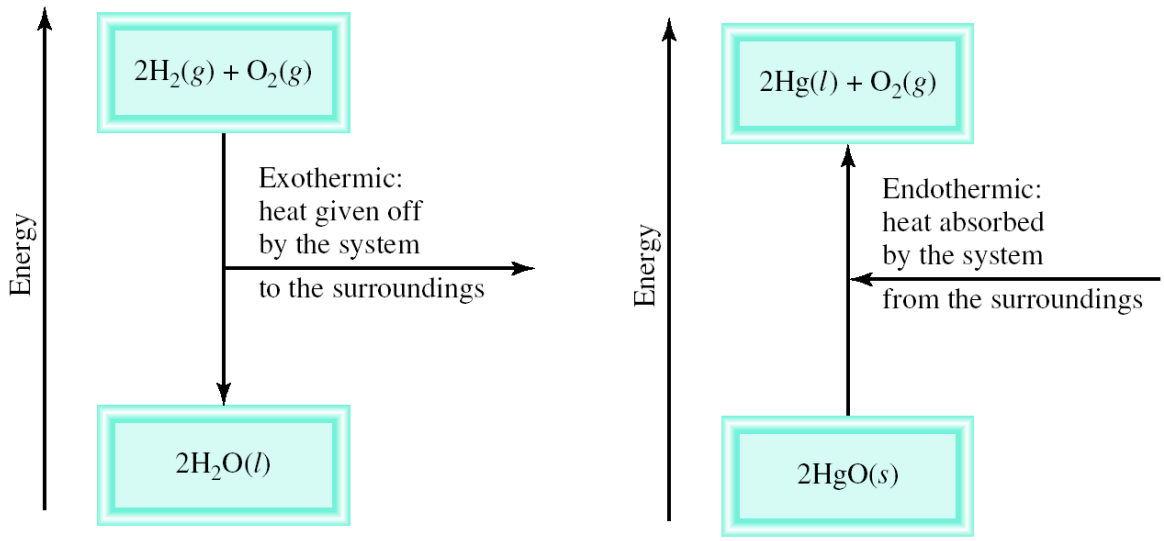
$$H_2O\,(g) \longrightarrow H_2O\,(l) + \text{energy}$$

Endothermic process is any process in which heat has to be supplied to the system from the surroundings.

$$\text{energy} + 2HgO\,(s) \longrightarrow 2Hg\,(l) + O_2\,(g)$$

$$\text{energy} + H_2O\,(s) \longrightarrow H_2O\,(l)$$

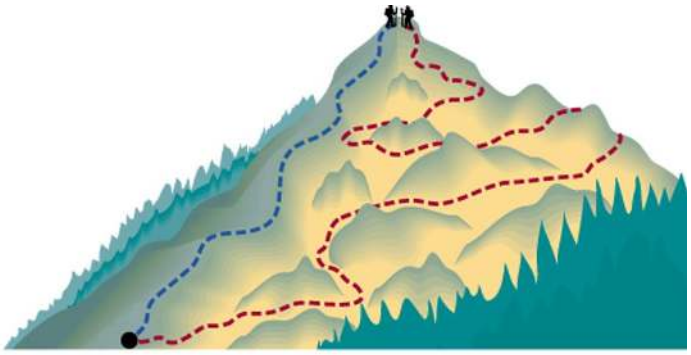
Schematic of Exothermic and Endothermic Processes



Thermodynamics is the scientific study of the interconversion of heat and other kinds of energy.

State functions are properties that are determined by the state of the system, regardless of how that condition was achieved.

energy, pressure, volume, temperature



$$\Delta E = E_{final} - E_{initial}$$

$$\Delta P = P_{final} - P_{initial}$$

$$\Delta V = V_{final} - V_{initial}$$

$$\Delta T = T_{final} - T_{initial}$$

Potential energy of hiker 1 and hiker 2 is the same even though they took different paths.

First law of thermodynamics – energy can be converted from one form to another, but cannot be created or destroyed.



$$\Delta E_{system} + \Delta E_{surroundings} = 0$$

or

$$\Delta E_{system} = -\Delta E_{surroundings}$$

$$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$$

Exothermic chemical reaction!

Chemical energy lost by combustion = Energy gained by the surroundings
system surroundings

Another form of the *first law* for ΔE_{system}

$\Delta E = q + w$

ΔE is the change in internal energy of a system

q is the heat exchange between the system and the surroundings

w is the work done on (or by) the system

$w = -P\Delta V$ when a gas expands against a constant external pressure

TABLE 6.1 Sign Conventions for Work and Heat

Process	Sign
Work done by the system on the surroundings	−
Work done on the system by the surroundings	+
Heat absorbed by the system from the surroundings (endothermic process)	+
Heat absorbed by the surroundings from the system (exothermic process)	−

Work Done On the System

$w = F \times d$

$w = -P \Delta V$

$P \times V = \frac{F}{d^2} \times d^3 = F \times d = w$

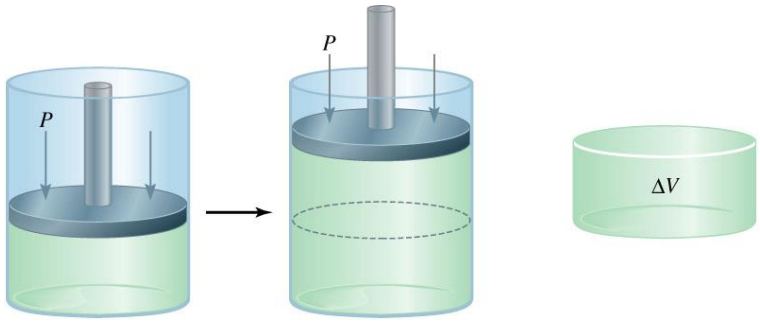
$\Delta V > 0$

$-P\Delta V < 0$

$w_{sys} < 0$

Work is not a state function.

$\Delta w \neq w_{final} - w_{initial}$



initial final

Example :

A sample of nitrogen gas expands in volume from 1.6 L to 5.4 L at constant temperature. What is the work done in joules if the gas expands (a) against a vacuum and (b) against a constant pressure of 3.7 atm?

$$w = -P\Delta V$$

(a) $\Delta V = 5.4\ \text{L} - 1.6\ \text{L} = 3.8\ \text{L}$ $P = 0$ atm

$$W = -0\ \text{atm} \times 3.8\ \text{L} = 0\ \text{L}\cdot\text{atm} = 0\ \text{joules}$$

(b) $\Delta V = 5.4\ \text{L} - 1.6\ \text{L} = 3.8\ \text{L}$ $P = 3.7$ atm

$$w = -3.7\ \text{atm} \times 3.8\ \text{L} = -14.1\ \text{L}\cdot\text{atm}$$

$$w = -14.1\ \cancel{\text{L}\cdot\text{atm}} \times \frac{101.3\ \text{J}}{1\ \cancel{\text{L}\cdot\text{atm}}} = -1430\ \text{J}$$

Chemistry in Action: Making Snow

$\Delta E = q + w$

$q = 0$

$w < 0$, $\Delta E < 0$

$\Delta E = C\Delta T$

$\Delta T < 0$, SNOW!



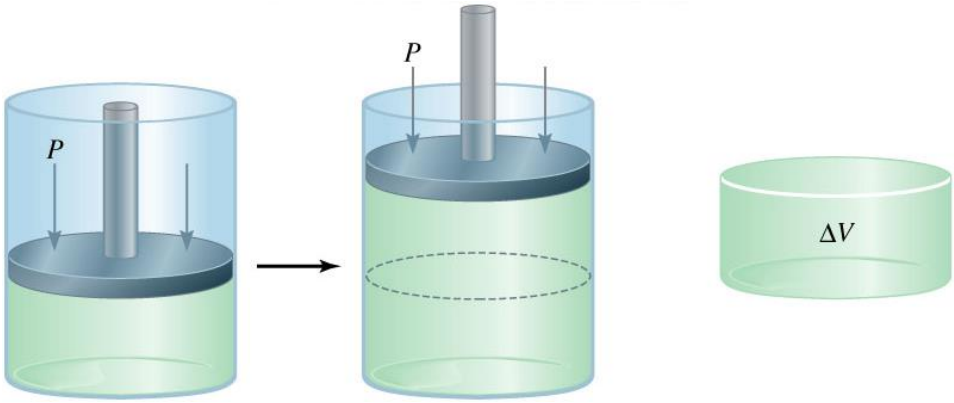
$\Delta E = q + w$

At constant pressure:

$q = \Delta H$ and $w = -P\Delta V$

$\Delta E = \Delta H - P\Delta V$

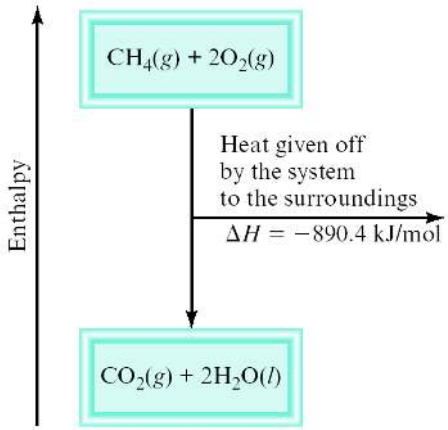
$\Delta H = \Delta E + P\Delta V$



Enthalpy (H) is used to quantify the heat flow into or out of a system in a process that occurs at constant pressure.

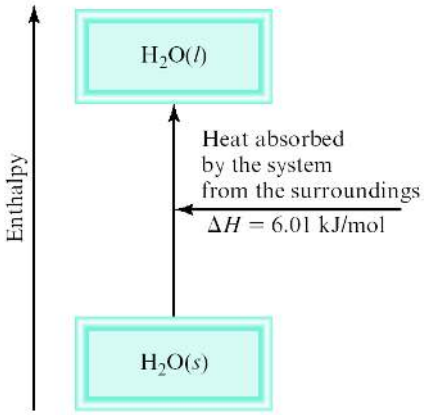
$$\Delta H = H(\text{products}) - H(\text{reactants})$$

ΔH = heat given off or absorbed during a reaction at constant pressure



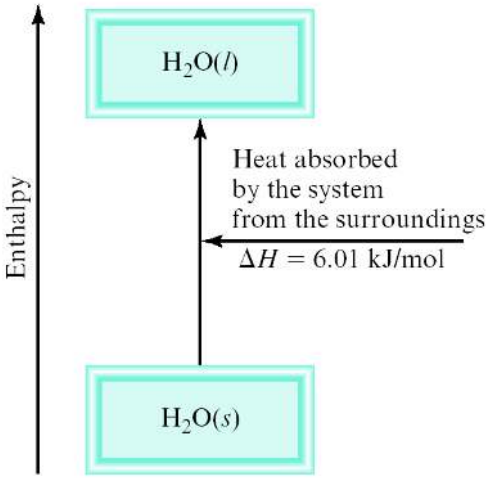
$H_{products} < H_{reactants}$

$\Delta H < 0$



$H_{products} > H_{reactants}$

$\Delta H > 0$



Is ΔH negative or positive?

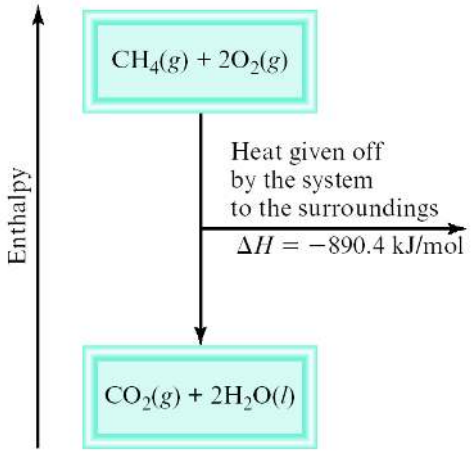
System absorbs heat

Endothermic

$\Delta H > 0$

6.01 kJ are absorbed for every 1 mole of ice that melts at 0^0C and 1 atm.

$$H_2O\ (s) \longrightarrow H_2O\ (l) \qquad \Delta H = 6.01 \text{ kJ/mol}$$



Is ΔH negative or positive?

System gives off heat

Exothermic

$\Delta H < 0$

890.4 kJ are released for every 1 mole of methane that is combusted at 25^0C and 1 atm.

$CH_4\ (g) + 2O_2\ (g) \longrightarrow CO_2\ (g) + 2H_2O\ (l)$ $\Delta H = -890.4$ kJ/mol

Thermochemical Equations

- The stoichiometric coefficients always refer to the number of moles of a substance

$H_2O\ (s) \longrightarrow H_2O\ (l) \qquad \Delta H = 6.01$ kJ/mol

- If you reverse a reaction, the sign of ΔH changes

$H_2O\ (l) \longrightarrow H_2O\ (s) \qquad \Delta H = -6.01$ kJ/mol

- If you multiply both sides of the equation by a factor *n*, then ΔH must change by the same factor *n*.

$2H_2O\ (s) \longrightarrow 2H_2O\ (l) \qquad \Delta H = 2 \times 6.01 = 12.0$ kJ

- The physical states of all reactants and products must be specified in thermochemical equations.

$$H_2O(s) \longrightarrow H_2O(l) \qquad \Delta H = 6.01 \text{ kJ/mol}$$

$$H_2O(l) \longrightarrow H_2O(g) \qquad \Delta H = 44.0 \text{ kJ/mol}$$

How much heat is evolved when 266 g of white phosphorus (P_4) burn in air?

$$P_4(s) + 5O_2(g) \longrightarrow P_4O_{10}(s) \qquad \Delta H = -3013 \text{ kJ/mol}$$

$$266 \text{ g } P_4 \times \frac{1 \text{ mol } P_4}{123.9 \text{ g } P_4} \times \frac{3013 \text{ kJ}}{1 \text{ mol } P_4} = 6470 \text{ kJ}$$

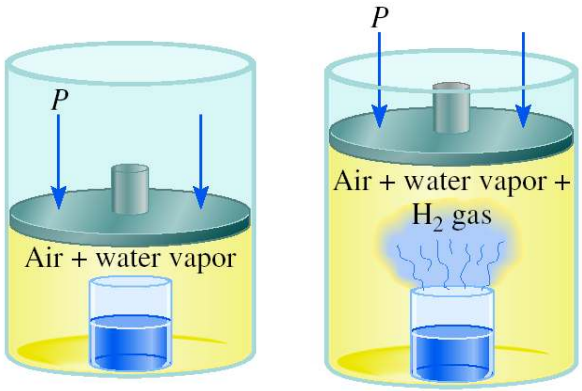
$2Na\ (s) + 2H_2O\ (l) \longrightarrow 2NaOH\ (aq) + H_2\ (g)\quad \Delta H = -367.5\ kJ/mol$

$\Delta E = \Delta H - P\Delta V$ At 25 °C, 1 mole H_2 = 24.5 L at 1 atm

$P\Delta V$ = 1 atm x 24.5 L = 2.5 kJ

ΔE = -367.5 kJ/mol – 2.5 kJ/mol = -370.0 kJ/mol





The *specific heat* (*s*) of a substance is the amount of heat (*q*) required to raise the temperature of one gram of the substance by one degree Celsius.

The *heat capacity* (*C*) of a substance is the amount of heat (*q*) required to raise the temperature of a given quantity (*m*) of the substance by one degree Celsius.

TABLE 6.2
The Specific Heats of Some Common Substances

Substance	Specific Heat (J/g · °C)
Al	0.900
Au	0.129
C (graphite)	0.720
C (diamond)	0.502
Cu	0.385
Fe	0.444
Hg	0.139
H_2O	4.184
C_2H_5OH (ethanol)	2.46

$$C = m \times s$$

Heat (*q*) absorbed or released:

$$q = m \times s \times \Delta t$$

$$q = C \times \Delta t$$

$$\Delta t = t_{final} - t_{initial}$$

Example:

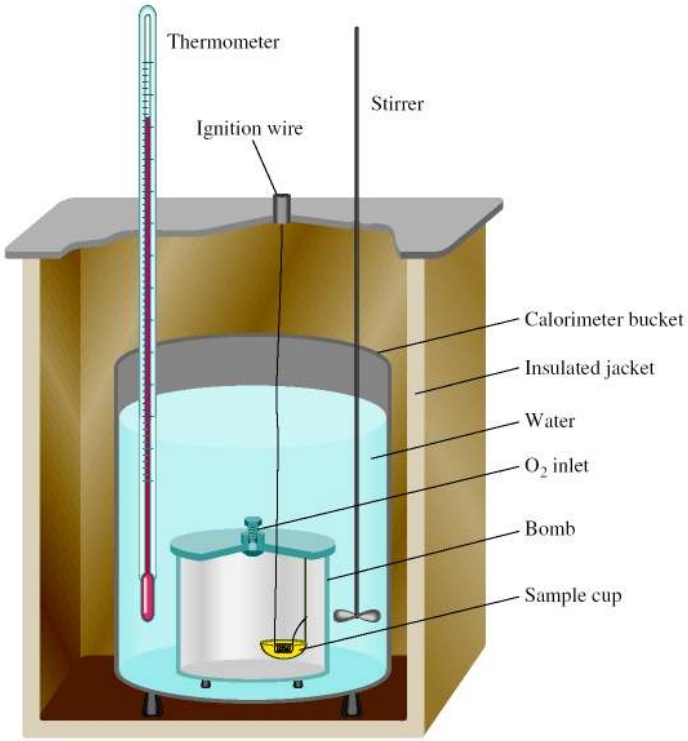
How much heat is given off when an 869 g iron bar cools from 94°C to 5°C?

s of Fe = 0.444 J/g • °C

$$\Delta t = t_{final} - t_{initial} = 5^{o}C - 94^{o}C = -89^{o}C$$

$$q = ms\Delta t = 869\ \cancel{g} \times 0.444\ J/\cancel{g} \cdot {}^{o}\cancel{C} \times -89^{o}\cancel{C} = -34{,}000\ J$$





No heat enters or leaves!

$q_{sys} = q_{water} + q_{bomb} + q_{rxn}$

$q_{sys} = 0$

$q_{rxn} = - (q_{water} + q_{bomb})$

$q_{water} = m \times s \times \Delta t$

$q_{bomb} = C_{bomb} \times \Delta t$

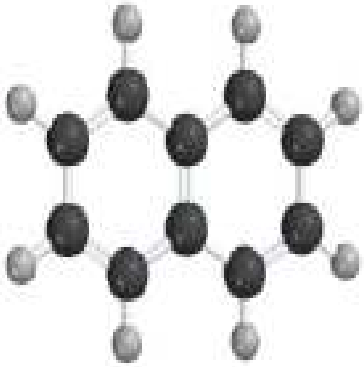
Reaction at Constant V

$\Delta H \neq q_{rxn}$

$\Delta H \sim q_{rxn}$

EXAMPLE 6.6

A quantity of 1.274 g of naphthalene ($C_{10}H_8$), a pungent-smelling substance used in moth repellants, was burned in a constant-volume bomb calorimeter. Consequently, the temperature of the water rose from 21.49°C to 26.52°C. If the heat capacity of the bomb plus water was 10.17 kJ/°C, calculate the heat of combustion of naphthalene on a molar basis; that is, find the molar heat of combustion.



$C_{10}H_8$

Strategy Knowing the heat capacity and the temperature rise, how do we calculate the heat absorbed by the calorimeter? What is the heat generated by the combustion of 1.274 g of naphthalene? What is the conversion factor between grams and moles of naphthalene?

(Continued)

Solution The heat absorbed by the bomb and water is equal to the product of the heat capacity and the temperature change. From Equation (6.16), assuming no heat is lost to the surroundings, we write

$$\begin{aligned} q_{cal} &= C_{cal}\Delta t \\ &= (10.17\ \text{kJ/°C})(26.52\text{°C} - 21.49\text{°C}) \\ &= 51.16\ \text{kJ} \end{aligned}$$

Because $q_{sys} = q_{cal} + q_{rxn} = 0$, $q_{cal} = -q_{rxn}$. The heat change of the reaction is −51.56 kJ. This is the heat released by the combustion of 1.274 g of $C_{10}H_8$; therefore, we can write the conversion factor as

$$\frac{-51.16\ \text{kJ}}{1.274\ \text{g}\ C_{10}H_8}$$

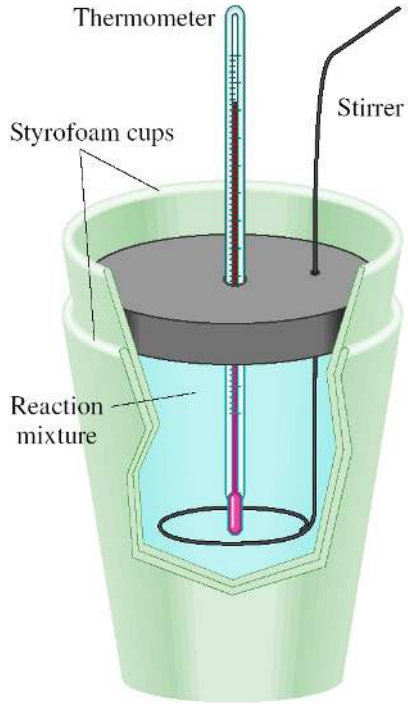
The molar mass of naphthalene is 128.2 g, so the heat of combustion of 1 mole of naphthalene is

$$\begin{aligned} \text{molar heat of combustion} &= \frac{-51.16\ \text{kJ}}{1.274\ \text{g}\ C_{10}H_8} \times \frac{128.2\ \text{g}\ C_{10}H_8}{1\ \text{mol}\ C_{10}H_8} \\ &= -5.148 \times 10^3\ \text{kJ/mol} \end{aligned}$$

Check Knowing that the combustion reaction is exothermic and that the molar mass of naphthalene is much greater than 1.2 g, is the answer reasonable? Under these conditions, can the heat change (−51.16 kJ) be equated to the enthalpy change of the reaction?

Practice Exercise A quantity of 1.922 g of methanol (CH_3OH) was burned in a constant-volume bomb calorimeter. Consequently, the temperature of the water rose by 4.20°C. If the heat capacity of the bomb plus water was 10.4 kJ/°C, calculate the molar heat of combustion of methanol.

Constant-Pressure Calorimetry



No heat enters or leaves!

$q_{sys} = q_{water} + q_{cal} + q_{rxn}$

$q_{sys} = 0$

$q_{rxn} = - (q_{water} + q_{cal})$

$q_{water} = m \, x \, s \times \Delta t$

$q_{cal} = C_{cal} \, x \, \Delta t$

Reaction at Constant P

$\Delta H = q_{rxn}$

TABLE 6.3 Heats of Some Typical Reactions Measured at Constant Pressure

Type of Reaction	Example	ΔH (kJ/mol)
Heat of neutralization	$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$	−56.2
Heat of ionization	$H_2O(l) \longrightarrow H^+(aq) + OH^-(aq)$	56.2
Heat of fusion	$H_2O(s) \longrightarrow H_2O(l)$	6.01
Heat of vaporization	$H_2O(l) \longrightarrow H_2O(g)$	44.0*
Heat of reaction	$MgCl_2(s) + 2Na(l) \longrightarrow 2NaCl(s) + Mg(s)$	−180.2

*Measured at 25°C. At 100°C, the value is 40.79 kJ.

Fuel Values of Foods and Other Substances

$$C_6H_{12}O_6\,(s) + 6O_2\,(g) \longrightarrow 6CO_2\,(g) + 6H_2O\,(l) \quad \Delta H = -2801\ kJ/mol$$

1 cal = 4.184 J

1 Cal = 1000 cal = 4184 J

Substance	$\Delta H_{combustion}$ (kJ/g)
Apple	-2
Beef	-8
Beer	-1.5
Gasoline	-34

Nutrition Facts
Serving Size 6 cookies (28g)
Servings Per Container about 11

Amount Per Serving

Calories 120	Calories from Fat 30
	% Daily Value*
Total Fat 4g	**6%**
Saturated Fat 0.5g	**4%**
Polyunsaturated Fat 0g	
Monounsaturated Fat 1g	
Cholesterol 5mg	**2%**
Sodium 105mg	**4%**
Total Carbohydrate 20g	**7%**
Dietary Fiber Less than 1gram	**2%**
Sugars 7g	
Protein 2g	

EXAMPLE 6.7

A lead (Pb) pellet having a mass of 26.47 g at 89.98°C was placed in a constant-pressure calorimeter of negligible heat capacity containing 100.0 mL of water. The water temperature rose from 22.50°C to 23.17°C. What is the specific heat of the lead pellet?

Strategy A sketch of the initial and final situation is as follows:



We know the masses of water and the lead pellet as well as the initial and final temperatures. Assuming no heat is lost to the surroundings, we can equate the heat lost by the lead pellet to the heat gained by the water. Knowing the specific heat of water, we can then calculate the specific heat of lead.

Solution Treating the calorimeter as an isolated system (no heat lost to the surroundings), we write

$$q_{Pb} + q_{H_2O} = 0$$

or

$$q_{Pb} = -q_{H_2O}$$

The heat gained by the water is given by

$$q_{H_2O} = msΔt$$

where m and s are the mass and specific heat and $Δt = t_{final} - t_{initial}$. Therefore,

$$q_{H_2O} = (100.0\ g)(4.184\ J/g \cdot °C)(23.17°C - 22.50°C)$$
$$= 280.3\ g$$

Because the heat lost by the lead pellet is equal to the heat gained by the water, so $q_{Pb} = -280.3$ J. Solving for the specific heat of Pb, we write

$$q_{Pb} = msΔt$$
$$-280.3\ J = (26.47\ g)(s)(23.17°C - 89.98°C)$$
$$s = 0.158\ J/g \cdot °C$$

Check The specific heat falls within the metals shown in Table 6.2.

Practice Exercise A 30.14-g stainless steel ball bearing at 117.82°C is placed in a constant-pressure calorimeter containing 120.0 mL of water at 18.44°C. If the specific heat of the ball bearing is 0.474 J/g · °C, calculate the final temperature of the water. Assume the calorimeter to have negligible heat capacity.

EXAMPLE 6.8

A quantity of 1.50×10^2 mL of 0.350 *M* HCl was mixed with 1.50×10^2 mL of 0.350 *M* NaOH in a constant-pressure calorimeter of negligible heat capacity. The initial temperature of the HCl and NaOH solutions was the same, 23.25°C, and the final temperature of the

(Continued)

mixed solution was 25.60°C. Calculate the heat change for the neutralization reaction on a molar basis

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

Assume that the densities and specific heats of the solutions are the same as for the water (1.00 g/mL and 4.184 J/g · °C, respectively).

Strategy Because the temperature rose, the neutralization reaction is exothermic. How do we calculate the heat absorbed by the combined solution? What is the heat of the reaction? What is the conversion factor for expressing the heat of reaction on a molar basis?

Solution Assuming no heat is lost to the surroundings, $q_{sys} = q_{soln} + q_{rxn} = 0$, so $q_{rxn} = -q_{soln}$, where q_{soln} is the heat absorbed by the combined solution. Because the density of the solution is 1.00 g/mL, the mass of a 150-mL solution is 150 g. Thus,

$$\begin{aligned} q_{soln} &= ms\Delta t \\ &= (1.50 \times 10^2 \text{ g} + 1.50 \times 10^2 \text{ g})(4.184 \text{ J/g} \cdot {}^\circ\text{C})(25.60{}^\circ\text{C} - 23.25{}^\circ\text{C}) \\ &= 2.95 \times 10^3 \text{ J} \\ &= 2.95 \text{ kJ} \end{aligned}$$

Because $q_{rxn} = -q_{soln}$, $q_{rxn} = -2.95$ kJ.

From the molarities given, the number of moles of both HCl and NaOH in 1.50×10^2 mL solution is

$$\frac{0.350 \text{ mol}}{1 \text{ L}} \times 0.150 \text{ L} = 0.0525 \text{ mol}$$

Therefore, the heat of neutralization when 1.00 mole of HCl reacts with 1.00 mole of NaOH is

$$\text{heat of neutralization} = \frac{-2.95 \text{ kJ}}{0.0525 \text{ mol}} = -56.2 \text{ kJ/mol}$$

Check Is the sign consistent with the nature of the reaction? Under the reaction condition, can the heat exchange be equated to the enthalpy change?

Practice Exercise A quantity of 4.00×10^2 mL of 0.600 *M* HNO_3 is mixed with 4.00×10^2 mL of 0.300 *M* $Ba(OH)_2$ in a constant-pressure calorimeter of negligible heat capacity. The initial temperature of both solutions is the same at 18.46°C. What is the final temperature of the solution? (Use the result in Example 6.8 for your calculation.)

Because there is no way to measure the absolute value of the enthalpy of a substance, must I measure the enthalpy change for every reaction of interest?

Establish an arbitrary scale with the standard enthalpy of formation (ΔH^0_f) as a reference point for all enthalpy expressions.

Standard enthalpy of formation (ΔH^0)$_f$ is the heat change that results when one mole of a compound is formed from its elements at a pressure of 1 atm.

The standard enthalpy of formation of any element in its most stable form is zero.

ΔH^0_f (O_2) = 0

ΔH^0_f (O_3) = 142 kJ/mol

ΔH^0_f (C, graphite) = 0

ΔH^0_f (C, diamond) = 1.90 kJ/mol

TABLE 6.4 Standard Enthalpies of Formation of Some Inorganic Substances at 25°C

Substance	ΔH_f° (kJ/mol)	Substance	ΔH_f° (kJ/mol)
$Ag(s)$	0	$H_2O_2(l)$	−187.6
$AgCl(s)$	−127.0	$Hg(l)$	0
$Al(s)$	0	$I_2(s)$	0
$Al_2O_3(s)$	−1669.8	$HI(g)$	25.9
$Br_2(l)$	0	$Mg(s)$	0
$HBr(g)$	−36.2	$MgO(s)$	−601.8
C(graphite)	0	$MgCO_3(s)$	−1112.9
C(diamond)	1.90	$N_2(g)$	0
$CO(g)$	−110.5	$NH_3(g)$	−46.3
$CO_2(g)$	−393.5	$NO(g)$	90.4
$Ca(s)$	0	$NO_2(g)$	33.85
$CaO(s)$	−635.6	$N_2O(g)$	81.56
$CaCO_3(s)$	−1206.9	$N_2O_4(g)$	9.66
$Cl_2(g)$	0	$O(g)$	249.4
$HCl(g)$	−92.3	$O_2(g)$	0
$Cu(s)$	0	$O_3(g)$	142.2
$CuO(s)$	−155.2	S(rhombic)	0
$F_2(g)$	0	S(monoclinic)	0.30
$HF(g)$	−271.6	$SO_2(g)$	−296.1
$H(g)$	218.2	$SO_3(g)$	−395.2
$H_2(g)$	0	$H_2S(g)$	−20.15
$H_2O(g)$	−241.8	$Zn(s)$	0
$H_2O(l)$	−285.8	$ZnO(s)$	−348.0

The standard enthalpy of reaction (ΔH^0_{rxn}) is the enthalpy of a reaction carried out at 1 atm.

$$aA + bB \longrightarrow cC + dD$$

$$\Delta H^0_{rxn} = [c\Delta H^0_f(C) + d\Delta H^0_f(D)] - [a\Delta H^0_f(A) + b\Delta H^0_f(B)]$$

$$\Delta H^0_{rxn} = \Sigma n\Delta H^0_f(\text{products}) - \Sigma m\Delta H^0_f(\text{reactants})$$

$$\Delta H^\circ_{rxn} = \Sigma n \Delta H^\circ_f(\text{products}) - \Sigma m \Delta H^\circ_f(\text{reactants}) \qquad (6.18)$$

where m and n denote the stoichiometric coefficients for the reactants and products, and Σ (sigma) means "the sum of." Note that in calculations, the stoichiometric coefficients are just numbers without units.

To use Equation (6.18) to calculate ΔH°_{rxn} we must know the ΔH°_f values of the compounds that take part in the reaction. To determine these values we can apply the direct method or the indirect method.

The Direct Method

This method of measuring ΔH°_f works for compounds that can be readily synthesized from their elements. Suppose we want to know the enthalpy of formation of carbon dioxide. We must measure the enthalpy of the reaction when carbon (graphite) and molecular oxygen in their standard states are converted to carbon dioxide in its standard state:

$$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g) \quad \Delta H^\circ_{rxn} = -393.5 \text{ kJ/mol}$$

We know from experience that this combustion easily goes to completion. Thus, from Equation (6.17) we can write

$$\begin{aligned}\Delta H^\circ_{rxn} &= \Delta H^\circ_f(CO_2, g) - [\Delta H^\circ_f(C, \text{graphite}) + \Delta H^\circ_f(O_2, g)] \\ &= -393.5 \text{ kJ/mol}\end{aligned}$$

Because both graphite and O_2 are stable allotropic forms of the elements, it follows that $\Delta H^\circ_f(C, \text{graphite})$ and $\Delta H^\circ_f(O_2, g)$ are zero. Therefore,

$$\Delta H^\circ_{rxn} = \Delta H^\circ_f(CO_2, g) = -393.5 \text{ kJ/mol}$$

or

$$\Delta H^\circ_f(CO_2, g) = -393.5 \text{ kJ/mol}$$

Benzene (C_6H_6) burns in air to produce carbon dioxide and liquid water. How much heat is released per mole of benzene combusted? The standard enthalpy of formation of benzene is 49.04 kJ/mol.

$$2C_6H_6\,(l) + 15O_2\,(g) \longrightarrow 12CO_2\,(g) + 6H_2O\,(l)$$

$$\Delta H^0_{rxn} = \Sigma n\Delta H^0_f\,(\text{products}) - \Sigma m\Delta H^0_f\,(\text{reactants})$$

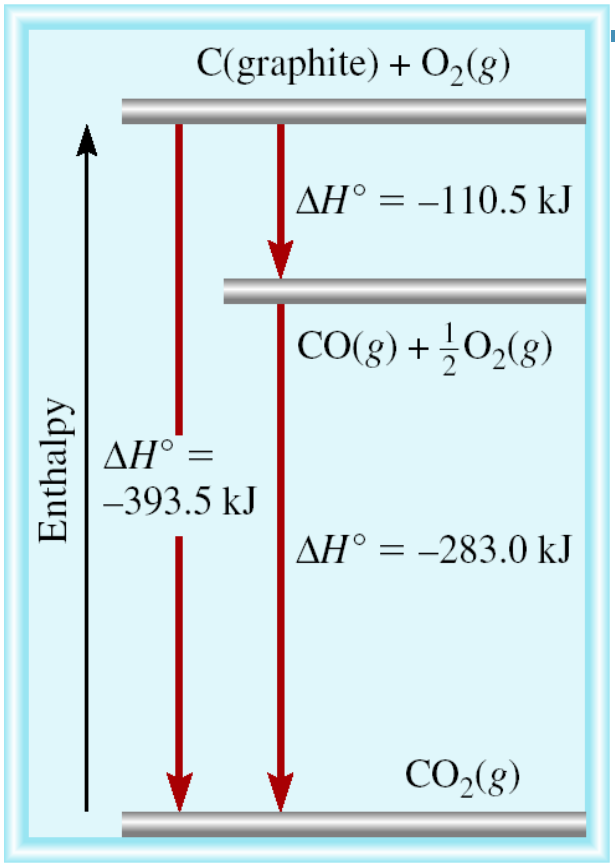
$$\Delta H^0_{rxn} = [12\Delta H^0_f\,(CO_2) + 6\Delta H^0_f\,(H_2O)] - [2\Delta H^0_f\,(C_6H_6)]$$

$$\Delta H^0_{rxn} = [12x{-}393.5 + 6x{-}187.6] - [2x49.04] = -5946\ \text{kJ}$$

$$\frac{-5946\ \text{kJ}}{2\ \text{mol}} = -2973\ \text{kJ/mol}\ C_6H_6$$

Hess's Law: When reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps.

(Enthalpy is a state function. It doesn't matter how you get there, only where you start and end.)



$$C\ (\text{graphite}) + 1/2O_2\ (g) \longrightarrow CO\ (g)$$

$$CO\ (g) + 1/2O_2\ (g) \longrightarrow CO_2\ (g)$$

$$C\ (\text{graphite}) + O_2\ (g) \longrightarrow CO_2\ (g)$$



Figure 6.9
The enthalpy change for the formation of 1 mole of CO_2 from graphite and O_2 can be broken into two steps according to Hess's law.



C_2H_2



An oxyacetylene torch has a high flame temperature (3000°C) and is used to weld metals.

Similar problems: 6.62, 6.63.

EXAMPLE 6.9

Calculate the standard enthalpy of formation of acetylene (C_2H_2) from its elements:

$$2C(\text{graphite}) + H_2(g) \longrightarrow C_2H_2(g)$$

The equations for each step and the corresponding enthalpy changes are

(a) $C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g)$ $\quad \Delta H^\circ_{rxn} = -393.5$ kJ/mol

(b) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$ $\quad \Delta H^\circ_{rxn} = -285.8$ kJ/mol

(c) $2C_2H_2(g) + 5O_2(g) \longrightarrow 4CO_2(g) + 2H_2O(l)$ $\quad \Delta H^\circ_{rxn} = -2598.8$ kJ/mol

Strategy Our goal here is to calculate the enthalpy change for the formation of C_2H_2 from its elements C and H_2. The reaction does not occur directly, however, so we must use an indirect route using the information given by Equations (a), (b), and (c).

Solution Looking at the synthesis of C_2H_2, we need 2 moles of graphite as reactant. So we multiply Equation (a) by 2 to get

(d) $2C(\text{graphite}) + 2O_2(g) \longrightarrow 2CO_2(g)$ $\quad \Delta H^\circ_{rxn} = 2(-393.5 \text{ kJ/mol}) = -787.0$ kJ/mol

Next, we need 1 mole of H_2 as a reactant and this is provided by Equation (b). Last, we need 1 mole of C_2H_2 as a product. Equation (c) has 2 moles of C_2H_2 as a reactant so we need to reverse the equation and divide it by 2:

(e) $2CO_2(g) + H_2O(l) \longrightarrow C_2H_2(g) + \frac{5}{2}O_2(g)$ $\quad \Delta H^\circ_{rxn} = \frac{1}{2}(2598.8 \text{ kJ/mol}) = 1299.4$ kJ/mol

Adding Equations (d), (b), and (e) together, we get

$$2C(\text{graphite}) + 2O_2(g) \longrightarrow 2CO_2(g) \qquad \Delta H^\circ_{rxn} = -787.0 \text{ kJ/mol}$$
$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O(l) \qquad \Delta H^\circ_{rxn} = -285.8 \text{ kJ/mol}$$
$$2CO_2(g) + H_2O(l) \longrightarrow C_2H_2(g) + \tfrac{5}{2}O_2(g) \qquad \Delta H^\circ_{rxn} = 1299.4 \text{ kJ/mol}$$
$$2C(\text{graphite}) + H_2(g) \longrightarrow C_2H_2(g) \qquad \Delta H^\circ_{rxn} = 226.6 \text{ kJ/mol}$$

Therefore, $\Delta H^\circ_f = \Delta H^\circ_{rxn} =$ 226.6 kJ/mol. The ΔH°_f value means that when 1 mole of C_2H_2 is synthesized from 2 moles of C (graphite) and 1 mole of H_2, 226.6 kJ of heat are absorbed by the reacting system from the surroundings. Thus, this is an endothermic process.

Practice Exercise Calculate the standard enthalpy of formation of carbon disulfide (CS_2) from its elements, given that

$$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H^\circ_{rxn} = -393.5 \text{ kJ/mol}$$
$$S(\text{rhombic}) + O_2(g) \longrightarrow SO_2(g) \qquad \Delta H^\circ_{rxn} = -296.4 \text{ kJ/mol}$$
$$CS_2(l) + 3O_2(g) \longrightarrow CO_2(g) + 2SO_2(g) \qquad \Delta H^\circ_{rxn} = -1073.6 \text{ kJ/mol}$$

Example:

Calculate the standard enthalpy of formation of CS_2 (l) given that:

$C(graphite) + O_2(g) \longrightarrow CO_2(g)$ $\quad \Delta H^0_{rxn} = -393.5$ kJ/mol

$S(rhombic) + O_2(g) \longrightarrow SO_2(g)$ $\quad \Delta H^0_{rxn} = -296.1$ kJ/mol

$CS_2(l) + 3O_2(g) \longrightarrow CO_2(g) + 2SO_2(g)$ $\quad \Delta H^0_{rxn} = -1072$ kJ/mol

1. Write the enthalpy of formation reaction for CS_2

$$C(graphite) + 2S(rhombic) \longrightarrow CS_2(l)$$

2. Add the given rxns so that the result is the desired rxn.

$C(graphite) + O_2(g) \longrightarrow CO_2(g)$ $\quad \Delta H^0_{rxn} = -393.5$ kJ/mol

$2S(rhombic) + 2O_2(g) \longrightarrow 2SO_2(g)$ $\quad \Delta H^0_{rxn} = -296.1$ kJ/mol x 2

$+ \ CO_2(g) + 2SO_2(g) \longrightarrow CS_2(l) + 3O_2(g)$ $\quad \Delta H^0_{rxn} = +1072$ kJ/mol

$C(graphite) + 2S(rhombic) \longrightarrow CS_2(l)$

$\Delta H^0_{rxn} = -393.5 + (2x-296.1) + 1072 = 86.3$ kJ/mol

Chemistry in Action: Bombardier Beetle Defense

$C_6H_4(OH)_2\ (aq) + H_2O_2\ (aq) \longrightarrow C_6H_4O_2\ (aq) + 2H_2O\ (l)\ \Delta H^0 = ?$

$C_6H_4(OH)_2\ (aq) \longrightarrow C_6H_4O_2\ (aq) + H_2\ (g)\ \Delta H^0 = 177$ kJ/mol

$H_2O_2\ (aq) \longrightarrow H_2O\ (l) + \frac{1}{2}O_2\ (g)\ \Delta H^0 = -94.6$ kJ/mol

$H_2\ (g) + \frac{1}{2}\ O_2\ (g) \longrightarrow H_2O\ (l)\ \Delta H^0 = -286$ kJ/mol

$\Delta H^0 = 177 - 94.6 - 286 = -204$ kJ/mol

Exothermic!



The *enthalpy of solution* (ΔH_{soln}) is the heat generated or absorbed when a certain amount of solute dissolves in a certain amount of solvent.

$$\Delta H_{soln} = H_{soln} - H_{components}$$

TABLE 6.5

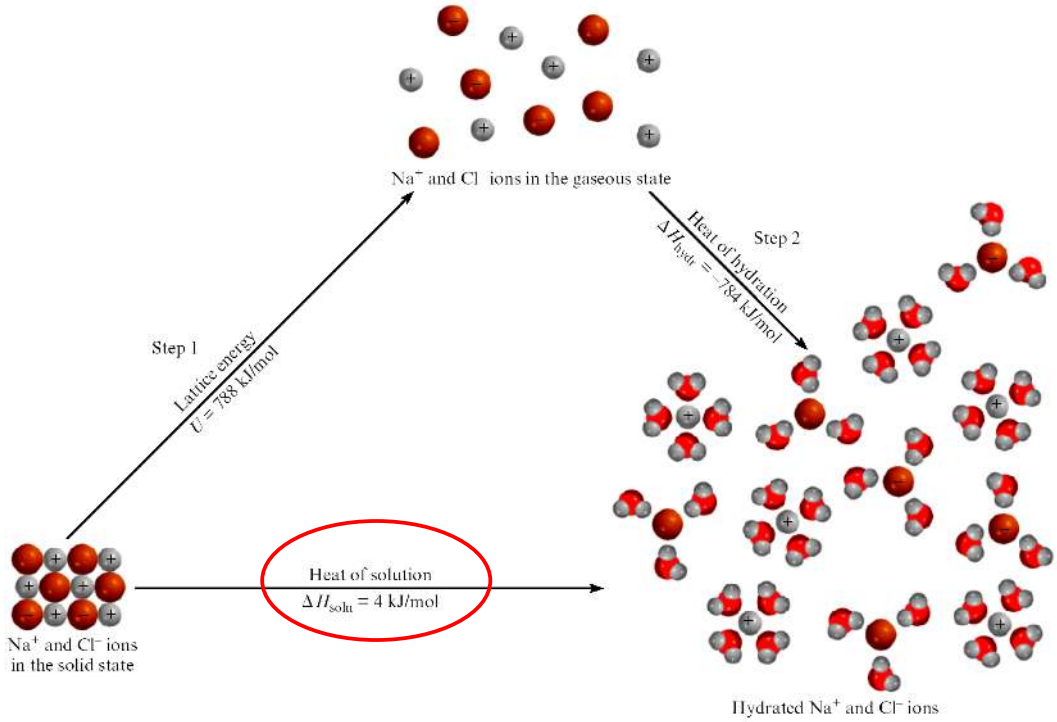
Heats of Solution of Some Ionic Compounds

Compound	ΔH_{soln} (kJ/mol)
LiCl	−37.1
$CaCl_2$	−82.8
NaCl	4.0
KCl	17.2
NH_4Cl	15.2
NH_4NO_3	26.2

Which substance(s) could be used for melting ice?

Which substance(s) could be used for a cold pack?

The Solution Process for NaCl



ΔH_{soln} = Step 1 + Step 2 = 788 – 784 = 4 kJ/mol

EXAMPLE 6.2

The work done when a gas is compressed in a cylinder like that shown in Figure 6.4 is 387 J. During this process, there is a heat transfer of 152 J from the gas to the surroundings. Calculate the energy change for this process.

(Continued)

Strategy Compression is work done on the gas, so what is the sign for w? Heat is released by the gas to the surroundings. Is this an endothermic or exothermic process? What is the sign for q?

Solution To calculate the energy change of the gas, we need Equation (6.1). Work of compression is positive and because heat is released by the gas, q is negative. Therefore, we have

$$\begin{aligned}\Delta U &= q + w \\ &= -152\ \text{J} + 387\ \text{J} \\ &= 235\ \text{J}\end{aligned}$$

As a result, the energy of the gas increases by 235 J.

Practice Exercise A gas expands and does P-V work on the surroundings equal to 279 J. At the same time, it absorbs 216 J of heat from the surroundings. What is the change in energy of the system?

EXAMPLE 6.3

Given the thermochemical equation

$$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g) \qquad \Delta H = -198.2 \text{ kJ/mol}$$

calculate the heat evolved when 93.4 g of SO_2 (molar mass = 64.07 g/mol) is converted to SO_3.

Strategy The thermochemical equation shows that for every 2 moles of SO_2 reacted, 198.2 kJ of heat are given off (note the negative sign). Therefore, the conversion factor is

$$\frac{-198.2 \text{ kJ}}{2 \text{ mol } SO_2}$$

How many moles of SO_2 are in 93.4 g of SO_2? What is the conversion factor between grams and moles?

Solution We need to first calculate the number of moles of SO_2 in 93.4 g of the compound and then find the number of kilojoules produced from the exothermic reaction. The sequence of conversions is as follows:

$$\text{grams of } SO_2 \longrightarrow \text{moles of } SO_2 \longrightarrow \text{kilojoules of heat generated}$$

Therefore, the enthalpy change for this reaction is given by

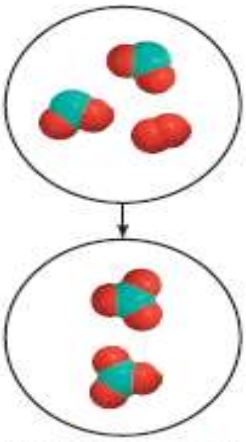
$$\Delta H = 93.4 \text{ g } \cancel{SO_2} \times \frac{1 \text{ mol } \cancel{SO_2}}{64.07 \text{ g } \cancel{SO_2}} \times \frac{-198.2 \text{ kJ}}{2 \text{ mol } \cancel{SO_2}} = -144 \text{ kJ}$$

and the heat released to the surroundings is 144 kJ.

Check Because 93.4 g is less than twice the molar mass of SO_2 (2×64.07 g) as shown in the preceding thermochemical equation, we expect the heat released to be smaller than 198.2 kJ.

Practice Exercise Calculate the heat evolved when 266 g of white phosphorus (P_4) burns in air according to the equation

$$P_4(s) + 5O_2(g) \longrightarrow P_4O_{10}(s) \qquad \Delta H = -3013 \text{ kJ/mol}$$



$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Similar problem: 6.26.

For reactions that do not result in a change in the number of moles of gases from reactants to products [for example, $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$], $\Delta U = \Delta H$.

This calculation shows that ΔU and ΔH are approximately the same. The reason ΔH is smaller than ΔU in magnitude is that some of the internal energy released is used to do gas expansion work, so less heat is evolved. For reactions that do not involve gases, ΔV is usually very small and so ΔU is practically the same as ΔH.

Another way to calculate the internal energy change of a gaseous reaction is to assume ideal gas behavior and constant temperature. In this case,

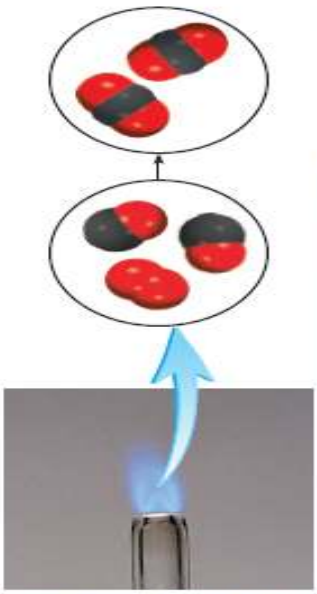
$$\begin{aligned}\Delta U &= \Delta H - \Delta(PV)\\ &= \Delta H - \Delta(nRT)\end{aligned}$$

or

$$\Delta U = \Delta H - RT\Delta n \qquad (6.10)$$

where Δn is defined as

$$\Delta n = \text{number of moles of product gases} - \text{number of moles of reactant gases}$$



Carbon monoxide burns in air to form carbon dioxide.

EXAMPLE 6.4

Calculate the change in internal energy when 2 moles of CO are converted to 2 moles of CO_2 at 1 atm and 25°C:

$$2CO(g) + O_2(g) \longrightarrow 2CO_2(g) \qquad \Delta H = -566.0 \text{ kJ/mol}$$

Strategy We are given the enthalpy change, ΔH, for the reaction and are asked to calculate the change in internal energy, ΔU. Therefore, we need Equation (6.10). What is the change in the number of moles of gases? ΔH is given in kilojoules, so what units should we use for R?

Solution From the chemical equation we see that 3 moles of gases are converted to 2 moles of gases so that

$$\begin{aligned}\Delta n &= \text{number of moles of product gases} - \text{number of moles of reactant gases}\\ &= 2 - 3\\ &= -1\end{aligned}$$

Using 8.314 J/K · mol for R and $T = 298$ K in Equation (6.10), we write

$$\begin{aligned}\Delta U &= \Delta H - RT\Delta n\\ &= -566.0 \text{ kJ/mol} - (8.314 \text{ J/K} \cdot \text{mol})\left(\frac{1 \text{ kJ}}{1000 \text{ J}}\right)(298 \text{ K})(-1)\\ &= -563.5 \text{ kJ/mol}\end{aligned}$$

(Continued)

EXAMPLE 6.10

The thermite reaction involves aluminum and iron(III) oxide

$$2Al(s) + Fe_2O_3(s) \longrightarrow Al_2O_3(s) + 2Fe(l)$$

This reaction is highly exothermic and the liquid iron formed is used to weld metals. Calculate the heat released in kilojoules per gram of Al reacted with Fe_2O_3. The ΔH_f° for Fe(l) is 12.40 kJ/mol.

Strategy The enthalpy of a reaction is the difference between the sum of the enthalpies of the products and the sum of the enthalpies of the reactants. The enthalpy of each species (reactant or product) is given by the product of the stoichiometric coefficient and the standard enthalpy of formation of the species.

Solution Using the given ΔH_f° value for Fe(l) and other ΔH_f° values in Appendix 2 and Equation (6.18), we write

$$\begin{aligned}\Delta H_{rxn}^\circ &= [\Delta H_f^\circ(Al_2O_3) + 2\Delta H_f^\circ(Fe)] - [2\Delta H_f^\circ(Al) + 2\Delta H_f^\circ(Fe_2O_3)] \\ &= [(-1669.8\ kJ/mol) + 2(12.40\ kJ/mol)] - [2(0) + (-822.2\ kJ/mol)] \\ &= -822.8\ kJ/mol\end{aligned}$$

This is the amount of heat released for two moles of Al reacted. We use the following ratio

$$\frac{-822.8\ kJ}{2\ mol\ Al}$$

to convert to kJ/g Al. The molar mass of Al is 26.98 g, so

$$\text{heat released per gram of Al} = \frac{-822.8\ kJ}{2\ \cancel{mol\ Al}} \times \frac{1\ \cancel{mol\ Al}}{26.98\ g\ Al}$$

$$= -15.25\ kJ/g$$

Check Is the negative sign consistent with the exothermic nature of the reaction? As a quick check, we see that 2 moles of Al weigh about 54 g and give off about 823 kJ of heat when reacted with Fe_2O_3. Therefore, the heat given off per gram of Al reacted is approximately −830 kJ/54 g or −15.4 kJ/g.

Practice Exercise Benzene (C_6H_6) burns in air to produce carbon dioxide and liquid water. Calculate the heat released (in kilojoules) per gram of the compound reacted with oxygen. The standard enthalpy of formation of benzene is 49.04 kJ/mol.



The molten iron formed in a thermite reaction is run down into a mold between the ends of two railroad rails. On cooling, the rails are welded together.

Similar problems: 6.54, 6.57.

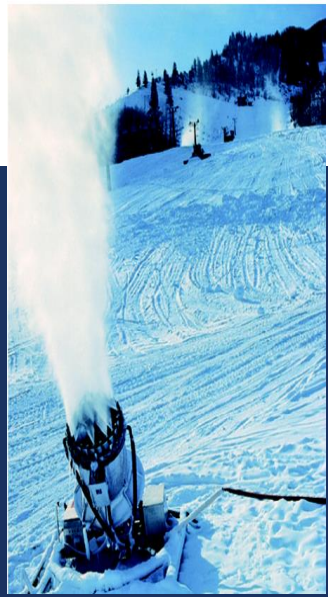
GENERAL CHEMISTRY I
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CHAPTER 6
ENERGY RELATIONSHIPS IN CHEMICAL REACTIONS

LECTURE 26

PROBLEM SOLVE



6.14 Consider these changes.

(a) $Hg(l) \longrightarrow Hg(g)$

(b) $3O_2(g) \longrightarrow 2O_3(g)$

(c) $CuSO_4 \cdot 5H_2O(s) \longrightarrow CuSO_4(s) + 5H_2O(g)$

(d) $H_2(g) + F_2(g) \longrightarrow 2HF(g)$

At constant pressure, in which of the reactions is work done by the system on the surroundings? By the surroundings on the system? In which of them is no work done?

6.15 A sample of nitrogen gas expands in volume from 1.6 L to 5.4 L at constant temperature. Calculate the work done in joules if the gas expands (a) against a vacuum, (b) against a constant pressure of 0.80 atm, and (c) against a constant pressure of 3.7 atm.

6.17 A gas expands and does P-V work on the surroundings equal to 325 J. At the same time, it absorbs 127 J of heat from the surroundings. Calculate the change in energy of the gas.

6.26 Determine the amount of heat (in kJ) given off when 1.26×10^4 g of NO_2 are produced according to the equation

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$$
$$\Delta H = -114.6 \text{ kJ/mol}$$

6.24 Consider this reaction:

$$2CH_3OH(l) + 3O_2(g) \longrightarrow 4H_2O(l) + 2CO_2(g)$$
$$\Delta H = -1452.8 \text{ kJ/mol}$$

What is the value of ΔH if (a) the equation is multiplied throughout by 2, (b) the direction of the reaction is reversed so that the products become the reactants and vice versa, (c) water vapor instead of liquid water is formed as the product?

6.28 Consider the reaction

$$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$$
$$\Delta H = -184.6 \text{ kJ/mol}$$

If 3 moles of H_2 react with 3 moles of Cl_2 to form HCl, calculate the work done against a pressure of 1.0 atm at 25°C. What is ΔU for this reaction? Assume the reaction goes to completion.

6.33 A piece of silver of mass 362 g has a heat capacity of 85.7 J/°C. What is the specific heat of silver?

6.36 A sheet of gold weighing 10.0 g and at a temperature of 18.0°C is placed flat on a sheet of iron weighing 20.0 g and at a temperature of 55.6°C. What is the final temperature of the combined metals? Assume that no heat is lost to the surroundings. (*Hint:* The heat gained by the gold must be equal to the heat lost by the iron. The specific heats of the metals are given in Table 6.2.)

6.37 A 0.1375-g sample of solid magnesium is burned in a constant-volume bomb calorimeter that has a heat capacity of 3024 J/°C. The temperature increases by 1.126°C. Calculate the heat given off by the burning Mg, in kJ/g and in kJ/mol.

6.38 A quantity of 2.00×10^2 mL of 0.862 *M* HCl is mixed with 2.00×10^2 mL of 0.431 *M* $Ba(OH)_2$ in a constant-pressure calorimeter of negligible heat capacity. The initial temperature of the HCl and $Ba(OH)_2$ solutions is the same at 20.48°C. For the process

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

the heat of neutralization is −56.2 kJ/mol. What is the final temperature of the mixed solution?

6.51 Calculate the heat of decomposition for this process at constant pressure and 25°C:

$$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$$

(Look up the standard enthalpy of formation of the reactant and products in Table 6.4.)

6.55 Methanol, ethanol, and *n*-propanol are three common alcohols. When 1.00 g of each of these alcohols is burned in air, heat is liberated as shown by the following data: (a) methanol (CH_3OH), −22.6 kJ; (b) ethanol (C_2H_5OH), −29.7 kJ; (c) *n*-propanol (C_3H_7OH), −33.4 kJ. Calculate the heats of combustion of these alcohols in kJ/mol.

6.61 From these data,

$$S(\text{rhombic}) + O_2(g) \longrightarrow SO_2(g) \qquad \Delta H^\circ_{rxn} = -296.06 \text{ kJ/mol}$$

$$S(\text{monoclinic}) + O_2(g) \longrightarrow SO_2(g) \qquad \Delta H^\circ_{rxn} = -296.36 \text{ kJ/mol}$$

calculate the enthalpy change for the transformation

$$S(\text{rhombic}) \longrightarrow S(\text{monoclinic})$$

(Monoclinic and rhombic are different allotropic forms of elemental sulfur.)

6.64 Calculate the standard enthalpy change for the reaction

$$2Al(s) + Fe_2O_3(s) \longrightarrow 2Fe(s) + Al_2O_3(s)$$

given that

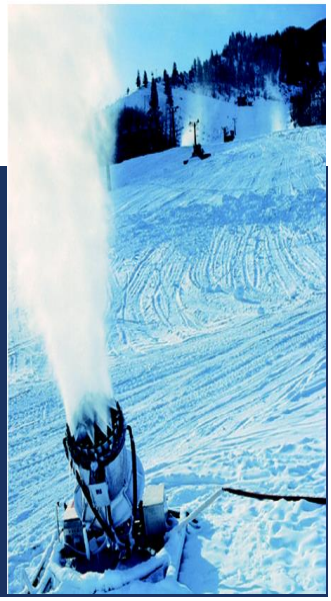
$$2Al(s) + \tfrac{3}{2}O_2(g) \longrightarrow Al_2O_3(s) \qquad \Delta H^\circ_{rxn} = -1669.8 \text{ kJ/mol}$$

$$2Fe(s) + \tfrac{3}{2}O_2(g) \longrightarrow Fe_2O_3(s) \qquad \Delta H^\circ_{rxn} = -822.2 \text{ kJ/mol}$$

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CHAPTER 18
THERMODYNAMICS
LECTURE 31-38



Thermodynamics

- Branch of physics that has to do with heat and temperature and their relation to energy and work.

- The behavior of these quantities is governed by the three laws of thermodynamics, irrespective of the composition or specific properties of the material or system in question.

First Law of Thermodynamics

- Energy cannot be created nor destroyed.
- Therefore, the total energy of the universe is a constant.
- Energy can, however, be converted from one form to another or transferred from a system to the surroundings or vice versa.
  - $\Delta E = q + w$

    $E_{\text{system}} = (\text{KE})_{\text{system}} + (\text{PE})_{\text{system}}$

First Law of Thermodynamics

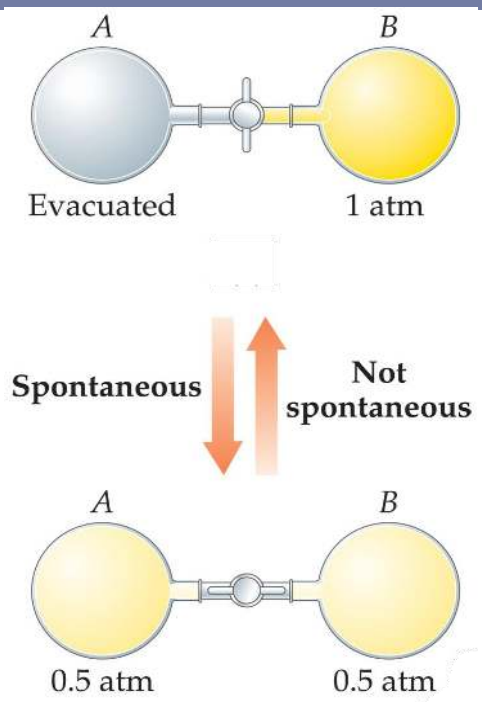
- Work in Chemical Systems :
  - Pressure-volume or $P\Delta V$
    - $w = -P\Delta V$ ➔ Where P = external pressure
    - If $P\Delta V$ only work in chemical system, then

$$\Delta E = q + (-P\Delta V) = q - P\Delta V$$

- Reaction done at constant V
  - $\Delta V = 0$ ➔ $P\Delta V = 0$, so ➔ $\Delta E = q_V$
  - Entire energy change due to heat absorbed or lost.

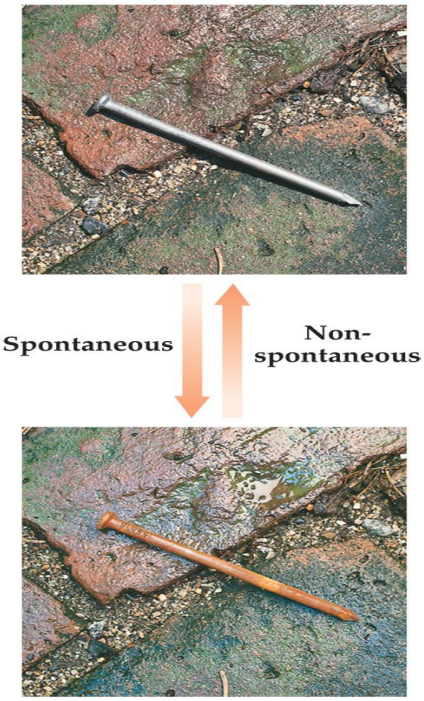
Spontaneous Processes

- Can proceed without any outside intervention.
- The gas in vessel *B* will spontaneously effuse into vessel *A*, but once the gas is in both vessels, it will *not* spontaneously.



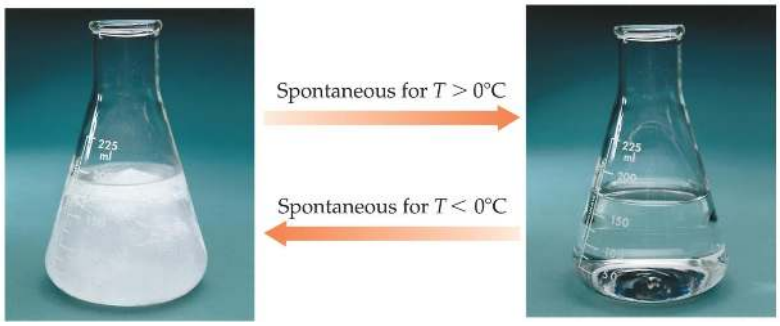
Spontaneous Processes

- Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.



Spontaneous Processes

- Processes that are spontaneous at one temperature may be nonspontaneous at other temperatures.
- Above 0°C it is spontaneous for ice to melt.
- Below 0°C the reverse process is spontaneous.



Example which of the following process is spontaneous or non spontaneous

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \quad \Delta H^\circ = -890.4\ kJ/mol$

Spontaneous highly exo

$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l) \quad \Delta H^\circ = -56.2\ kJ/mol$

Spontaneous

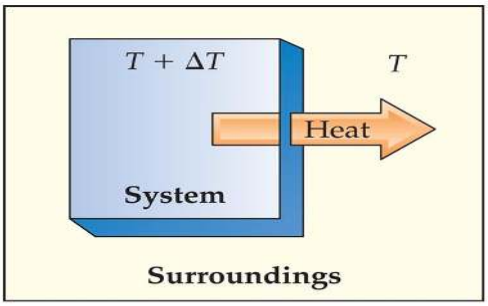
$H_2O(s) \longrightarrow H_2O(l) \quad \Delta H^\circ = 6.01\ kJ/mol$

Phase change spont

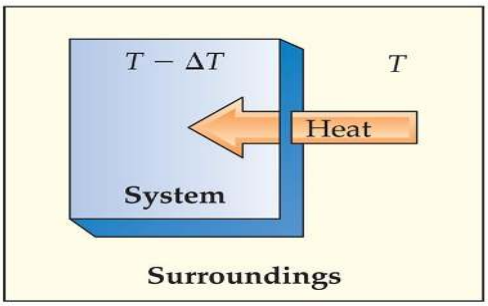
$NH_4NO_3(s) \xrightarrow{H_2O} NH_4^+(aq) + NO_3^-(aq) \quad \Delta H^\circ = 25\ kJ/mol$

$2HgO(s) \longrightarrow 2Hg(l) + O_2(g) \quad \Delta H^\circ = 90.7\ kJ/mol$

Reversible Processes

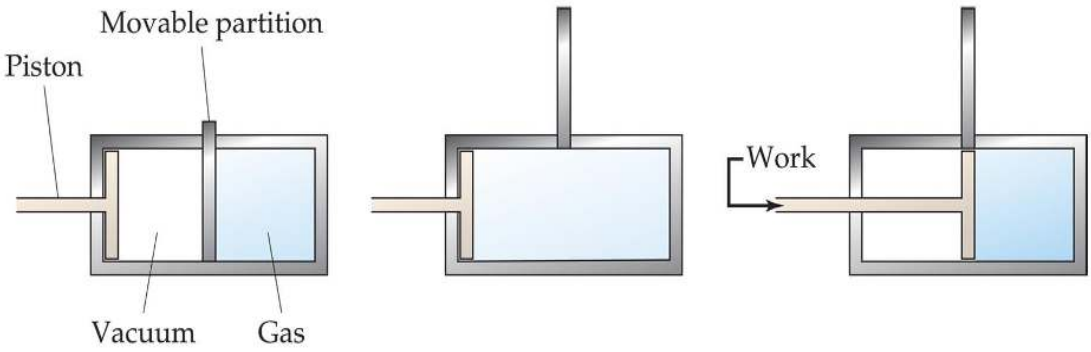


In a reversible process the system changes in such a way that the system and surroundings can be put back in their original states by exactly reversing the process.



Changes are infinitesimally **small** in a reversible process.

Irreversible Processes



- Irreversible processes cannot be undone by exactly reversing the change to the system.
- All **Spontaneous** processes are **irreversible**.
- All **Real** processes are **irreversible**.

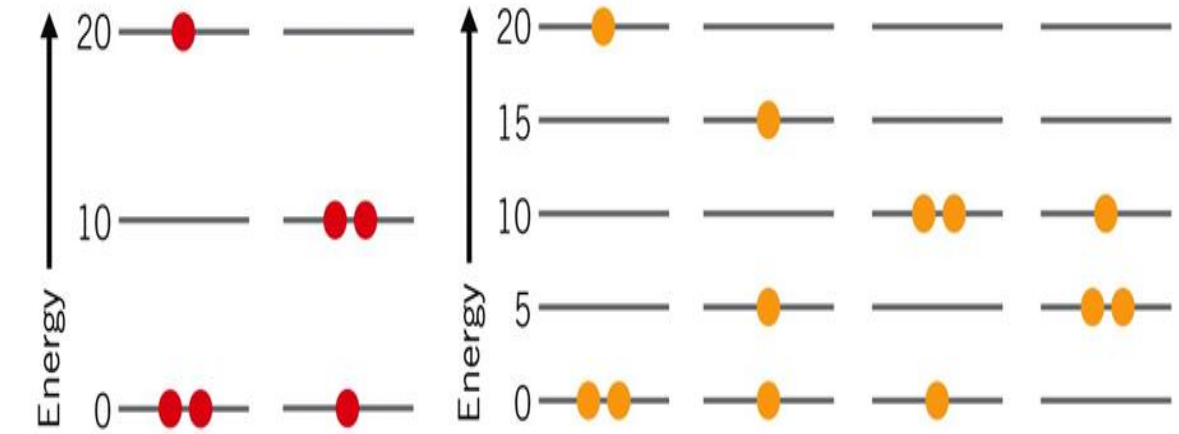
- Exothermicity favors the spontaneity of a reaction but does not guarantee it. Just as it is possible for an endothermic reaction to be spontaneous, it is possible for an exothermic reaction to be nonspontaneous.
- we cannot decide whether or not a chemical reaction will occur spontaneously solely on the basis of energy changes in the system. To make this kind of prediction we need another thermodynamic quantity, which turns out to be *entropy.*

Entropy

- Entropy can be thought of as a measure of the randomness of a system.
- It is related to the various modes of motion in molecules.
- Therefore,

$$\Delta S = S_{final} - S_{initial}$$

Entropy



(*a*) Low entropy (fewer ways to distribute energy) Red dots represent energies of molecules of *A*.

(*b*) High entropy (more ways to distribute energy) Yellow dots represent energies of molecules of *B*.

Entropy

- For a process occurring at constant temperature (an isothermal process):

$$\Delta S = \frac{q_{rev}}{T}$$

q_{rev} = the heat that is transferred when the process is carried out **reversibly** at a constant temperature.
T = temperature in Kelvin.

Second Law of Thermodynamics

The second law of thermodynamics: The entropy of the universe does not change for reversible processes and increases for spontaneous processes.

Reversible (ideal):

$$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surroundings} = 0$$

Irreversible (real, spontaneous):

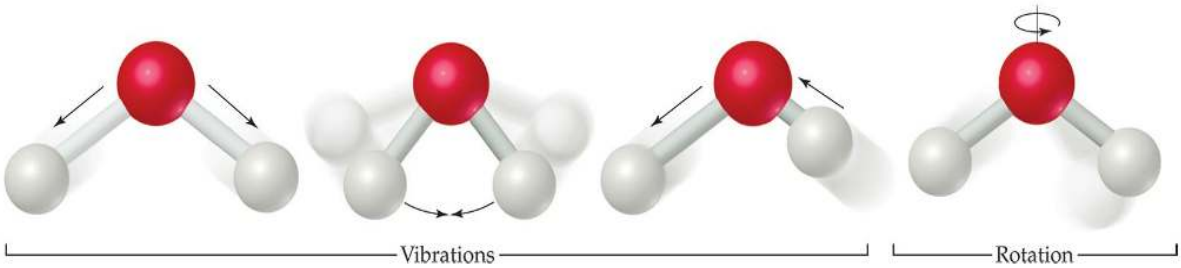
$$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surroundings} > 0$$

Entropy on the Molecular Scale

- Molecules exhibit several types of motion:
  - Translational, vibrational, rotational.
- Each thermodynamic state has a specific number of microstates, *W*, associated with it.

$$S = k \ln W$$

where k is the Boltzmann constant, 1.38×10^{-23} J/K.

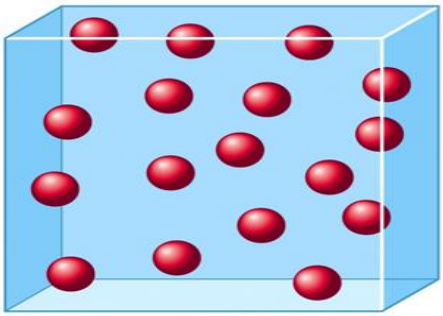


Entropy on the Molecular Scale

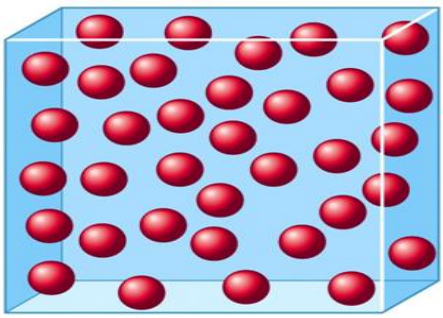
- Implications:

- more particles

-> more states -> more entropy



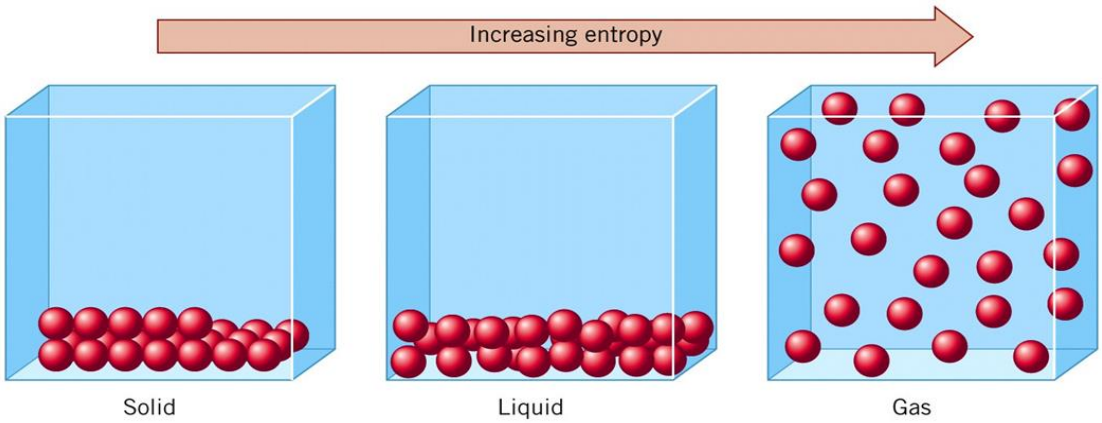
Lower entropy



Higher entropy

Entropy on the Molecular Scale

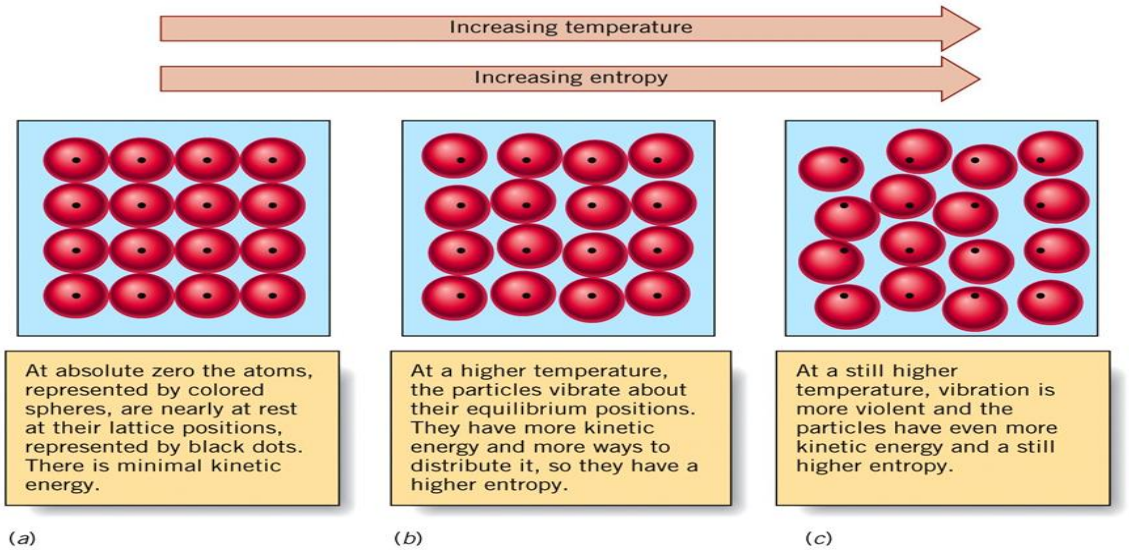
- Physical state (gas vs solid)
  -> more freedom of motion of molecules
  -> more entropy



Entropy on the Molecular Scale

- higher T

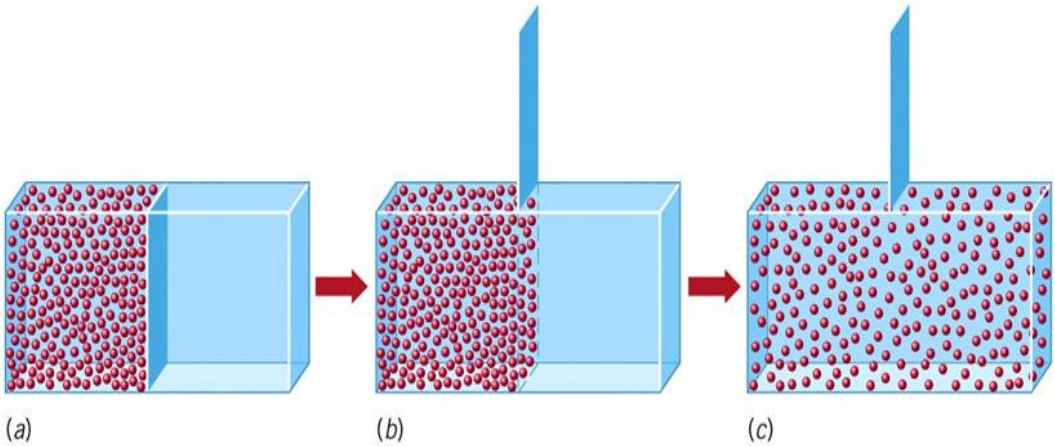
  -> more energy states -> more entropy

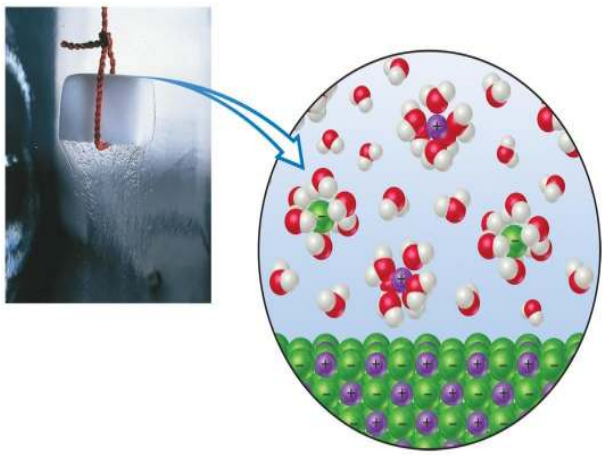


Entropy on the Molecular Scale

- Volume (for gases)

  -> more states -> more entropy





Dissolution of a solid:

Ions have more entropy (more states)

But,

Some water molecules have less entropy (they are grouped around ions).

Usually, there is an overall increase in S.
(The exception is very highly charged ions that make a lot of water molecules align around them.)

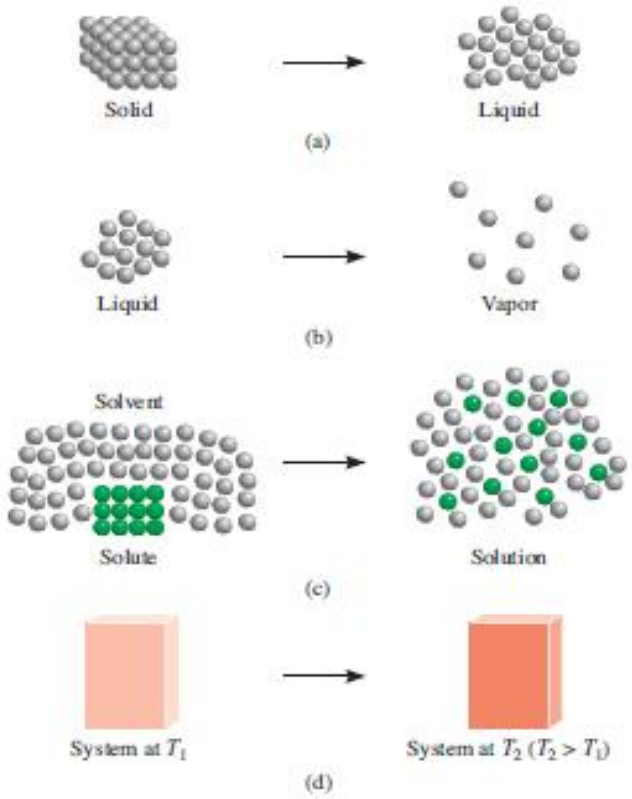
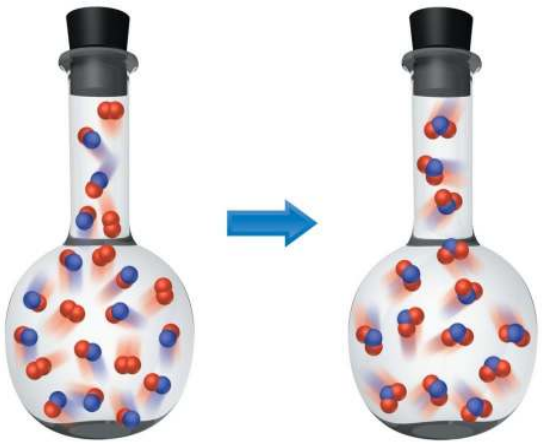


Figure 18.3
Processes that lead to an increase in entropy of the system: (a) melting: $S_{liquid} > S_{solid}$; (b) vaporization: $S_{vapor} > S_{liquid}$; (c) dissolving: $S_{soln} > S_{solute} + S_{solvent}$; (d) heating: $S_{T_2} > S_{T_1}$

Entropy Changes

- In general, entropy *increases* when
  - Gases are formed from liquids and solids.
  - Liquids or solutions are formed from solids.
  - The number of gas molecules increases.
  - The number of moles increases.



EXAMPLE 18.1

Predict whether the entropy change is greater or less than zero for each of the following processes: (a) freezing ethanol, (b) evaporating a beaker of liquid bromine at room temperature, (c) dissolving glucose in water, (d) cooling nitrogen gas from 80°C to 20°C.

(Continued)

Solution

(a) Upon freezing, the ethanol molecules are held rigidly in position. This phase transition reduces the number of microstates and therefore the entropy decreases; that is, $\Delta S < 0$.

(b) Evaporating bromine increases the number of microstates because the Br_2 molecules can occupy many more positions in nearly empty space. Therefore $\Delta S > 0$.

(c) Glucose is a nonelectrolyte. The solution process leads to a greater dispersal of matter due to the mixing of glucose and water molecules so we expect $\Delta S > 0$.

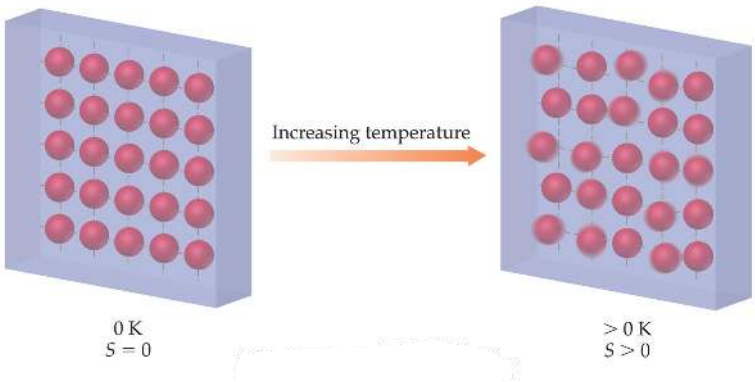
(d) The cooling process decreases various molecular motions. This leads to a decrease in microstates and so $\Delta S < 0$.

Practice Exercise How does the entropy of a system change for each of the following processes? (a) condensing water vapor, (b) forming sucrose crystals from a supersaturated solution, (c) heating hydrogen gas from 60°C to 80°C, and (d) subliming dry ice.

Third Law of Thermodynamics

The entropy of a pure crystalline substance at absolute zero is 0.

$$S = k\ lnW = k\ ln1 = 0$$



Standard Entropies

- These are molar entropy values of substances in their standard states.
- Standard entropies tend to increase with increasing molar mass.

$$\Delta S = S_{product} - S_{reactant}$$

TABLE Standard Molar Entropies ot Selected Substances at 298 K

Substance	$S°$, J/mol-K
Gases	
$H_2(g)$	130.6
$N_2(g)$	191.5
$O_2(g)$	205.0
$H_2O(g)$	188.8
$NH_3(g)$	192.5
$CH_3OH(g)$	237.6
$C_6H_6(g)$	269.2
Liquids	
$H_2O(l)$	69.9
$CH_3OH(l)$	126.8
$C_6H_6(l)$	172.8
Solids	
$Li(s)$	29.1
$Na(s)$	51.4
$K(s)$	64.7
$Fe(s)$	27.23
$FeCl_3(s)$	142.3
$NaCl(s)$	72.3

Entropy Changes

- Entropy changes for a reaction can be calculated:

$$\Delta S_{rxn} = \Sigma S^{\circ}_{reactants} - \Sigma S^{\circ}_{products}$$

$$\Delta S^\circ_{rxn} = \Sigma nS^\circ(\text{products}) - \Sigma mS^\circ(\text{reactants}) \qquad (18.7)$$

The standard entropy values of a large number of compounds have been measured in J/K · mol. To calculate ΔS°_{rxn} (which is ΔS_{sys}), we look up their values in Appendix 2 and proceed according to Example 18.2.

EXAMPLE 18.2

From the standard entropy values in Appendix 2, calculate the standard entropy changes for the following reactions at 25°C.

(a) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

(b) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

(c) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$

Strategy To calculate the standard entropy of a reaction, we look up the standard entropies of reactants and products in Appendix 2 and apply Equation (18.7). As in the calculation of enthalpy of reaction [see Equation (6.18)], the stoichiometric coefficients have no units, so ΔS°_{rxn} is expressed in units of J/K · mol.

Solution

(a) $\Delta S^\circ_{rxn} = [S^\circ(CaO) + S^\circ(CO_2)] - [S^\circ(CaCO_3)]$
$= [(39.8 \text{ J/K} \cdot \text{mol}) + (213.6 \text{ J/K} \cdot \text{mol})] - (92.9 \text{ J/K} \cdot \text{mol})$
$= 160.5 \text{ J/K} \cdot \text{mol}$

Thus, when 1 mole of $CaCO_3$ decomposes to form 1 mole of CaO and 1 mole of gaseous CO_2, there is an increase in entropy equal to 160.5 J/K · mol.

(b) $\Delta S^\circ_{rxn} = [2S^\circ(NH_3)] - [S^\circ(N_2) + 3S^\circ(H_2)]$
$= (2)(193 \text{ J/K} \cdot \text{mol}) - [(192 \text{ J/K} \cdot \text{mol}) + (3)(131 \text{ J/K} \cdot \text{mol})]$
$= -199 \text{ J/K} \cdot \text{mol}$

This result shows that when 1 mole of gaseous nitrogen reacts with 3 moles of gaseous hydrogen to form 2 moles of gaseous ammonia, there is a decrease in entropy equal to −199 J/K · mol.

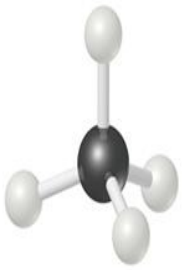
(c) $\Delta S^\circ_{rxn} = [2S^\circ(HCl)] - [S^\circ(H_2) + S^\circ(Cl_2)]$
$= (2)(187 \text{ J/K} \cdot \text{mol}) - [(131 \text{ J/K} \cdot \text{mol}) + (223 \text{ J/K} \cdot \text{mol})]$
$= 20 \text{ J/K} \cdot \text{mol}$

Thus, the formation of 2 moles of gaseous HCl from 1 mole of gaseous H_2 and 1 mole of gaseous Cl_2 results in a small increase in entropy equal to 20 J/K · mol.

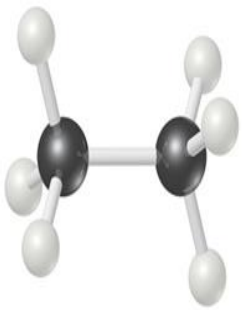
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Standard Entropies

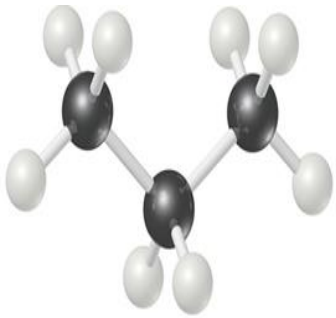
Larger and more complex molecules have greater entropies.



Methane, CH_4
$S° = 186.3\ J\ mol^{-1}\ K^{-1}$



Ethane, C_2H_6
$S° = 229.6\ J\ mol^{-1}\ K^{-1}$



Propane, C_3H_8
$S° = 270.3\ J\ mol^{-1}\ K^{-1}$

EXAMPLE 18.3

Predict whether the entropy change of the system in each of the following reactions is positive or negative.

(a) $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$

(b) $NH_4Cl(s) \longrightarrow NH_3(g) + HCl(g)$

(c) $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$

Entropy vs. Enthalpy

- Both entropy and enthalpy affect Reaction Spontaneity
- They work together or against each other.

- When ΔH and ΔS have same sign, T determines whether spontaneous or nonspontaneous

Entropy Changes in Surroundings

- Heat that flows into or out of the system also changes the entropy of the surroundings.
- For an isothermal process:

$$\Delta S_{surr} = \frac{-q_{sys}}{T}$$

- At constant pressure, q_{sys} is simply ΔH° for the system.

$$\Delta S_{surr} = \frac{-q_{sys}}{T} = \frac{-\Delta H^\circ}{T}$$

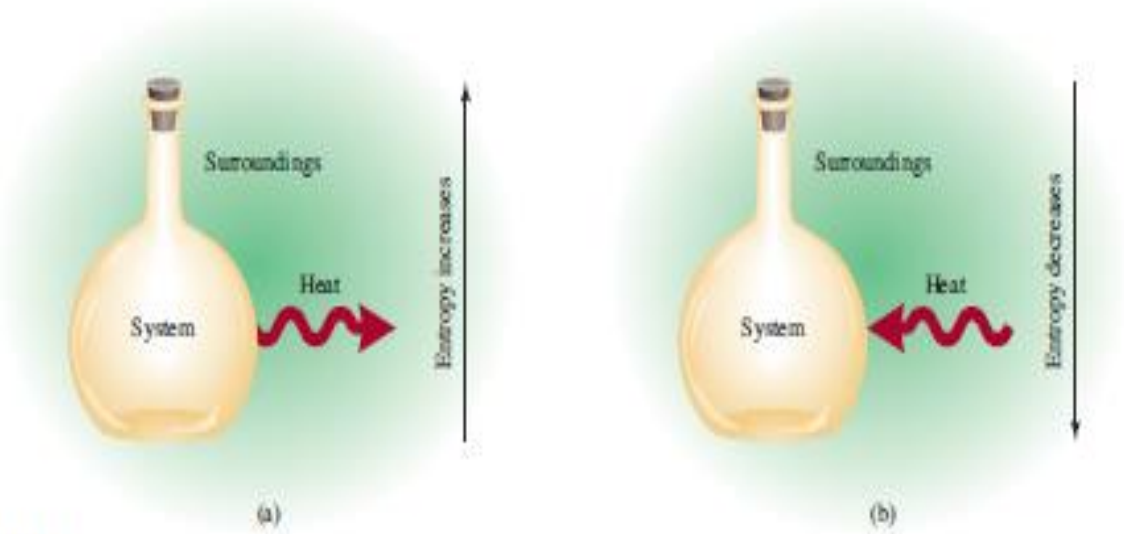


Figure 18.5

(a) An exothermic process transfers heat from the system to the surroundings and results in an increase in the entropy of the surroundings. (b) An endothermic process absorbs heat from the surroundings and thereby decreases the entropy of the surroundings.

Link S and ΔH: Phase changes

$$\Delta S_{surr} = \frac{-q_{sys}}{T} = \frac{-\Delta H^{\circ}_{sys}}{T}$$

A phase change is isothermal (no chang in T).

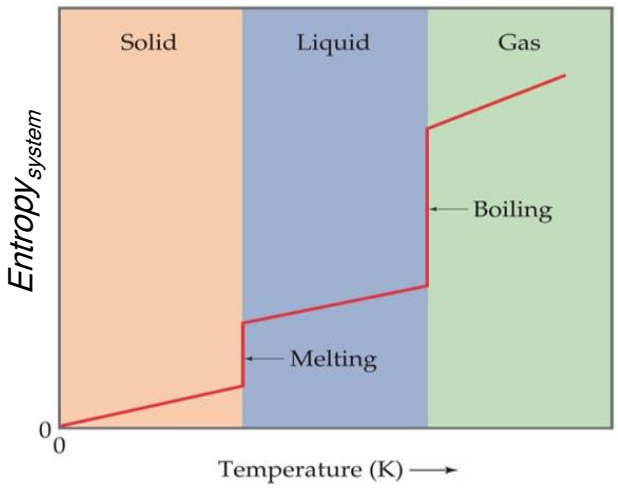
For water:

ΔH_{fusion} = 6 kJ/mol

ΔH_{vap} = 41 kJ/mol

If we do this reversibly:

$\Delta S_{surr} = -\Delta S_{sys}$



Surroundings & System

Entropy Change in the Universe

- The universe is composed of the system and the surroundings.

  Therefore,

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- For spontaneous processes

$$\Delta S_{\text{universe}} > 0$$

Let us now apply the procedure for calculating ΔS_{sys} and ΔS_{surr} to the synthesis of ammonia and ask whether the reaction is spontaneous at 25°C:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g) \quad \Delta H^\circ_{rxn} = -92.6 \text{ kJ/mol}$$

From Example 18.2(b) we have $\Delta S_{sys} = -199$ J/K · mol, and substituting ΔH_{sys} (−92.6 kJ/mol) in Equation (18.8), we obtain

$$\Delta S_{surr} = \frac{-(-92.6 \times 1000) \text{ J/mol}}{298 \text{ K}} = 311 \text{ J/K} \cdot \text{mol}$$

The change in entropy of the universe is

$$\begin{aligned} \Delta S_{univ} &= \Delta S_{sys} + \Delta S_{surr} \\ &= -199 \text{ J/K} \cdot \text{mol} + 311 \text{ J/K} \cdot \text{mol} \\ &= 112 \text{ J/K} \cdot \text{mol} \end{aligned}$$



The synthesis of NH_3 from H_2 and N_2.

Surroundings & System

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surround}$$

$$\Delta S_{surr} = \frac{-\Delta H^{\circ}_{sys}}{T}$$

$$\Delta S_{universe} = \Delta S_{system} + \frac{-\Delta H^{\circ}_{sys}}{T}$$

$$T\Delta S_{universe} = T\Delta S_{system} + -\Delta H^{\circ}_{sys}$$

= – Gibbs Free Energy

Surroundings & System

$$\underbrace{T\Delta S_{universe}} = T\Delta S_{system} + -\Delta H^{\circ}_{sys}$$

= – Gibbs Free Energy

Make this equation nicer:

$$-T\Delta S_{universe} = \Delta H^{\circ}_{sys} - T\Delta S_{system}$$

$$\Delta G = \Delta H^{\circ}_{sys} - T\Delta S_{system}$$

Standard Free Energy Changes

Standard free energies of formation, ΔG_f°, can be calculated.

$$\Delta G_f^\circ = \Sigma \Delta G_{reactants}^\circ - \Sigma \Delta G_{products}^\circ$$

$$\Delta G = \Delta H_{sys}^\circ - T\Delta S_{system}$$

Appendix 2 lists the values of ΔG_f° for a number of compounds.

EXAMPLE 18.4

Calculate the standard free-energy changes for the following reactions at 25°C.

(a) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

(b) $2MgO(s) \longrightarrow 2Mg(s) + O_2(g)$

Strategy To calculate the standard free-energy change of a reaction, we look up the standard free energies of formation of reactants and products in Appendix 2 and apply

(Continued)

Solution

(a) According to Equation (18.12), we write

$$\Delta G^\circ_{rxn} = [\Delta G_f^\circ(CO_2) + 2\Delta G_f^\circ(H_2O)] - [\Delta G_f^\circ(CH_4) + 2\Delta G_f^\circ(O_2)]$$

We insert the appropriate values from Appendix 2:

$$\Delta G^\circ_{rxn} = [(-394.4\ kJ/mol) + (2)(-237.2\ kJ/mol)] - [(-50.8\ kJ/mol) + (2)(0\ kJ/mol)]$$

$$= -818.0\ kJ/mol$$

(b) The equation is

$$\Delta G^\circ_{rxn} = [2\Delta G_f^\circ(Mg) + \Delta G_f^\circ(O_2)] - [2\Delta G_f^\circ(MgO)]$$

From data in Appendix 2 we write

$$\Delta G^\circ_{rxn} = [(2)(0\ kJ/mol) + (0\ kJ/mol)] - [(2)(-569.6\ kJ/mol)]$$

$$= 1139\ kJ/mol$$

Practice Exercise Calculate the standard free-energy changes for the following reactions at 25°C:

(a) $H_2(g) + Br_2(l) \longrightarrow 2HBr(g)$

(b) $2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(l)$

Free Energy and Temperature

- There are two parts to the free energy equation:
  - ΔH°— the enthalpy term
  - $T\Delta S^\circ$ — the entropy term

- The temperature dependence of free energy comes from the entropy term.

Free Energy and Temperature

- Gibbs free energy: Maximum energy in reaction that is "free" or available to do useful work.

$$\Delta G = \Delta H - T\Delta S$$

ΔG	< 0	Spontaneous process
ΔG	$= 0$	At equilibrium
ΔG	> 0	Nonspontaneous

Free Energy and Temperature

- At constant P and T, process spontaneous only if it is accompanied by decrease in free energy of system.

ΔH	ΔS		Spontaneous?
–	+	$\Delta G = (-) - [T(+)] = -$	Always, regardless of T
+	–	$\Delta G = (+) - [T(-)] = +$	Never, regardless of T
+	+	$\Delta G = (+) - [T(+)] = ?$	Depends; spontaneous at high T, $-\Delta G$
–	–	$\Delta G = (-) - [T(-)] = ?$	Depends; spontaneous at low T, $-\Delta G$

Table 18.3 Factors Affecting the Sign of ΔG in the Relationship $\Delta G = \Delta H - T\Delta S$

ΔH	ΔS	ΔG	Example
+	+	Reaction proceeds spontaneously at high temperatures. At low temperatures, reaction is spontaneous in the reverse direction.	$2HgO(s) \longrightarrow 2Hg(l) + O_2(g)$
+	−	ΔG is always positive. Reaction is spontaneous in the reverse direction at all temperatures.	$3O_2(g) \longrightarrow 2O_3(g)$
−	+	ΔG is always negative. Reaction proceeds spontaneously at all temperatures.	$2H_2O_2(l) \longrightarrow 2H_2O(l) + O_2(g)$
−	−	Reaction proceeds spontaneously at low temperatures. At high temperatures, the reverse reaction becomes spontaneous.	$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$



Liquid and solid benzene in equilibrium at 5.5°C.

EXAMPLE 18.5

The molar heats of fusion and vaporization of benzene are 10.9 kJ/mol and 31.0 kJ/mol, respectively. Calculate the entropy changes for the solid ⟶ liquid and liquid ⟶ vapor transitions for benzene. At 1 atm pressure, benzene melts at 5.5°C and boils at 80.1°C.

Strategy At the melting point, liquid and solid benzene are at equilibrium, so $\Delta G = 0$. From Equation (18.10) we have $\Delta G = 0 = \Delta H - T\Delta S$ or $\Delta S = \Delta H/T$. To calculate the entropy change for the solid benzene ⟶ liquid benzene transition, we write $\Delta S_{fus} = \Delta H_{fus}/T_f$. Here ΔH_{fus} is positive for an endothermic process, so ΔS_{fus} is also positive, as expected for a solid to liquid transition. The same procedure applies to the liquid benzene ⟶ vapor benzene transition. What temperature unit should be used?

Solution The entropy change for melting 1 mole of benzene at 5.5°C is

$$\Delta S_{fus} = \frac{\Delta H_{fus}}{\Delta T_f}$$

$$= \frac{(10.9\ \text{kJ/mol})(1000\ \text{J/1 kJ})}{(5.5 + 273)\ \text{K}}$$

$$= 39.1\ \text{J/K} \cdot \text{mol}$$

(Continued)

Similarly, the entropy change for boiling 1 mole of benzene at 80.1°C is

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T_{bp}}$$

$$= \frac{(31.0\ \text{kJ/mol})(1000\ \text{J/1 kJ})}{(80.1 + 273)\ \text{K}}$$

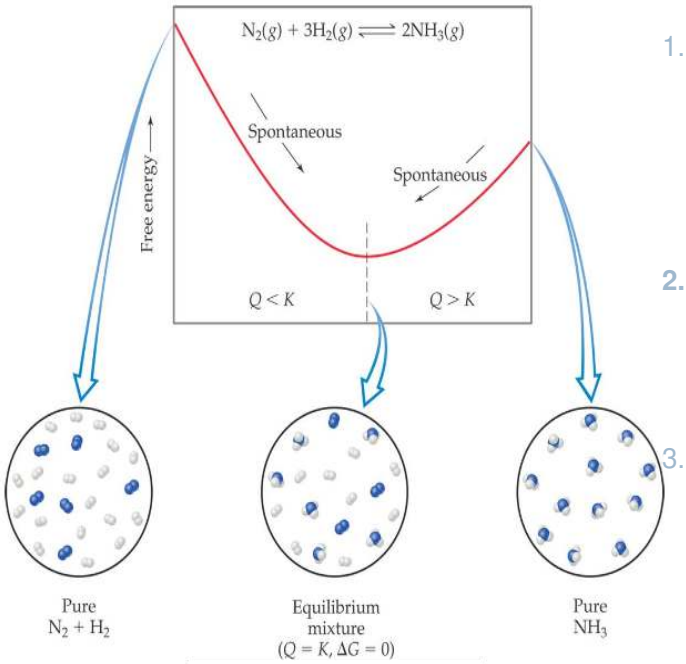
$$= 87.8\ \text{J/K} \cdot \text{mol}$$

Check Because vaporization creates more microstates than the melting process, $\Delta S_{vap} > \Delta S_{fus}$.

Practice Exercise The molar heats of fusion and vaporization of argon are 1.3 kJ/mol and 6.3 kJ/mol, and argon's melting point and boiling point are −190°C and −186°C, respectively. Calculate the entropy changes for fusion and vaporization.

Similar problems: 18.58.

Gibbs Free Energy



1. If ΔG is negative, the forward reaction is spontaneous.
2. **If ΔG is 0, the system is at equilibrium.**
3. If ΔG is positive, the reaction is spontaneous in the reverse direction.

Free Energy and Equilibrium

From above:

If ΔG is 0, the system is at equilibrium.

So ΔG must be related to the equilibrium constant, K. The ***standard*** free energy, $\Delta G°$, is directly linked to K_{eq} by:

$$\Delta G° = -RT \ln K$$

Free Energy and Equilibrium

Under non-standard conditions, we need to use ΔG instead of $\Delta G°$.

$$\Delta G° = -RT \ln K$$

$$\Delta G = \Delta G° + RT \ln Q$$

Note: at equilibrium: $\Delta G = 0$.
away from equil, sign of ΔG tells which way rxn goes spontaneously.

Two cases:

- **Case 1:** A large negative value of ΔG° will tend to make ΔG also negative. Thus, the net reaction will proceed from left to right until a signifi cant amount of product has been formed. At that point, the $RT\ln Q$ term will become positive enough to match the negative ΔG° term.

- **Case 2:** A large positive ΔG° term will tend to make ΔG also positive. Thus, the net reaction will proceed from right to left until a signifi cant amount of reactant has been formed. At that point, the $RT\ln Q$ term will become negative enough to match the positive ΔG° term.

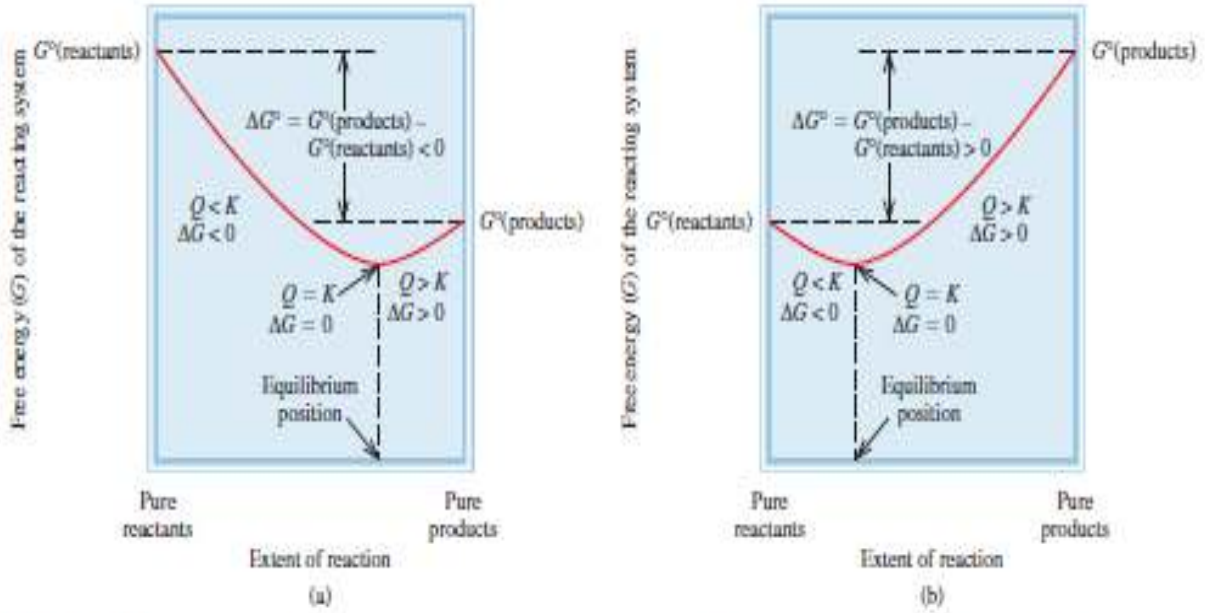


Figure 18.9

(a) $\Delta G^\circ < 0$. At equilibrium, there is a significant conversion of reactants to products. (b) $\Delta G^\circ > 0$. At equilibrium, reactants are favored over products. In both cases, the net reaction toward equilibrium is from left to right (reactants to products) if $Q < K$ and right to left (products to reactants) if $Q > K$.

At equilibrium, by definition, $\Delta G = 0$ and $Q = K$, where K is the equilibrium constant.

$$0 = \Delta G^\circ + RT \ln K$$

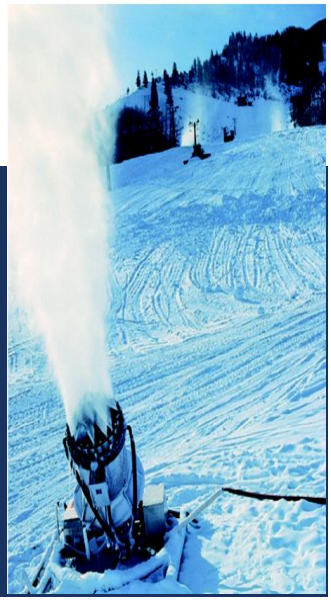
or

$$\Delta G^\circ = -RT \ln K$$

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CHAPTER 18
THERMODYNAMICS
LECTURE 39-40



Sooner or later a reversible reaction will reach equilibrium.

At equilibrium, by definition, $\Delta G = 0$ and $Q = K$, where K is the equilibrium constant. Thus,

$$0 = \Delta G^\circ + RT \ln K$$

or

$$\Delta G^\circ = -RT \ln K \tag{18.14}$$

In this equation, K_p is used for gases and K_c for reactions in solution. Note that the larger the K is, the more negative ΔG° is. For chemists, Equation (18.14) is one of the most important equations in thermodynamics because it enables us to find the equilibrium constant of a reaction if we know the change in standard free energy and vice versa.

It is significant that Equation (18.14) relates the equilibrium constant to the *standard* free-energy change ΔG° rather than to the *actual* free-energy change ΔG. The actual free-energy change of the system varies as the reaction progresses and becomes zero at equilibrium. On the other hand, ΔG° is a constant for a particular reaction at a given temperature. Figure 18.9 shows plots of the free energy of a reacting system versus the extent of the reaction for two types of reactions. As you can see, if $\Delta G^\circ < 0$, the products are favored over reactants at equilibrium. Conversely, if $\Delta G^\circ > 0$, there will be more reactants than products at equilibrium. Table 18.4 summarizes the three possible

Table 18.4 Relation Between ΔG° and K as Predicted by the Equation $\Delta G^\circ = -RT \ln K$

K	$\ln K$	ΔG°	Comments
> 1	Positive	Negative	Products are favored over reactants at equilibrium.
$= 1$	0	0	Products and reactants are equally favored at equilibrium.
< 1	Negative	Positive	Reactants are favored over products at equilibrium.

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

At 25°C, the equilibrium constant K_P is

$$K_P = \frac{P_{NO}^2}{P_{N_2}P_{O_2}} = 4.0 \times 10^{-31}$$

The very small value of K_P means that the concentration of NO at equilibrium will be exceedingly low. In such a case the equilibrium constant is more conveniently obtained from $\Delta G°$. (As we have seen, $\Delta G°$ can be calculated from $\Delta H°$ and $\Delta S°$.) On the other hand, the equilibrium constant for the formation of hydrogen iodide from molecular hydrogen and molecular iodine is near unity at room temperature:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

For this reaction it is easier to measure K_P and then calculate $\Delta G°$ using Equation (18.14) than to measure $\Delta H°$ and $\Delta S°$ and use Equation (18.10).

EXAMPLE 18.6

Using data listed in Appendix 2, calculate the equilibrium constant (K_p) for the following reaction at 25°C:

$$2H_2O(l) \rightleftharpoons 2H_2(g) + O_2(g)$$

Strategy According to Equation (18.14), the equilibrium constant for the reaction is related to the standard free energy-change; that is, $\Delta G^\circ = -RT \ln K$. Therefore, we first need to calculate ΔG° by following the procedure in Example 18.4. Then we can calculate K_p. What temperature unit should be used?

Solution According to Equation (18.12),

$$\begin{aligned}\Delta G^\circ_{rxn} &= [2\Delta G^\circ_f(H_2) + \Delta G^\circ_f(O_2)] - [2\Delta G^\circ_f(H_2O)] \\ &= [(2)(0\ \text{kJ/mol}) + (0\ \text{kJ/mol})] - [(2)(-237.2\ \text{kJ/mol})] \\ &= 474.4\ \text{kJ/mol}\end{aligned}$$

Using Equation (18.14)

$$\Delta G^\circ_{rxn} = -RT \ln K_p$$

$$474.4\ \text{kJ/mol} \times \frac{1000\ \text{J}}{1\ \text{kJ}} = -(8.314\ \text{J/K} \cdot \text{mol})(298\ \text{K}) \ln K_p$$

$$\ln K_p = -191.5$$

$$K_p = e^{-191.5} = 7 \times 10^{-84}$$

Comment This extremely small equilibrium constant is consistent with the fact that water does not decompose into hydrogen and oxygen gases at 25°C. Thus, a large positive ΔG° favors reactants over products at equilibrium.

Practice Exercise Calculate the equilibrium constant (K_p) for the reaction at 25°C

$$2O_3(g) \longrightarrow 3O_2(g)$$

EXAMPLE 18.7

In Chapter 17 we discussed the solubility product of slightly soluble substances. Using the solubility product of silver chloride at 25°C (1.6×10^{-10}), calculate $\Delta G°$ for the process

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

Strategy According to Equation (18.14), the equilibrium constant for the reaction is related to standard free-energy change; that is, $\Delta G° = -RT \ln K$. Because this is a heterogeneous equilibrium, the solubility product (K_{sp}) is the equilibrium constant. We calculate the standard free energy change from the K_{sp} value of AgCl. What temperature unit should be used?

Solution The solubility equilibrium for AgCl is

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$
$$K_{sp} = [Ag^+][Cl^-] = 1.6 \times 10^{-10}$$

Using Equation (18.14) we obtain

$$\begin{aligned}\Delta G° &= -(8.314\ \text{J/K} \cdot \text{mol})(298\ \text{K}) \ln (1.6 \times 10^{-10}) \\ &= 5.6 \times 10^4\ \text{J/mol} \\ &= 56\ \text{kJ/mol}\end{aligned}$$

Check The large, positive $\Delta G°$ indicates that AgCl is slightly soluble and that the equilibrium lies mostly to the left.

Practice Exercise Calculate $\Delta G°$ for the following process at 25°C:

$$BaF_2(s) \rightleftharpoons Ba^{2+}(aq) + 2F^-(aq)$$

The K_{sp} of BaF_2 is 1.7×10^{-6}.

EXAMPLE 18.8

The equilibrium constant (K_p) for the reaction

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

is 0.113 at 298 K, which corresponds to a standard free-energy change of 5.40 kJ/mol. In a certain experiment, the initial pressures are $P_{NO_2} = 0.122$ atm and $P_{N_2O_4} = 0.453$ atm. Calculate ΔG for the reaction at these pressures and predict the direction of the net reaction.

Strategy From the information given we see that neither the reactant nor the product is at its standard state of 1 atm. To determine the direction of the net reaction, we need to calculate the free-energy change under nonstandard-state conditions (ΔG) using Equation (18.13) and the given $\Delta G°$ value. Note that the partial pressures are expressed as dimensionless quantities in the reaction quotient Q_p.

(Continued)

Solution Equation (18.13) can be written as

$$\begin{aligned}\Delta G &= \Delta G° + RT \ln Q_p \\ &= \Delta G° + RT \ln \frac{P_{NO_2}^2}{P_{N_2O_4}} \\ &= 5.40 \times 10^3 \text{ J/mol} + (8.314 \text{ J/K} \cdot \text{mol})(298 \text{ K}) \times \ln \frac{(0.122)^2}{0.453} \\ &= 5.40 \times 10^3 \text{ J/mol} - 8.46 \times 10^3 \text{ J/mol} \\ &= -3.06 \times 10^3 \text{ J/mol} = -3.06 \text{ kJ/mol}\end{aligned}$$

Because $\Delta G < 0$, the net reaction proceeds from left to right to reach equilibrium.

Check Note that although $\Delta G° > 0$, the reaction can be made to favor product formation initially by having a small concentration (pressure) of the product compared to that of the reactant. Confirm the prediction by showing that $Q_p < K_p$.

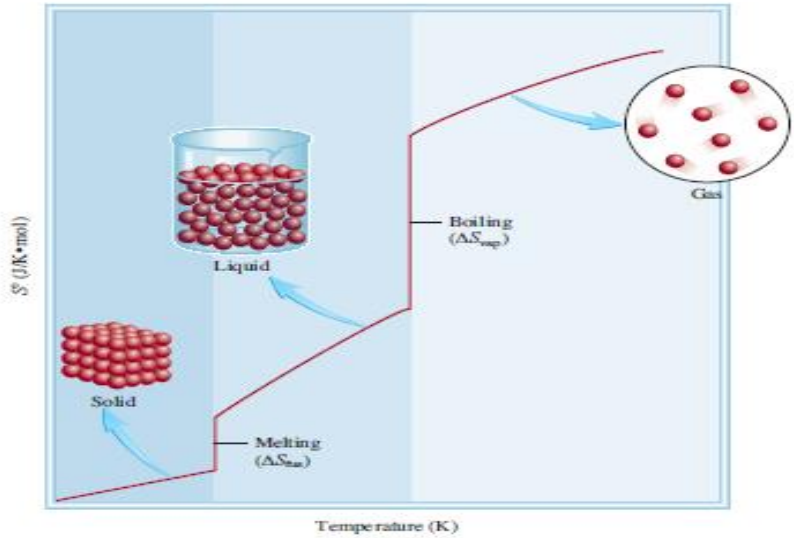
Practice Exercise The $\Delta G°$ for the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

is 2.60 kJ/mol at 25°C. In one experiment, the initial pressures are $P_{H_2} = 4.26$ atm, $P_{I_2} = 0.024$ atm, and $P_{HI} = 0.23$ atm. Calculate ΔG for the reaction and predict the direction of the net reaction.

PHASE CHANGE

Figure 18.6
Entropy increase of a substance as the temperature rises from absolute zero.



PHASE CHANGE EQUILIBRIUM

Phase change	name	Temperature	eqn
Solid-liquid	Fusion	Melting point T_m	$T_m=\Delta H_{Fus}/\Delta S_{Fus}$
Liquid-gas	vaporization	Boiling point T_b	$T_B=\Delta H_{vap}/\Delta S_{vap}$
Liquid-solid	freezing	Freezing point T_f	$T_f=\Delta H_{freez}/\Delta S_{freez}$
Solid- gas	sublimation	Sublimation point T_s	$T_s=\Delta H_{sub}/\Delta S_{sub}$

EXAMPLE:

- Calculate the melting of Cl_2 if the enthalpy of fusion $\Delta H_{Fus} = 10.25$ & $\Delta S_{Fus} = 130.7$
- $T_m = \Delta H_{Fus} / \Delta S_{Fus}$

18.5 How does the entropy of a system change for each of the following processes?

(a) A solid melts.

(b) A liquid freezes.

(c) A liquid boils.

(d) A vapor is converted to a solid.

(e) A vapor condenses to a liquid.

(f) A solid sublimes.

(g) Urea dissolves in water.

18.10 Arrange the following substances (1 mole each) in order of increasing entropy at 25°C: (a) $Ne(g)$, (b) $SO_2(g)$, (c) $Na(s)$, (d) $NaCl(s)$, (e) $H_2(g)$. Give the reasons for your arrangement.

c) , (d) , (e) , (a) , (b). Solids have smaller entropies than gases

18.12 Using the data in Appendix 2, calculate the standard entropy changes for the following reactions at 25°C:

(a) $H_2(g) + CuO(s) \longrightarrow Cu(s) + H_2O(g)$

(b) $2Al(s) + 3ZnO(s) \longrightarrow Al_2O_3(s) + 3Zn(s)$

(c) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

18.13 Without consulting Appendix 2, predict whether the entropy change is positive or negative for each of the following reactions. Give reasons for your predictions.

(a) $2KClO_4(s) \longrightarrow 2KClO_3(s) + O_2(g)$

(b) $H_2O(g) \longrightarrow H_2O(l)$

(c) $2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$

(d) $N_2(g) \longrightarrow 2N(g)$

18.18 Calculate ΔG° for the following reactions at 25°C:

(a) $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$

(b) $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

(c) $2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(l)$

See Appendix 2 for thermodynamic data.

18.24 For the autoionization of water at 25°C,

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

K_w is 1.0×10^{-14}. What is ΔG° for the process?

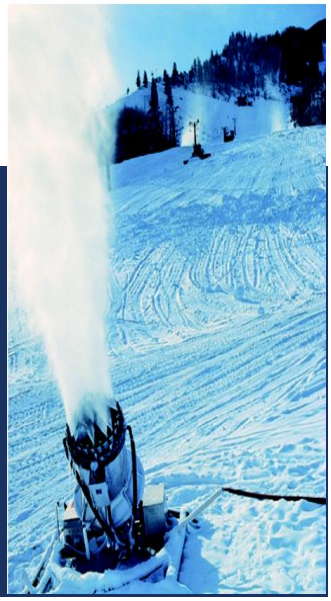
GENERAL CHEMISTRY 1
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CHAPTER 19

OXIDATION
REDUCTION
REACTION

LECTURE 40





OXIDATION-REDUCTION REACTIONS

Electron transfer reactions

- Electrons transferred from one substance to another
- Originally only combustion of fuels or reactions of metal with oxygen
- Important class of chemical reactions that occur in all areas of chemistry & biology
- Also called **redox reactions**

OXIDATION–REDUCTION REACTIONS

Involves 2 processes:

Oxidation = Loss of Electrons (LEO)

$Na \longrightarrow Na^+ + e^-$ Oxidation Half-Reaction

Reduction = Gain of electrons (GER)

$Cl_2 + 2e^- \longrightarrow 2Cl^-$ Reduction Half-Reaction

Net reaction:

$$2Na + Cl_2 \longrightarrow 2Na^+ + 2Cl^-$$

- Oxidation & reduction **always** occur together
- Can't have one without the other

REDEFINE OXIDATION-REDUCTION IN TERMS OF OXIDATION NUMBER

- A redox reaction occurs when there is a change in oxidation number.

Oxidation

- **Increase in oxidation number**
- e^- loss

Reduction

- **Decrease in oxidation number**
- e^- gain

OXIDATION REDUCTION REACTION

Oxidizing Agent

- Substance that accepts **e⁻'s**
  - Accepts **e⁻'s** from another substance
  - Substance that is reduced
  - $Cl_2 + 2e^- \longrightarrow 2Cl^-$

Reducing Agent

- Substance that donates e⁻'s
  - Releases **e⁻'s** to another substance
  - Substance that is oxidized
  - $Na \longrightarrow Na^+ + e^-$

EXAMPLE

Which species functions as the ***oxidizing agent*** in the following oxidation-reduction reaction?

$$Zn(s) + Pt^{2+}(aq) \longrightarrow Pt(s) + Zn^{2+}(aq)$$

A. $Pt(s)$
B. $Zn^{2+}(aq)$
C. $Pt^{2+}(aq)$
D. $Zn(s)$
E. None of these, as this is not a redox reaction.

GUIDELINES FOR REDOX REACTIONS

- Oxidation & reduction always occur simultaneously
- Total number of electrons lost by one substance = total number of electrons gained by second substance
- For a redox reaction to occur, something must accept electrons that are lost by another substance

OXIDATION NUMBERS

Bookkeeping Method

- Way to keep track of electrons
  - Not all redox reactions contain O_2 & give ions
  - Covalent molecules & ions often involved

    Ex. CH_4, SO_2, MnO_4^-, etc.
- Defined by set of rules
  - How to divide up shared electrons in compounds with covalent bonds
- Change in oxidation number of element during reaction indicates **redox reaction** has occurred

RULES FOR ASSIGNING OXIDATION NUMBERS

1. Oxidation numbers must add up to charge on molecule, formula unit or ion.
2. Atoms of free elements have oxidation numbers of zero.
3. Metals in Groups 1A, 2A, and Al have +1, +2, and +3 oxidation numbers, respectively.
4. H & F in compounds have +1 & –1 oxidation numbers, respectively.
5. Oxygen has –2 oxidation number.
6. Group 7A elements have –1 oxidation number.

HIERARCHY OF RULES FOR ASSIGNING OXIDATION NUMBERS

7. Group 6A elements have –2 oxidation number.
8. Group 5A elements have –3 oxidation number.
9. When there is a conflict between 2 of these rules or ambiguity in assigning an oxidation number, apply rule with lower number & ignore conflicting rule.

Oxidation State

- **Used interchangeably with oxidation number**
- **Indicates charge on monatomic ions**
- **Iron (III) means +3 oxidation state of Fe or Fe^{3+}**

EX. ASSIGNING OXIDATION NUMBER

1. Li_2O

Li (2 atoms) × (+1) = +2 (Rule 3)

O (1 atom) × (–2) = –2 (Rule 5)

sum = 0 (Rule 1)

+2 –2 = 0 so the charges are balanced to zero

2. CO_2

C (1 atom) × (x) = x

O (2 atoms) × (–2) = –4 (Rule 5)

sum = 0 (Rule 1)

$x - 4 = 0$ or $x = +4$

C is in +4 oxidation state

LEARNING CHECK

Assign oxidation numbers to all atoms:

Ex. ClO_4^-

O (4 atoms) × (–2) = –8

Cl (1 atom) × (–1) = –1

(molecular ion) sum ≠ –1 (violates Rule 1)

Rule 5 for O comes before Rule 6 for halogens

O (4 atoms) × (–2) = –8

Cl (1 atom) × (x) = x

sum = –1 (Rule 1)

$-8 + x = -1$ or $x = 8 - 1$

So x = +7; Cl is oxidation state +7

LEARNING CHECK

Assign Oxidation States To All Atoms:

- $MgCr_2O_7$

  Mg =+2; O = –2; and Cr = x (unknown)

  +2 + 2x + {7 × (–2)} = 0

  2x – 12 = 0 x = +3

  Cr is oxidation # of +3

- $KMnO_4$

  K =+1; O = – 2; so Mn = x

  +1 + x + {4 × (–2)} = 0

  x – 7 = 0 x = +7

  Mn is oxidation # of +7

YOUR TURN!

What is the oxidation number of each atom in H_3PO_4?

A. H = –1; P = +5; O = –2

B. H = 0; P = +3; O = –2

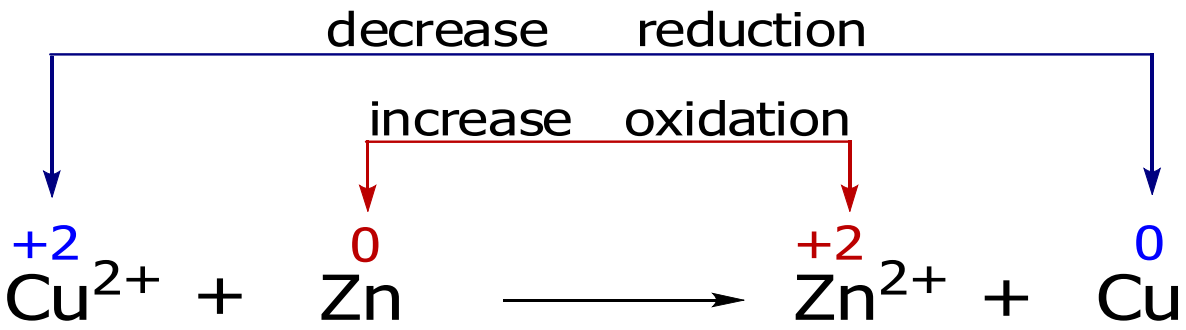
C. H = +1; P = +7; O = –2

D. H = +1; P = +1; O = –1

E. H = +1; P = +5; O = –2

USING OXIDATION NUMBERS TO RECOGNIZE REDOX REACTIONS

- Sometimes literal electron transfer:

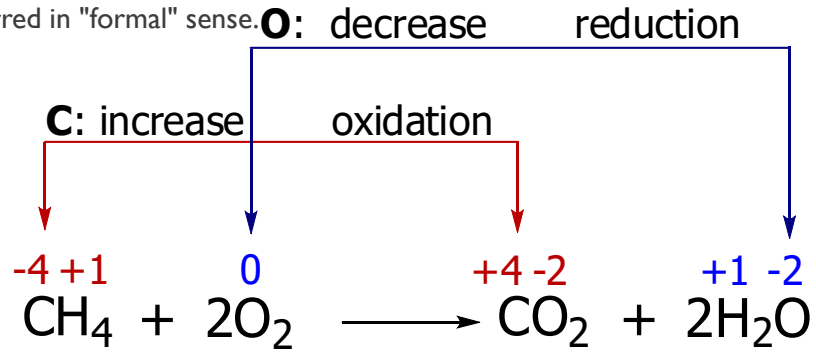


Cu: oxidation number decreases by 2
⇔ reduction

Zn: oxidation number increases by 2
⇔ oxidation

USING OXIDATION NUMBERS TO RECOGNIZE REDOX REACTIONS

- Sometimes electron transferred in "formal" sense.



  - **O:** oxidation number decreases by 2
    ⇔ reduction
  - **C:** oxidation number increases by 8
    ⇔ oxidation

OXIDATION NUMBER

The charge the atom would have in a molecule (or an ionic compound) if electrons were completely transferred.

1. Free elements (uncombined state) have an oxidation number of zero.

$$Na, Be, K, Pb, H_2, O_2, P_4 = 0$$

2. In monatomic ions, the oxidation number is equal to the charge on the ion.

$$Li^+, Li = +1; Fe^{3+}, Fe = +3; O^{2-}, O = -2$$

3. The oxidation number of oxygen is **usually** –2. In H_2O_2 and O_2^{2-} it is –1.

4. The oxidation number of hydrogen is +1 *except* when it is bonded to metals in binary compounds. In these cases, its oxidation number is –1.
5. Group IA metals are +1, IIA metals are +2, Al is +3 and fluorine is always –1.
6. The sum of the oxidation numbers of all the atoms in a molecule or ion is equal to the charge on the molecule or ion.
7. Oxidation numbers do not have to be integers. Oxidation number of oxygen in the superoxide ion, O_2^-, is -½.



Oxidation numbers of all the elements in HCO_3^- ?

HCO_3^-

O = -2 H = +1

3x(-2) + 1 + ? = -1

C = +4

1 1A	2 2A	3 3B	4 4B	5 5B	6 6B	7 7B	8	9 8B	10	11 1B	12 2B	13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
1 **H** +1 −1																	2 **He**
3 **Li** +1	4 **Be** +2											5 **B** +3	6 **C** +4 +2 −4	7 **N** +5 +4 +3 +2 +1 −3	8 **O** +2 −½ −1 −2	9 **F** −1	10 **Ne**
11 **Na** +1	12 **Mg** +2											13 **Al** +3	14 **Si** +4 −4	15 **P** +5 +3 −3	16 **S** +6 +4 +2 −2	17 **Cl** +7 +6 +5 +4 +3 +1 −1	18 **Ar**
19 **K** +1	20 **Ca** +2	21 **Sc** +3	22 **Ti** +4 +3 +2	23 **V** +5 +4 +3 +2	24 **Cr** +6 +5 +4 +3 +2	25 **Mn** +7 +6 +4 +3 +2	26 **Fe** +3 +2	27 **Co** +3 +2	28 **Ni** +2	29 **Cu** +2 +1	30 **Zn** +2	31 **Ga** +3	32 **Ge** +4 −4	33 **As** +5 +3 −3	34 **Se** +6 +4 −2	35 **Br** +5 +3 +1 −1	36 **Kr** +4 +2
37 **Rb** +1	38 **Sr** +2	39 **Y** +3	40 **Zr** +4	41 **Nb** +5 +4	42 **Mo** +6 +4 +3	43 **Tc** +7 +6 +4	44 **Ru** +8 +6 +4 +3	45 **Rh** +4 +3 +2	46 **Pd** +4 +2	47 **Ag** +1	48 **Cd** +2	49 **In** +3	50 **Sn** +4 +2	51 **Sb** +5 +3 −3	52 **Te** +6 +4 −2	53 **I** +7 +5 +1 −1	54 **Xe** +6 +4 +2
55 **Cs** +1	56 **Ba** +2	57 **La** +3	72 **Hf** +4	73 **Ta** +5	74 **W** +6 +4	75 **Re** +7 +6 +4	76 **Os** +8 +4	77 **Ir** +4 +3	78 **Pt** +4 +2	79 **Au** +3 +1	80 **Hg** +2 +1	81 **Tl** +3 +1	82 **Pb** +4 +2	83 **Bi** +5 +3	84 **Po** +2	85 **At** −1	86 **Rn**



Oxidation numbers of all the elements in the following ?

IF_7

F = -1

7x(-1) + ? = 0

I = +7

$NaIO_3$

Na = +1 O = -2

3x(-2) + 1 + ? = 0

I = +5

$K_2Cr_2O_7$

O = -2 K = +1

7x(-2) + 2x(+1) + 2x(?) = 0

Cr = +6

ION ELECTRON METHOD

- Way to balance redox equations
- Must balance both mass & charge
- Write ***skeleton equation***
  - Only ions & molecules involved in reaction
- Break into 2 half-reactions
  - Oxidation
  - Reduction
- Balance each half-reaction separately
- Recombine to get balanced net ionic equation

BALANCING REDOX REACTIONS

Some Redox reactions are simple:

Ex. 1 $Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2+}(aq)$

Break into half-reactions

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ **oxidation**

↑ **LEO**

Reducing agent

$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$ **reduction**

↑ **GER**

Oxidizing agent

EXAMPLE 1

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ **oxidation**

$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$ **reduction**

- Each half-reaction is balanced for atoms
  - Same # atoms of each type on each side
- Each half-reaction is balanced for charge
  - Same sum of charges on each side
- Add both equations algebraically, canceling e^-'s
- **NEVER have e^-'s in net ionic equation**

$Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2+}(aq)$

BALANCING REDOX EQUATIONS IN AQUEOUS SOLUTIONS

- Many redox reactions in aqueous solution involve H_2O and H^+ or OH^-
- Balancing the equation cannot be done by inspection.
- Need method to balance equation correctly
- Start with **acidic** solution then work to basic conditions

REDOX IN AQUEOUS SOLUTION

Ex. 2 Mix solutions of $K_2Cr_2O_7$ & $FeSO_4$

- Dichromate ion, $Cr_2O_7^{2-}$, oxidizes Fe^{2+} to Fe^{3+}
- $Cr_2O_7^{2-}$ is reduced to form Cr^{3+}
- Acidity of mixture decreases as H^+ reacts with oxygen to form water

Skeletal Eqn. $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$

Ox. # Cr = +6 Fe = +2 Cr = +3 Fe = +3



ION-ELECTRON METHOD – ACIDIC SOLUTION

1. Divide equation into 2 half-reactions
2. Balance atoms other than H & O
3. Balance O by adding H_2O to side that needs O
4. Balance H by adding H^+ to side that needs H
5. Balance net charge by adding e^-
6. Make e^- gain equal e^- loss; then add half-reactions
7. Cancel anything that is the same on both sides

ION ELECTRON METHOD

Ex. 2 Balance in Acidic Solution

$$Cr_2O_7^{2-} + Fe^{2+} \longrightarrow Cr^{3+} + Fe^{3+}$$

1. Break into half-reactions

$$Cr_2O_7^{2-} \longrightarrow Cr^{3+}$$

$$Fe^{2+} \longrightarrow Fe^{3+}$$

2. Balance atoms other than H & O

$$Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$$

- Put in 2 coefficient to balance Cr

$$Fe^{2+} \longrightarrow Fe^{3+}$$

- Fe already balanced

EX. 2 ION-ELECTRON METHOD IN ACID

3. Balance O by adding H_2O to the side that needs O.

$$Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7\ H_2O$$

- Right side has 7 O atoms
- Left side has none
- Add 7 H_2O to left side

$$Fe^{2+} \longrightarrow Fe^{3+}$$

- No O to balance

EX. 2 ION-ELECTRON METHOD IN ACID

4. Balance H by adding H^+ to side that needs H

$$14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$$

- Left side has 14 H atoms
- Right side has none
- Add 14 H^+ to right side

$$Fe^{2+} \longrightarrow Fe^{3+}$$

- No H to balance

EX. 2 ION-ELECTRON METHOD IN ACID

5- Balance net charge by adding electrons

$$6e^- + 14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$$

Net Charge = 14(+1) (−2) = 12

Net Charge = 2(+3)+7(0) = 6

- 6 electrons must be added to reactant side

$Fe^{2+} \longrightarrow Fe^{3+}$

- 1 electron must be added to product side $+ e^-$
- Now both half-reactions balanced for mass & charge

EX. 2 ION-ELECTRON METHOD IN ACID

6. Make e^- gain equal e^- loss; then add half-reactions

$$6e^- + 14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$$

$$6\times[\ Fe^{2+} \longrightarrow Fe^{3+} + e^-\ ]$$

$$\cancel{6e^-} + 6Fe^{2+} + 14H^+ + Cr_2O_7^{2-} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O + \cancel{6e^-}$$

7. Cancel anything that's the same on both sides

$$6Fe^{2+} + 14H^+ + Cr_2O_7^{2-} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

ION-ELECTRON IN BASIC SOLUTION

- The simplest way to balance an equation in **basic solution**

Use steps 1-7 above, then

8. Add the same number of OH^- to ***both*** sides of the equation as there are H^+.

9. Combine H^+ & OH^- to form H_2O

10. Cancel any H_2O that you can from both sides

EX.2 ION-ELECTRON METHOD IN BASE

Returning to our example of $Cr_2O_7^{2-}$ & Fe^{2+}

8. Add to ***both*** sides of equation the same number of OH^- as there are H^+.

$$6Fe^{2+} + 14H^+ + Cr_2O_7^{2-} + 14\,OH^- \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O + 14\,OH^-$$

9. Combine H^+ and OH^- to form H_2O.

$$6Fe^{2+} + 14H_2O + Cr_2O_7^{2-} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O + 14OH^-$$

(7 above $14H_2O$)

10. Cancel any H_2O that you can

$$6Fe^{2+} + 7H_2O + Cr_2O_7^{2-} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 14OH^-$$

YOUR TURN!

Which of the following is a correctly balanced **reduction** half-reaction?

A. $Fe^{3+} + e^- \longrightarrow Fe^\circ$

B. $2Fe + 6HNO_3 \longrightarrow 2Fe(NO_3)_3 + 3H_2$

C. $Mn^{2+} + 4H_2O \longrightarrow MnO_4^- + 8H^+ + 5e^-$

D. $2O^{2-} \longrightarrow O_2 + 4e^-$

E. $Mg^{2+} + 2e^- \longrightarrow Mg^\circ$

EX. 3 ION-ELECTRON METHOD

Balance the following equation in basic solution:

$$MnO_4^- + HSO_3^- \longrightarrow Mn^{2+} + SO_4^{2-}$$

1. Break it into half-reactions

$$MnO_4^- \longrightarrow Mn^{2+}$$

$$HSO_3^- \longrightarrow SO_4^{2-}$$

2. Balance atoms other than H & O

$$MnO_4^- \longrightarrow Mn^{2+}$$

- Balanced for Mn

$$HSO_3^- \longrightarrow SO_4^{2-}$$

- Balanced for S

EX. 3 ION-ELECTRON METHOD

3. Add H_2O to balance O

$$MnO_4^- \longrightarrow Mn^{2+} + 4H_2O$$

$$H_2O + HSO_3^- \longrightarrow SO_4^{2-}$$

4. Add H^+ to balance H

$$8H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4H_2O$$

$$H_2O + HSO_3^- \longrightarrow SO_4^{2-} + 3H^+$$

EX. 3 ION-ELECTRON METHOD

5. Balance net charge by adding e^-.

$$5e^- + 8H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4H_2O$$

$8\times(+1) + (-1) = +7$ $\quad +2 + 0 = +2$

Add 5 e^- to reactant side

$$H_2O + HSO_3^- \longrightarrow SO_4^{2-} + 3H^+ + 2\,e^-$$

$0 + (-1) = -1$ $\quad -2 + 3(+1) = +1$

Add 2 e^- to product side

EX. 3 ION-ELECTRON METHOD

6. Make e^- gain equal e^- loss

2×[$5e^- + 8H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4H_2O$ **]**

5×[$H_2O + HSO_3^- \longrightarrow SO_4^{2-} + 3H^+ + 2e^-$ **]**

- Must multiply Mn half-reaction by 2
- Must multiply S half-reaction by 5
- Now have 10 e^- on each side

EX. 3 ION-ELECTRON METHOD

6. Then add the two half-reactions

1 3

$$10e^- + 16H^+ + 2MnO_4^- \longrightarrow 2Mn^{2+} + 8H_2O$$

$$5H_2O + 5HSO_3^- \longrightarrow 5SO_4^{2-} + 15H^+ + 10e^-$$

$$10e^- + 16H^+ + 2MnO_4^- + 5H_2O + 5HSO_3^- \longrightarrow 2Mn^{2+} + 8H_2O + 5SO_4^{2-} + 15H^+ + 10e^-$$

7. Cancel anything that is the same on both sides.

$$H^+ + 2MnO_4^- + 5HSO_3^- \longrightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-}$$

Balanced in acid.

EX 3 ION-ELECTRON METHOD IN BASE

8. Add same number of OH^- to *both* sides of equation as there are H^+

$$H^+ + 2MnO_4^- + OH^- + 5HSO_3^- \longrightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-} + OH^-$$

9. Combine H^+ and OH^- to form H_2O

$$H_2O + 2MnO_4^- + 5HSO_3^- \longrightarrow 2Mn^{2+} + 2 3H_2O + 5SO_4^{2-} + OH^-$$

10. Cancel any H_2O that you can

$$2MnO_4^- + 5HSO_3^- \rightarrow 2Mn^{2+} + 2H_2O + OH^- + 5SO_4^{2-}$$

YOUR TURN!

Balance the following in Acid & Base using the Ion Electron Method.

$MnO_4^- + C_2O_4^{2-} \longrightarrow MnO_2 + CO_3^{2-}$

Acid: $2MnO_4^- + 3C_2O_4^{2-} + 2H_2O \rightarrow 2MnO_2 + 4H^+ + 6CO_3^{2-}$

Base: $2MnO_4^- + 3C_2O_4^{2-} + 4OH^- \rightarrow 2MnO_2 + 2H_2O + 6CO_3^{2-}$

$ClO^- + VO_3^- \longrightarrow ClO_3^- + V(OH)_3$

Acid: $ClO^- + 2H_2O + 2VO_3^- + 2H^+ \rightarrow ClO_3^- + 2V(OH)_3$

Base: $ClO^- + 4H_2O + 2VO_3^- \rightarrow ClO_3^- + 2V(OH)_3 + 2OH^-$